Two Rankings from One Posterior: A Bayesian Power Rating and a Wins-Above-Benchmark Resume for College Football

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Abstract

A college football ranking is asked two questions: which team is better, and which has earned more. The first is predictive and should use margin of victory; the second is retrodictive and should reward only results. We answer both from one Bayesian model. Team strength, in points on a neutral field, is observed through the regulation margin and an efficiency margin built from expected points added, modelled jointly so that one performance is not counted twice, pooled within conferences and divisional tiers, carried over between seasons with roster information, and allowed to drift from week to week as a random walk. The shape of the margin distribution is chosen among four families by leave-one-out cross-validation, and a parallel model for the total score splits strength into offence and defence. The power rating is the posterior of strength. The resume, Wins Above Benchmark, counts the wins a team has beyond those the twelfth-best team would expect against the same schedule, computed per posterior draw. We evaluate the system on every week of every season since 2014 with convergence gates, posterior predictive and prior-sensitivity checks, simulation-based calibration, ablations, and a rolling out-of-sample backtest against results, closing betting lines and totals, and baseline ratings.

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1 Introduction

Every autumn, some one hundred and thirty teams in the Football Bowl Subdivision (FBS) play about twelve games each, against opponents they largely do not choose, and at the end a committee selects twelve of them for a playoff. The schedule graph is sparse: a team plays fewer than a tenth of the other teams in its division, most of them from its own conference, and conferences are linked by a few dozen non-conference games played in September. Any ranking has to propagate information about strength through that graph, and the history of the problem is largely the history of ways to do so: least squares on scoreboard margin [1, 2], paired-comparison models for wins and losses [3, 4], eigenvector methods [5], and linear-algebraic record-only schemes such as the Colley matrix [6], one of the computer components of the Bowl Championship Series era [7, 8].

These methods disagree partly because they answer different questions. A *predictive* rating asks which team would win a game played next week, and by how much. Margin of victory is informative about that, so a predictive rating should use it, together with any other signal of how well a team played. A *retrodictive* ranking asks what a team has earned by its results. Here margin is usually excluded on purpose, so that a team is neither rewarded for running up a score nor penalised for resting its starters, and a win over a strong opponent is worth more than a win over a weak one. Combining the two into one number, as weighted composites of standardised scores do, gives a quantity that answers neither question and whose weights cannot be validated against anything observable.

This paper keeps the two apart but derives both from one posterior distribution. The contributions are four.

1. A **power model** (Section 3) for team strength in points on a neutral field, observed through the regulation scoreboard margin and through an efficiency margin built from expected points added per play [9]. The margin is modelled *conditionally* on the efficiency residual, so the two signals share one latent strength without being treated as independent evidence, and their relative weight is estimated rather than set. Strength is pooled within conferences and within three divisional tiers, carried over from the two previous seasons by estimated coefficients, shifted before the season by roster information (returning production and recruited talent), and moves through the season as a random walk whose step size is estimated. The shape of the margin distribution is one of four families, chosen by cross-validation on every complete season (Section 7.4); the published one is the scale-t. A parallel model for the total score, observed through the final score and summed offensive efficiency, splits each rating into offence and defence without changing it (Section 3.3).
2. A result on how much the efficiency signal adds (Section 4). Under a Gaussian version of the model the gain in per-game Fisher information about strength has a closed form, and at the fitted parameters it is 4.2–10.4% — small, and explained by the estimated scale of efficiency being close to the value at which the gain vanishes. An ablation backtest measures the same thing out of sample (Section 8.4).
3. A **resume** measure, Wins Above Benchmark (Section 5): for each posterior draw, the number of wins a team has beyond those the draw's twelfth-best FBS team would be expected to earn against the same opponents at the same venues. It is additive over games, never

rewards a loss, is centred at zero for a team exactly as strong as the benchmark, and inherits uncertainty in the opponents, the benchmark and the ranks directly from the posterior.

4. A **validation** of the whole system (Section 6, Section 7, Section 8) on every week of every season from 2014 to 2026: 242 weekly posteriors fitted by the No-U-Turn sampler, of which 242 pass the convergence gates; posterior predictive checks, which reproduce the spread, the blowouts and home-field advantage but miss the share of one-score games in 13 of 13 seasons; prior-sensitivity diagnostics, which flag 31 of 31 structural parameters in at least one season; simulation-based calibration; ablations and baselines; and a rolling backtest of 19,408 games, each predicted from the posterior fitted through the week before, scored against the result and the closing betting line.

Figure 1 shows how these pieces fit together. Section 2 describes the data, Section 3 and Section 4 the power model, and Section 5 the resume and the reference rankings. Section 6 describes fitting and the pipeline, Section 7 model checking, Section 8 the out-of-sample evaluation, Section 9 the 2025 season as a worked example, and Section 10 the limitations.

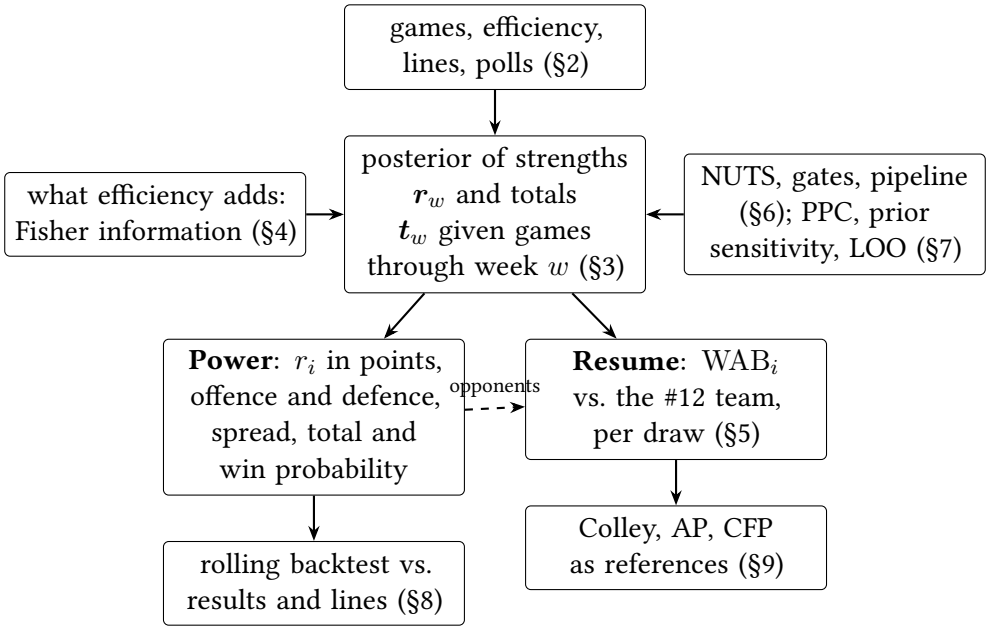


Figure 1: Structure of the paper. One posterior per week feeds both products; the dashed arrow is the only link between them: the Resume values results using opponent and benchmark strengths drawn from the Power posterior, but never uses margin. Each box names the section where it is developed.

Related work. The margin model descends from Harville [1] and Harville [10], who fit mixed linear models to scoreboard margin with random team effects, and from Stern [11], who showed that National Football League margins are close to normal around the betting line with a standard deviation near fourteen points, so that a spread determines a win probability. Glickman and Stern [12] made team strengths evolve over time in a Bayesian state-space model; the weekly random walk here is a version of theirs. Using the t distribution for robustness to blowouts follows Lange et al. [13]. On the retrodictive side, Mease [4] and Colley [6] give record-only rankings; Stern

[7] discusses why the Bowl Championship Series excluded margin from its computer rankings, and Lopez et al. [14] quantify how much of a season's outcome is noise, which is the reason a resume should report uncertainty. Dynamic paired-comparison models [15, 16] let strengths drift between games, and Annis and Craig [17] combine win–loss and margin information for college football specifically. Among published football ratings, Elo systems [18, 19] update a rating after each game, SP+ [20] and ESPN's FPI [21] are proprietary efficiency-based power ratings, and Massey's composite [22] aggregates about a hundred computer rankings. Wins Above Benchmark is the football analogue of the "wins above bubble" measure on the team sheets of the college basketball selection committee [23], with the benchmark placed at the playoff boundary. The workflow follows Gelman et al. [24].

1.1 Notation

Table 1 collects the notation. Games are indexed by g, teams by i and j, weeks by w, conferences by c and divisional tiers by $\ell \in \{1, 2, 3\}$ for FBS, the Football Championship Subdivision (FCS) and all lower divisions. Game g is played in week $w(g)$. Posterior draws are indexed by $s = 1, \ldots, S$, and a superscript (s) marks a quantity evaluated at draw s. A team's rating without a week index, r_i, is its rating at the week of the snapshot being published. The code keeps its own variable names; Table 1 gives both, and the total model of Section 3.3 has its own, listed there.

2 Data

All data come from the CollegeFootballData.com API [25]: every game involving an FBS or FCS team from 2014 to 2026, with line scores, venue, divisional classification and conference of each team; per-game advanced statistics; closing betting lines (the spread and the over/under); and the AP, Coaches and College Football Playoff (CFP) polls. Teams are keyed by the provider's identifiers throughout. The following derived quantities enter the model.

Regulation margin. y_g is the home score minus the away score after four quarters, computed from the line scores. A game decided in overtime (4.1% of games) therefore contributes a margin of zero. This is deliberate: overtime is a different game, with possession alternating from the opponent's 25-yard line, and its points say little about strength. The win still counts as a win in the resume.

Efficiency margin. For each side the provider reports the mean predicted points added (PPA) per play and the number of plays, both with garbage time excluded. PPA is the change in expected points across a play, where expected points are estimated from down, distance and field position [9]. We use the net total,

$$e_g = \overline{\mathrm{PPA}}_{h(g)}\, n_{h(g)} - \overline{\mathrm{PPA}}_{a(g)}\, n_{a(g)},$$

which is on the scale of points. It measures how well each offence moved the ball regardless of how drives ended, so it is less exposed than the scoreboard to turnovers returned for touchdowns, missed field goals and fourth-down variance. It is used only when both offences have at least 20 non-garbage plays, and is missing otherwise — in practice for most games against lower-division opponents, which the provider does not chart.

Table 1: Notation. The last column names the corresponding variable in `stan/power.stan` or `R/03_rankings.R`.

Symbol	Meaning	Code
$h(g), a(g)$	home and away team of game g	`home`, `away`
δ_g	1 for a true home game, 0 at a neutral site	`home_adv`
y_g	regulation scoreboard margin, home minus away	`margin`
e_g	efficiency margin: net expected points added	`eff`
$r_{i,w}$	strength of team i in week w, points on a neutral field	`r_state`, `r`
r_i^0	preseason (week-1) level of team i	`r0`
ϕ_r	weekly random-walk step sd of strength	`tau_r`
η	home-field advantage, points	`hfa`
μ_g	$r_{h(g),w(g)} - r_{a(g),w(g)} + \eta\delta_g$, expected margin	`mu`
λ, σ_e	scale and noise of the efficiency margin	`lambda`, `sigma_eff`
κ, σ_m	loading and scale of the margin	`kappa`, `sigma_m`
$\nu, \gamma, p_w, \sigma_w$	shape parameters of the margin family	`nu`, `gamma_m`, `p_wide`, `sigma_w`
$\bar{\sigma}$	$(\sigma_m^2 + \kappa^2\sigma_e^2)^{1/2}$, marginal margin scale	`sigma_marg`
$F(\cdot \mid \mu)$	predictive distribution function of the margin at expected margin μ	`margin_cdf`
m_ℓ, ω	tier mean ($m_1 = 0$) and conference spread	`mu_tier`, `sigma_conf`
α_c	mean strength of conference c	`mu_conf`
τ_ℓ	within-conference spread of tier ℓ	`sigma_team`
d_i, s_i, β	last season's relative rating, its posterior sd, and the carry-over	`prev_dev`, `prev_sd`, `b_prev`
$q_i, a_i, d_i^{(2)}$	returning production, talent composite, and the relative rating two seasons back	`ret`, `tal`, `prev2_dev`
$\beta_q, \gamma_q, \gamma_a, \beta_2$	their coefficients in the preseason level	`b_ret`, `c_ret`, `b_tal`, `b_prev2`
$B^{(s)}$	strength of the benchmark team in draw s	`bench`
π_g	probability that the benchmark wins game g's fixture	`p_bench`
$R_i^{(s)}$	rank of team i within its tier in draw s	`rank_draws`

Total. For the total model (Section 3.3), x_g is the final combined score and $e_g^+ = \overline{\mathrm{PPA}}_{h(g)}\, n_{h(g)} + \overline{\mathrm{PPA}}_{a(g)}\, n_{a(g)}$ the summed offensive efficiency, available exactly when e_g is. The total of a game decided in overtime is inflated by the extra possessions, so for those games x_g is treated as missing and only e_g^+ is used.

Preseason information. Three quantities from before the season enter the preseason level of each FBS team (Section 3.2): the share of last season's offensive production, measured in PPA, that returns (the provider's *returning production*); the 247Sports team talent composite, a recruiting-based measure of the roster [25], available from 2015; and the team's final rating two seasons back. Returning production is centred and the talent composite standardised within FBS each season; both are zero for FCS and lower-division teams, which the provider does not cover. A head-coach change was also built as a candidate input and not adopted (Section 8.5).

Tiers and conferences. Each team is assigned to FBS, FCS or a single pooled tier for Division II, III and NAIA opponents, and to its conference in that season. An independent is not a conference, so each independent (8 FBS programmes over the period, among them Notre Dame, Army, BYU, Massachusetts) gets a conference effect of its own, drawn from its tier like any conference's; the lower-division opponents share one.

Missing efficiency. The efficiency margin is missing for 30% of games, almost all of them involving an FCS or lower-division team that the provider does not chart. Missingness therefore depends on the tiers of the two teams, which are observed, and not on how the game went, so the data are missing at random in the sense of Rubin [26] and the likelihood of the observed data (Equation 2 for these games) is the right one to use without modelling why efficiency is missing.

Weeks. Postseason games are relabelled as one week after the last regular-season week. In the season being played, a week counts as concluded once at least 90% of its games are final and six hours have passed since its last kickoff; in a finished season, every week with at least one result counts, since the remaining games were cancelled. Only a contiguous prefix of weeks can be concluded. A posterior fitted "through week w" uses exactly the games of concluded weeks $\leq w$.

The 2020 season. The 2020 season was shortened and partly cancelled, and most FCS conferences moved their season to the spring of 2021. Those 257 spring games belong to the 2020 season in the provider's data but are numbered from week 1 again; left as they are, a posterior "through week 3" of 2020 would contain games played five months later. They are therefore placed after the fall postseason in date order, as spring weeks 1 to 11 followed by a spring postseason, and fitted and backtested like any other weeks.

Table 2 summarises the data.

* In progress: through week 9.

3 The Power Model

3.1 Likelihood

For game g let

$$\mu_g = r_{h(g),w(g)} - r_{a(g),w(g)} + \eta\, \delta_g$$

Table 2: Completed games by season. FBS–FBS games carry most of the information about FBS strength; FBS–FCS games link the two tiers. *Eff.* is the share of games with an efficiency margin; *Snap.* is the number of weekly posteriors fitted, including the preseason one from 2015.

Season	Games	FBS–FBS	FBS–FCS	FCS–FCS	Other	Eff.	Neutral	OT	Snap.
2014	1,584	760	108	639	77	54%	8%	4%	18
2015	1,536	765	105	611	55	56%	8%	4%	18
2016	1,547	760	113	620	54	55%	5%	5%	18
2017	1,549	776	98	629	46	56%	5%	4%	18
2018	1,554	772	112	611	59	57%	5%	3%	18
2019	1,621	774	114	666	67	55%	4%	4%	19
2020	850	534	36	259	21	67%	5%	4%	31
2021	1,570	770	117	635	48	56%	4%	4%	18
2022	1,598	776	120	645	57	91%	5%	5%	18
2023	1,603	792	118	635	58	93%	5%	5%	18
2024	1,669	798	121	692	58	96%	6%	4%	19
2025	1,687	808	126	691	62	98%	5%	4%	19
2026*	1,041	444	122	409	66	100%	2%	3%	10

be the expected regulation margin: the difference between the two teams' strengths in the week of the game, plus the home-field advantage when the game is not at a neutral site. Both observed signals depend on the teams only through μ_g. How strengths move from week to week is the subject of Section 3.2.

Power model likelihood. For a game with an efficiency margin,

$$e_g \mid \mu_g \sim \mathcal{N}\big(\lambda\mu_g,\ \sigma_e^2\big), \qquad y_g \mid e_g, \mu_g \sim \mathcal{D}\big(\mu_g + \kappa(e_g - \lambda\mu_g),\ \sigma_m;\ \mu_g\big), \tag{1}$$

where $\mathcal{D}(\xi, \sigma; \mu)$ is a law with location ξ and scale σ from one of the four families of Table 3. For a game without one,

$$y_g \mid \mu_g \sim \mathcal{D}\big(\mu_g,\ \bar{\sigma};\ \mu_g\big), \qquad \bar{\sigma}^2 = \sigma_m^2 + \kappa^2\sigma_e^2, \tag{2}$$

with the mixture's wide scale widened in the same way, $\bar{\sigma}_w^2 = \sigma_w^2 + \kappa^2\sigma_e^2$. Games are conditionally independent given the parameters.

Three features of Equation 1 matter. First, both signals load on the same μ_g, so strength is measured in points of scoreboard margin: λ converts it to the efficiency scale, and it is estimated. Second, the margin is modelled *given* the efficiency residual $e_g - \lambda\mu_g$. A team that outplays its expected efficiency usually outscores its expected margin too, and the loading κ absorbs that shared component. Were the two modelled as conditionally independent given μ_g, one good performance would be counted twice. Third, the margin is not normal. The Student-t lets a 63–0 result inform strength without dominating it [13]. The scale mixture puts most games in a narrow component and a minority in a wide one, which bunches margins near the expected value more than the t does. The scale-t lets the noise grow with the expected margin $|\mu_g|$: a mismatch is less

Table 3: The four margin families. $t_\nu(\xi, \sigma)$ is the Student-t law with ν degrees of freedom, location ξ and scale σ. The published family, chosen by cross-validation (Section 7.4), is the scale-t.

Family	$\mathcal{D}(\xi, \sigma; \mu)$	Extra parameters
Normal	$\mathcal{N}(\xi, \sigma^2)$	none
Student-t	$t_\nu(\xi, \sigma)$	ν
Scale mixture	$(1-p_w)\,\mathcal{N}(\xi, \sigma^2) + p_w\,\mathcal{N}(\xi, \sigma_w^2)$, $\sigma_w > \sigma$	p_w, σ_w
Scale-t	$t_\nu\big(\xi, (\sigma^2 + \gamma^2\mu^2)^{1/2}\big)$	ν, γ

predictable in size than an even game, because the favourite may empty its bench at any score. The family is a modelling choice made once, by leave-one-out cross-validation on every complete season (Section 7.4), and its parameters are estimated in every fit.

The scale $\bar{\sigma}$ in Equation 2 is chosen so that the two cases agree for the Gaussian families.

Marginal Margin Lemma. Replace $\mathcal{D}$ in Equation 1 by the normal law with the same location and scale. Then, marginally over e_g, $y_g \mid \mu_g \sim \mathcal{N}(\mu_g, \bar{\sigma}^2)$, and (e_g, y_g) is bivariate normal with mean $(\lambda\mu_g, \mu_g)$ and covariance

$$\Sigma = \begin{pmatrix} \sigma_e^2 & \kappa\sigma_e^2 \\ \kappa\sigma_e^2 & \sigma_m^2 + \kappa^2\sigma_e^2 \end{pmatrix}.$$

Proof. Write $e_g = \lambda\mu_g + \varepsilon_e$ and $y_g = \mu_g + \kappa\varepsilon_e + \varepsilon_m$ with $\varepsilon_e \sim \mathcal{N}(0, \sigma_e^2)$ and $\varepsilon_m \sim \mathcal{N}(0, \sigma_m^2)$ independent. Then (e_g, y_g) is a linear image of a Gaussian vector, with the stated mean, $\mathrm{Var}(y_g) = \kappa^2\sigma_e^2 + \sigma_m^2$ and $\mathrm{Cov}(e_g, y_g) = \kappa\sigma_e^2$. □

The same argument applied to each component shows that Equation 2 is exact for the normal and the scale mixture. Under the two t families the marginal of y_g is not exactly of the form Equation 2, so there it is an approximation that matches the location and the scale; it affects only the games without an efficiency margin, which are mostly lopsided games against lower-division teams. The same marginal law defines the predictive win probability. Write $F(\cdot \mid \mu)$ for the distribution function of $\mathcal{D}(\mu, \bar{\sigma}; \mu)$, the predictive law of a margin whose expected value is μ.

Win probability. The probability that team i beats team j at venue $v \in \{-1, 0, 1\}$ (away, neutral, home for i) is the posterior mean

$$p_{ij}(v) = \frac{1}{S}\sum_{s=1}^{S}\Big[1 - F^{(s)}\big(0 \mid \mu_{ij}^{(s)}(v)\big)\Big], \qquad \mu_{ij}^{(s)}(v) = r_i^{(s)} - r_j^{(s)} + \eta^{(s)}v,$$

where $F^{(s)}$ uses the family's parameters in draw s. The predicted spread is the posterior mean of $\mu_{ij}(v)$.

For the Student-t family $1 - F(0 \mid \mu) = F_\nu(\mu/\bar{\sigma})$, with F_ν the distribution function of the standard t_ν law: the rule of Stern [11] with the normal replaced by the fitted t and averaged over the posterior, which widens the predictive distribution by the uncertainty in the ratings. For the

scale-t it is $F_\nu\big(\mu/(\bar\sigma^2+\gamma^2\mu^2)^{1/2}\big)$, which still increases with μ but approaches $F_\nu(1/\gamma)$ rather than one as μ grows: no favourite is certain. A game h weeks after the snapshot adds the random-walk drift of both teams over those weeks (Section 3.2) to $\mu_{ij}^{(s)}$ before the average is taken.

3.2 Hierarchy, Preseason Information and the Weekly Walk

A team's strength starts the season at a preseason level, pooled at two levels — teams within conferences and conferences within tiers — and then moves from week to week.

Strength hierarchy. For team i in conference $c(i)$ of tier $\ell(i)$, the preseason (week-1) level is

$$\begin{aligned} r_i^0 &= \alpha_{c(i)} + \beta_i\, d_i + \gamma_q\, q_i + \gamma_a\, a_i + \beta_2\, d_i^{(2)} + \big(\tau^2_{\ell(i)} + \beta_i^2 s_i^2\big)^{1/2} z_i, \\ \beta_i &= \beta + \beta_q\, q_i, \qquad \alpha_c = m_{\ell(c)} + \omega\, u_c, \qquad z_i, u_c \overset{\text{iid}}{\sim} \mathcal{N}(0,1), \end{aligned} \tag{3}$$

with $m_1 = 0$. Here d_i is team i's final power rating in the previous season minus the mean final rating of its current tier, s_i is the posterior standard deviation of that rating, and $d_i = s_i = 0$ when there is no previous rating (the first season, and teams new to the data); $d_i^{(2)}$ is the same deviation two seasons back. For FBS teams, q_i is the share of last season's production that returns and a_i the standardised talent composite (Section 2), both zero for other teams and when missing. Returning production enters twice: as a level, $\gamma_q q_i$, and through the carry-over β_i, so that a team returning more of its roster may keep more of last season's deviation.

Weekly random walk. Let $w_1 < w_2 < \cdots$ be the weeks in which team i plays, and $w_0 = 1$. Its strength in those weeks is

$$r_{i,w_k} = r_{i,w_{k-1}} + \phi_r\,(w_k - w_{k-1})^{1/2}\,\varepsilon_{i,k}, \qquad r_{i,w_0} = r_i^0, \qquad \varepsilon_{i,k} \overset{\text{iid}}{\sim} \mathcal{N}(0,1), \tag{4}$$

so a team that does not play in a week accumulates that week's variance at its next game. Its rating at the week w of a snapshot is its last state plus the drift since, $r_i = r_{i,w_K} + \phi_r\,(w - w_K)^{1/2}\varepsilon_i$, with w_K its last game week up to w.

Only the weeks in which a team plays carry a state, so a season adds about a dozen parameters per team rather than one per week. The step ϕ_r is shared by all teams and estimated. It is the model's answer to how quickly a rating should forget: with $\phi_r = 0$ every game of the season counts equally, as in a static model, and the larger ϕ_r is relative to the noise of a game, the more the latest games dominate. A prediction for a game h weeks after the snapshot adds independent drift for both teams, so the variance of the expected margin grows by $2h\phi_r^2$.

The parameterisation is non-centred [27] throughout, in z_i, u_c and the steps $\varepsilon_{i,k}$: in early weeks each team has one or two games and the data barely constrain z_i, which is the regime in which a centred hierarchy produces the funnel geometry that defeats Hamiltonian samplers [28]. The random walk has a funnel of its own, between ϕ_r and the steps, and the data identify ϕ_r only weakly, which is why its prior is informative and keeps it away from zero (Section 3.4).

The carry-over β does most of the work of a preseason rating. In week 0 no game has been played and the posterior of r_i is centred on $\beta_i d_i + \gamma_q q_i + \gamma_a a_i + \beta_2 d_i^{(2)}$ plus its conference and tier means;

as games accumulate, the team effect z_i and the steps take over. The roster terms and the second season back were each adopted because they improve out-of-sample predictions in the first weeks (Section 8.5). Because d_i is measured against the tier and not the conference, it carries conference strength over as well, so preseason predictions of non-conference games are informative from the first week.

Figure 2 shows the estimated step by season. At the posterior mean of a typical complete season, $\phi_r = 1.72$ points per week: over a twelve-week regular season a team's strength drifts by about 6.0 points (one sd), small next to the spread of FBS strengths but not negligible next to the margin of a close game. Whether that drift improves predictions over static ratings is tested directly in Section 8.4.

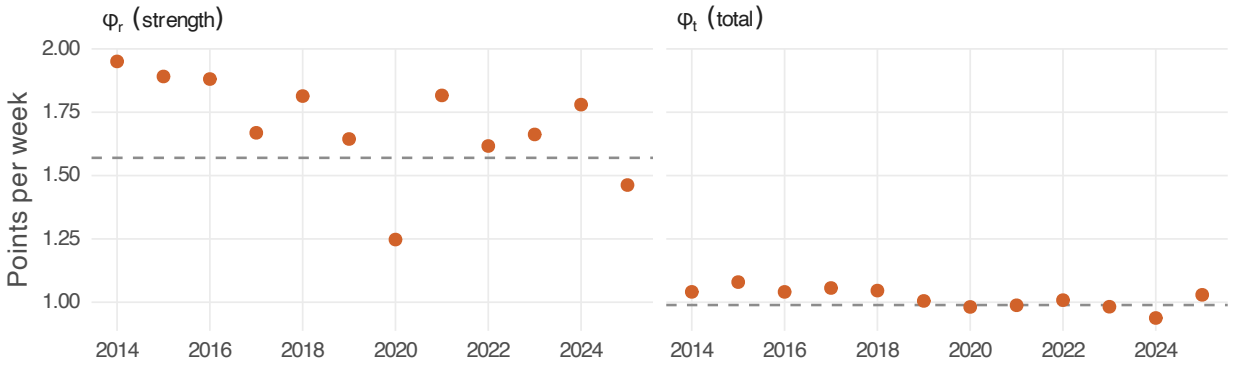


Figure 2: Weekly random-walk step sd by season, posterior mean of the published final fit: ϕ_r for strength and ϕ_t for the total tendency of Section 3.3. The dashed lines are the prior means.

Uncertainty in last season's rating. d_i is built from a posterior mean, and last season's rating is itself uncertain: the final posterior standard deviation of an FBS rating is about 5.6 points, against a spread of 13 points in d_i. Treating the true previous rating as $\mathcal{N}(d_i, s_i^2)$ and integrating it out adds $\beta^2 s_i^2$ to the team's variance, which is the extra term in Equation 3; a team whose previous rating was poorly determined is shrunk further towards its conference. Using the posterior mean as the covariate is regression calibration [29]: the posterior mean is already the best predictor of the true rating given last season's data, so the regression on it is not attenuated in the way a regression on a noisy measurement would be, and β needs no correction. The reliability $1 - \bar{s}^2 / \mathrm{Var}(d)$ is 0.71–0.85 across seasons, so even a classical measurement-error correction would change β by only a few percent.

Teams that change tier. A programme moving from FCS to FBS (11 team-seasons since 2014, among them Charlotte, Coastal Carolina, Liberty, James Madison, Jacksonville State) has a previous rating on the FBS scale far below the FBS mean, so its d_i is strongly negative and its preseason rating keeps its absolute level rather than jumping to an average FBS team's. Its conference effect and τ_1 are the new tier's. In weeks 1–4 of a transition season the out-of-sample error on these teams' FBS games is 13.7 points (24 games), against 13.1 for other FBS games in the same weeks: larger, as expected for a team whose schedule changes completely, but not badly biased (mean error -2.0 points).

One conference spread for all tiers. The team spreads τ_ℓ differ by tier, but the conference spread ω is shared. FBS and FCS each have only about a dozen conferences a season, and the lower divisions are one node, so a tier-specific ω_ℓ would be estimated from a dozen values and would lean on its prior. The fitted conference means support pooling: at the end of 2025 their standard deviation within FBS is 1.2 points and within FCS 1.1.

One home-field advantage. η is one number per season for every true home game, FBS or FCS, near or far. Travel, altitude and crowd size surely vary, but the data identify one average well and many small effects poorly. Figure 3 shows the estimate by season: it is stable at about 2.5 points, except in 2020, when most games were played without crowds and it falls to 1.5.

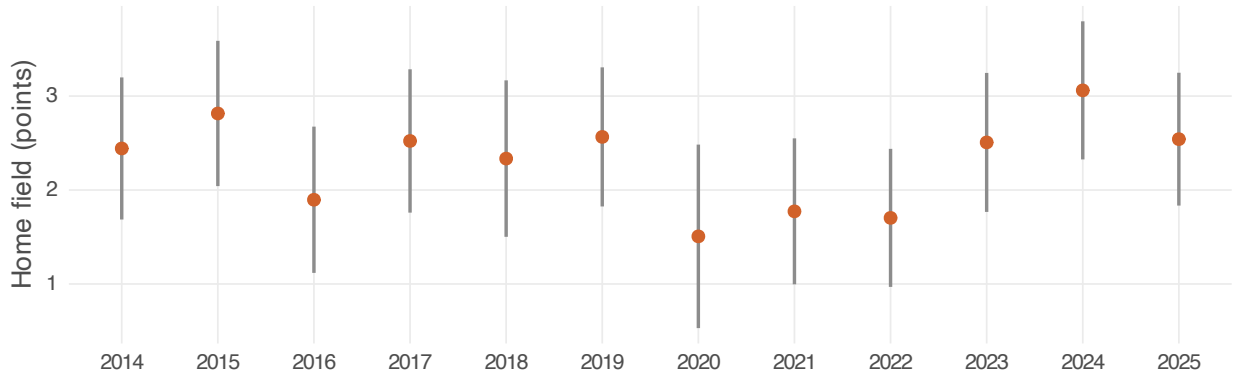


Figure 3: Home-field advantage η by season: posterior mean and ± 2 sd of the full-season fit under vague priors (Section 3.4). The 2020 season was played largely without crowds.

Remark (identification). The likelihood depends on the strengths only through differences $r_{i,w} - r_{j,w}$ within a connected component of the schedule graph, so it is flat along one direction per component. Every prior in the model is proper (Table 5), so the posterior is proper regardless, and the published FBS-centred ratings $\tilde{r}_i = r_i - \bar{r}_{\text{FBS}}$ do not depend on the anchor $m_1 = 0$. Between components the relative strengths come from the prior alone — from the carry-over and the tier and conference structure — which is what happens in the first weeks.

Table 4 counts the components. In week 1 the FBS teams fall into dozens of pieces; the graph of teams that have played becomes connected within the FBS by week 4 in a typical season, and by the end of every season except 2020 the whole graph, all three tiers, is one component. In 2020 a few teams (1 small component(s)) played only among themselves, and their ratings relative to everyone else rest on the prior. Figure 4 draws one season's graph.

Table 4: Connected components of the schedule graph among teams that have played, by season: after week 1, at the first snapshot in which all FBS teams are connected, and at the end of the season (all tiers).

Season	Week 1	FBS connected from week	Final, all tiers
2014	0 FBS	4	1
2015	0 FBS	4	1
2016	1 FBS	1	1
2017	5 FBS	4	1
2018	4 FBS	4	1
2019	2 FBS	4	1
2020	0 FBS	18	2
2021	5 FBS	4	1
2022	11 FBS	4	1
2023	7 FBS	4	1
2024	4 FBS	4	1
2025	5 FBS	4	1
2026	8 FBS	3	1

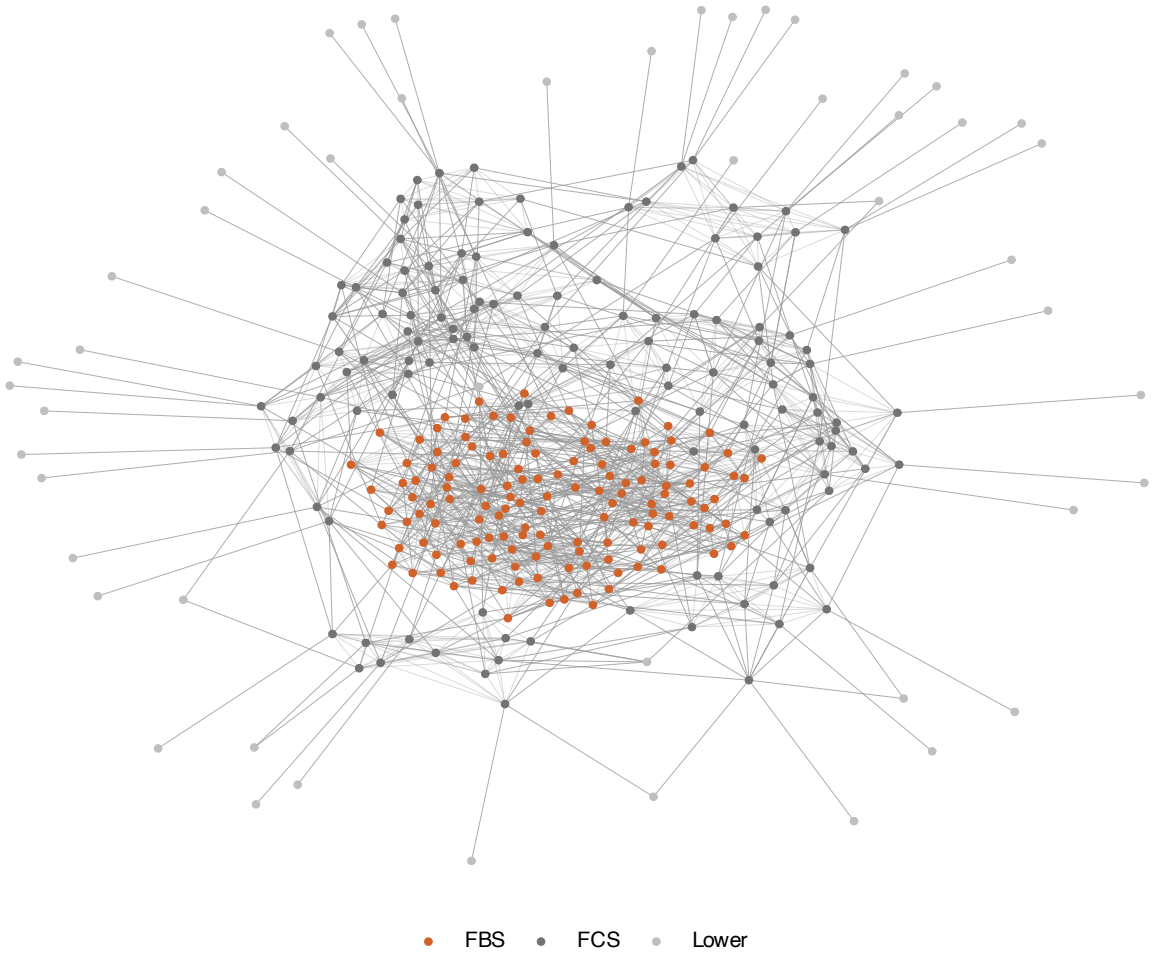


Figure 4: The 2025 schedule graph, all completed games. Points are teams (FBS in orange, FCS in grey, lower divisions in light grey), placed by a force-directed layout so that conferences form clusters; lines are games. Non-conference games are the few long edges that tie the clusters together.

Published ratings are the centred $\tilde{r}_i$, computed draw by draw, so a rating of $+10$ means ten points better than an average FBS team on a neutral field.

Figure 5 draws the full model as a directed acyclic graph.

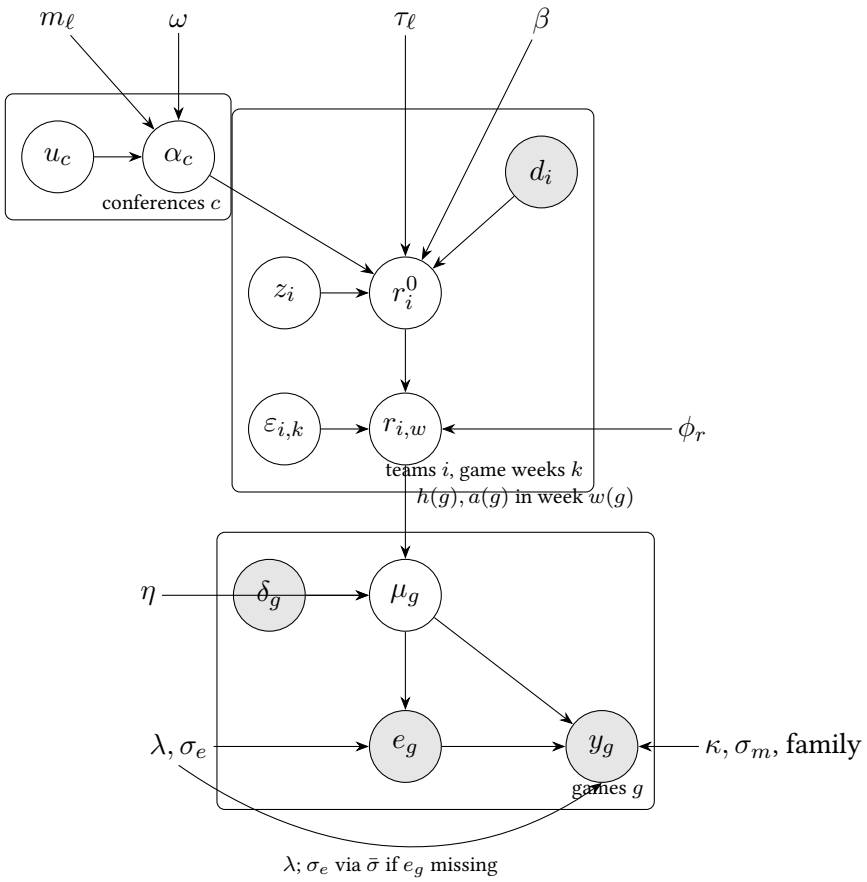


Figure 5: The power model. Shaded nodes are observed; plates repeat over conferences, teams (and, for $\varepsilon_{i,k}$ and $r_{i,w}$, the weeks in which a team plays) and games; s_i, the posterior sd of d_i, enters r_i^0 alongside d_i and is not drawn. "Family" stands for the shape parameters of the margin family (ν, γ, or p_w and σ_w). The total model of Section 3.3 has the same structure with its own parameters and is not drawn. The efficiency margin e_g is a parent of the scoreboard margin y_g, which is how the conditional likelihood of Equation 1 avoids counting one performance twice. The curved edge carries two dependences: the residual $e_g - \lambda\mu_g$ enters y_g through λ, and when e_g is missing the edge $e_g \to y_g$ is marginalised as in Equation 2, so σ_e enters y_g through $\bar{\sigma}$.

3.3 Totals, Offence and Defence

The margin says which team is better but not how many points a game will produce. A second latent quantity per team supplies that: a *total tendency* $t_{i,w}$, how far above or below the average the scores of team i's games run. It has the same structure as strength — a preseason level t_i^0 pooled within conferences (spread ω_t) and tiers (offsets m_2^t, m_3^t), a within-conference spread τ^t, a carry-over β_t of last season's final total tendency measured against its tier, and a weekly random

walk with its own step ϕ_t — and is observed in the same way as strength, through a scoreboard quantity conditioned on an efficiency one.

Total model. With $T_g = t_{h(g),w(g)} + t_{a(g),w(g)}$, for a game with an efficiency margin

$$e_g^+ \mid T_g \sim \mathcal{N}\big(\alpha_s + \lambda_t T_g,\ \sigma_s^2\big), \qquad x_g \mid e_g^+, T_g \sim \mathcal{N}\big(2\theta + T_g + \kappa_t(e_g^+ - \alpha_s - \lambda_t T_g),\ \sigma_x^2\big), \quad (5)$$

and for a game without one $x_g \mid T_g \sim \mathcal{N}(2\theta + T_g, \sigma_x^2 + \kappa_t^2\sigma_s^2)$. Here θ is the points each team scores in an average game.

Strength and total tendency together give each team an offence and a defence. Writing $\mathrm{off}_i = (r_i + t_i)/2$ and $\mathrm{def}_i = (r_i - t_i)/2$, the expected scores of game g are

$$\text{home: } \theta + \tfrac{1}{2}(\mu_g + T_g), \qquad \text{away: } \theta + \tfrac{1}{2}(T_g - \mu_g),$$

whose difference is the expected margin μ_g and whose sum is the expected total $2\theta + T_g$; the home-field advantage, part of μ_g, adds $\eta/2$ to the home score and takes $\eta/2$ from the away score. The published **Off** and **Def** are the points a team would score and allow against an average FBS team on a neutral field, $\mathrm{Off}_i = \theta + \frac{1}{2}(\tilde{r}_i + t_i + \bar{t})$ and $\mathrm{Def}_i = \theta + \frac{1}{2}(t_i + \bar{t} - \tilde{r}_i)$, with $\bar{t}$ the mean FBS total tendency, computed draw by draw. By construction $\mathrm{Off}_i - \mathrm{Def}_i = \tilde{r}_i$, the power rating.

The total model shares no parameter with the margin model of Equation 1, and its likelihood involves only the total tendencies, so the joint posterior factorises: adding it changes neither the power rating nor the resume, only what is reported alongside them. The price of that separation is that the noise of the margin and the noise of the total are modelled as independent, and that overtime games inform the total only through e_g^+. The total is scored against the closing over/under in Section 8.2.

3.4 Priors

The structural parameters — tier offsets, spreads, the carry-over and the preseason terms, home field, the likelihood scales, the random-walk steps and the total model — describe the sport rather than any team. In the first weeks of a season, with about one game per team, the data cannot estimate them, and a vague prior leaves the sampler in a funnel. They are therefore given informative priors, set from history. Each complete season's final snapshot (about 1,550 games) is refitted under deliberately vague priors (Section 11.6); the refits are the `prior_refits` target of the pipeline. The season-by-season posterior means $\hat{\theta}_s$, with posterior standard deviations v_s, are then pooled as in a random-effects meta-analysis. Each prior is centred at the weighted mean of the $\hat{\theta}_s$, with weights $w_s = 2^{-|s-s^*|/h}$ that halve every $h = 3$ seasons away from the season s^* the prior is for, and its standard deviation is the predictive standard deviation for a new season,

$$\sigma_{\text{prior}}^2 = \max\big(\hat{\sigma}_b^2 - \bar{v}^2,\ 0\big) + \hat{\sigma}_b^2/n_{\text{eff}}, \quad (6)$$

where $\hat{\sigma}_b$ is the weighted between-season standard deviation of the $\hat{\theta}_s$, $\bar{v}^2$ the weighted mean of the v_s^2 and $n_{\text{eff}} = (\sum_s w_s)^2 / \sum_s w_s^2$. The first term is the season-to-season variation in excess of what estimation noise alone produces; the second is the uncertainty of the centre. The recency

Table 5: Priors (normal, truncated to the support where a parameter is constrained) and the range over complete seasons (2014–2025) of the full-season posterior mean, refitted under vague priors and as published. The last column is the median posterior sd of the vague-prior refits. The FBS tier offset is fixed at zero.

Parameter	Prior	Vague priors	Published	Post. sd
η (home field)	$\mathcal{N}(2.44, 0.37^2)$	1.51 to 3.06	2.05 to 2.76	0.38
β (carry-over)	$\mathcal{N}(0.59, 0.04^2)$	0.48 to 0.68	0.57 to 0.64	0.07
λ	$\mathcal{N}(0.79, 0.02^2)$	0.68 to 0.82	0.74 to 0.81	0.02
κ	$\mathcal{N}(0.75, 0.02^2)$	0.72 to 0.82	0.73 to 0.79	0.02
σ_e	$\mathcal{N}(14.00, 0.17^2)$	12.46 to 14.73	13.68 to 14.17	0.34
σ_m	$\mathcal{N}(6.60, 0.40^2)$	5.89 to 8.27	6.28 to 7.53	0.35
m_2 (FCS)	$\mathcal{N}(-28.40, 2.15^2)$	-30.12 to -26.70	-29.16 to -27.10	2.08
m_3 (lower)	$\mathcal{N}(-60.40, 4.85^2)$	-66.63 to -56.01	-63.79 to -57.91	4.78
ω	$\mathcal{N}(2.43, 0.33^2)$	1.43 to 10.01	2.31 to 3.59	1.14
τ_1 (FBS)	$\mathcal{N}(5.00, 1.25^2)$	2.00 to 8.55	3.56 to 7.50	1.10
τ_2 (FCS)	$\mathcal{N}(4.98, 0.48^2)$	2.51 to 10.61	4.73 to 6.55	1.20
τ_3 (lower)	$\mathcal{N}(11.40, 1.71^2)$	5.51 to 19.31	10.51 to 12.87	3.57

weights let the priors follow trends, such as the fall in scoring after the clock rules of 2023. An earlier version used equal weights and twice the between-season standard deviation, which counts estimation noise as variation and made the priors wider than the history supports; both choices were made on out-of-sample predictions of the first weeks (Section 8.5).

Two seasons are left out of the pooling. In 2014, the first season of the data, there is no carry-over ($d_i = 0$ for every team), so the conference effects absorb team strength that later seasons attribute to βd_i, and ω is 10.0 points against 2.8 in a typical later season; it measures a different quantity. And 2020, played largely without crowds, is left out for η (Figure 3). Table 5 lists the priors with the range of the full-season posterior means under the vague priors and under the published priors.

Some parameters vary between seasons and others hardly at all, and the table cannot distinguish real change from estimation noise by itself. Comparing the two standard deviations can: for 7 of the 12 parameters (β, m_2, m_3, ω, σ_e, τ_2 and τ_3) the between-season standard deviation of the posterior means is less than 1.3 times the typical within-season posterior standard deviation, which is about what sampling noise alone would produce. For these the first term of Equation 6 vanishes and the prior standard deviation is the uncertainty of the centre alone: the history gives no evidence that the parameter changes from season to season.

The remaining parameters (Table 6) are pooled in the same way. The degrees of freedom ν get the Gamma prior with the pooled mean and standard deviation, truncated to $\nu > 1$. The random-walk steps are refitted under a wide prior centred away from zero, $\phi_r \sim \mathcal{N}^+(0.8, 0.8^2)$ and $\phi_t \sim \mathcal{N}^+(1, 0.8^2)$: one with its mode at zero puts the sampler in the funnel between the step size and the steps (Section 3.2). Before this version the refits kept the published step prior, $\phi_r \sim \mathcal{N}^+(0.6, 0.25^2)$; a season's posterior standard deviation of ϕ_r is about as large as that prior's,

so pooling under it largely returned the prior. Under the wide prior the full-season estimates of ϕ_r range from 0.95 to 2.17 points per week. The total-model priors and the coefficients of the preseason terms are pooled likewise.

The priors are empirical Bayes at the level of the sport, and two consequences need checking. First, a season's own data help set its prior, so saying that its posterior lies inside that prior would be circular. Section 11.6 therefore also builds, for each season, the prior pooled from the *other* seasons only, and asks whether the season's own estimate lies inside it. Second, the early-week backtest of Section 8 for, say, 2016 uses priors informed by later seasons. Section 8.5 refits those weeks under leave-one-season-out priors and measures the difference. Section 11.7 gives the priors they induce on team strengths, win probabilities and margins, and Section 11.11 refits the model under widened and shifted priors.

4 What the Efficiency Signal Adds

The efficiency margin is expensive — it needs play-by-play data — and a natural question is how much it sharpens the estimate of strength once the scoreboard is known. The Gaussian version of the model answers it in closed form.

Information Proposition. In the Gaussian model of the Marginal Margin Lemma, the Fisher information about μ_g carried by one game is

$$\mathcal{I}_y = \frac{1}{\Sigma_{22}} = \frac{1}{\sigma_m^2 + \kappa^2 \sigma_e^2}$$

from the margin alone, and

$$\mathcal{I}_{e,y} = \frac{\lambda^2}{\sigma_e^2} + \frac{(1-\kappa\lambda)^2}{\sigma_m^2} = \mathcal{I}_y + \frac{(\lambda - \lambda^*)^2}{\Sigma_{11} - \Sigma_{12}^2/\Sigma_{22}}, \qquad \lambda^* = \frac{\Sigma_{12}}{\Sigma_{22}} = \frac{\kappa\sigma_e^2}{\sigma_m^2 + \kappa^2\sigma_e^2}, \tag{7}$$

from both signals. Hence $\mathcal{I}_{e,y} \geq \mathcal{I}_y$, with equality exactly when $\lambda = \lambda^*$.

Proof. For a Gaussian vector with mean $\boldsymbol{a}\mu$ and known covariance Σ the information about μ is $\boldsymbol{a}^\top \Sigma^{-1} \boldsymbol{a}$. With $\boldsymbol{a} = (\lambda, 1)^\top$, factor the joint density as $p(y)\, p(e \mid y)$: the margin contributes $1/\Sigma_{22}$, and $e \mid y$ is normal with mean $\lambda\mu + (\Sigma_{12}/\Sigma_{22})(y - \mu)$ and variance $\Sigma_{11} - \Sigma_{12}^2/\Sigma_{22}$, whose derivative in μ is $\lambda - \lambda^*$. Adding the two informations gives the second form of Equation 7. The first form follows by factoring instead as $p(e)\, p(y \mid e)$: e contributes λ^2/σ_e^2 and $y \mid e$ has mean $(1 - \kappa\lambda)\mu + \kappa e$ and variance σ_m^2. □

The proposition says that the efficiency margin helps exactly insofar as its scale λ differs from λ^*, the slope of the regression of e on y. If efficiency were nothing but a noisy copy of the scoreboard, its mean would move with μ exactly as the regression predicts from y, and it would add nothing.

Example. At the end of 2025 the posterior means are $\lambda = 0.780$, $\kappa = 0.726$, $\sigma_e = 13.94$ and $\sigma_m = 6.90$. The margin alone then has scale $\bar{\sigma} = (\sigma_m^2 + \kappa^2\sigma_e^2)^{1/2} = 12.25$ points and carries $\mathcal{I}_y = 1/\bar{\sigma}^2 = 6.66 \times 10^{-3}$ per game. The slope at which efficiency would add nothing is

Table 6: Priors of the margin family, the random walk, the total model and the preseason terms, pooled from the refits like those of Table 5: normal, truncated to the support where a parameter is constrained, unless stated. The last column is the range over complete seasons of the full-season posterior mean under the vague priors of Section 11.6. Shape parameters of families other than the published scale-t and preseason terms the published model does not use are omitted.

Parameter	Meaning	Prior	Vague-prior refits
ν	degrees of freedom	$\mathrm{Gamma}(56.2, 1.83)$, $\nu > 1$	16.21 to 55.76
γ	growth of the scale with $\lvert\mu\rvert$	$\mathcal{N}(0.28, 0.01^2)$	0.20 to 0.33
ϕ_r	weekly step of strength	$\mathcal{N}(1.57, 0.28^2)$	0.95 to 2.17
ϕ_t	weekly step of the total tendency	$\mathcal{N}(0.99, 0.11^2)$	0.52 to 1.70
θ	points per team, average game	$\mathcal{N}(27.20, 0.64^2)$	26.27 to 28.78
m_2^t	FCS total offset	$\mathcal{N}(-0.98, 0.82^2)$	-3.62 to 0.23
m_3^t	lower-division total offset	$\mathcal{N}(0.10, 4.98^2)$	-12.22 to 6.93
ω_t	conference spread of totals	$\mathcal{N}(0.77, 0.08^2)$	0.54 to 1.19
τ^t	team spread of totals	$\mathcal{N}(4.05, 0.24^2)$	2.64 to 6.72
β_t	carry-over of totals	$\mathcal{N}(0.48, 0.12^2)$	0.33 to 0.80
λ_t	scale of summed efficiency	$\mathcal{N}(1.08, 0.11^2)$	0.86 to 1.29
α_s	intercept of summed efficiency	$\mathcal{N}(14.30, 1.41^2)$	12.80 to 20.41
σ_s	noise of summed efficiency	$\mathcal{N}(16.60, 0.54^2)$	15.38 to 17.84
κ_t	loading of the total on it	$\mathcal{N}(0.46, 0.03^2)$	0.42 to 0.58
σ_x	noise of the total	$\mathcal{N}(12.80, 0.32^2)$	12.16 to 13.38
β_q	carry-over per unit returning production	$\mathcal{N}(0.11, 0.07^2)$	-0.23 to 0.42
γ_q	level per unit returning production	$\mathcal{N}(7.04, 4.11^2)$	1.43 to 16.36
γ_a	level per sd of talent	$\mathcal{N}(1.87, 0.35^2)$	-0.03 to 3.77
β_2	carry-over from two seasons back	$\mathcal{N}(0.14, 0.04^2)$	-0.00 to 0.39

Table 7: Per-game Fisher information about the expected margin under the Gaussian model, at each season's final posterior means. R^2 is the share of the margin's variance explained by the efficiency residual, $\kappa^2\sigma_e^2/\bar{\sigma}^2$.

Season	λ	λ^*	R^2	$10^3\mathcal{I}_y$	$10^3\mathcal{I}_{e,y}$	Gain
2014	0.738	0.869	0.65	6.23	6.49	4.2%
2015	0.783	0.964	0.73	6.48	7.11	9.8%
2016	0.762	0.927	0.70	6.26	6.73	7.4%
2017	0.780	0.933	0.71	6.32	6.75	6.7%
2018	0.766	0.915	0.72	5.88	6.29	7.0%
2019	0.771	0.955	0.71	6.54	7.15	9.2%
2020	0.792	0.978	0.74	6.55	7.24	10.4%
2021	0.786	0.914	0.69	6.22	6.49	4.4%
2022	0.784	0.979	0.71	6.71	7.36	9.7%
2023	0.805	0.962	0.74	6.39	6.86	7.3%
2024	0.798	0.968	0.72	6.68	7.20	7.8%
2025	0.780	0.941	0.68	6.66	7.08	6.3%

$\lambda^* = \kappa\sigma_e^2/\bar{\sigma}^2 = 0.941$, 0.160 from the fitted λ. The second term of Equation 7 is $(\lambda - \lambda^*)^2/(\sigma_e^2 - \kappa^2\sigma_e^4/\bar{\sigma}^2) = 0.417 \times 10^{-3}$, so both signals together carry $\mathcal{I}_{e,y} = 7.08 \times 10^{-3}$: a gain of 6.3%, or about 0.75 extra games' worth of information over a twelve-game season.

Table 7 evaluates Equation 7 at the posterior means of each complete season's final fit.

The efficiency residual explains 65%–74% of the margin's variance, so the two signals are strongly related, and the fitted λ lies within about 0.19 of λ^*. The per-game information gain is correspondingly small: 4.2% to 10.4%, about 0.89 games' worth over a twelve-game season. Three caveats limit how far this should be read. The calculation is Gaussian, while the published margin law is the scale-t, with fitted ν from 29 to 32, whose scale grows with the expected margin, so that σ_m is the scale of an even game and a lopsided one carries less information than the calculation assigns it; under a heavy-tailed likelihood the efficiency residual also helps decide *which* large margins are outliers, which a Fisher-information calculation at the mean does not capture. It holds λ, κ and the scales fixed, whereas in the fit they are estimated jointly with strength. And it is per game: it does not measure any robustness the efficiency margin brings to the early weeks. The backtest is the arbiter. Section 8.4 refits the model without the efficiency likelihood on every week of 2022–2025: on the same 6,557 games, the margin-only model's mean absolute error is -0.05 points (standard error 0.01) lower and its Brier score -0.9 $\times 10^{-3}$ (se 0.3) lower. At each season's final posterior the expected log predictive density of the margin itself (Section 7.3) favours the full model by 11 in a typical season (full minus margin-only : -2 to 16 across seasons, standard errors about 4).

5 The Resume

5.1 Wins Above Benchmark

A resume should value results, not performances, and should value them relative to a standard that matters for selection. The playoff takes twelve teams, so the natural standard is the twelfth-best team.

Benchmark. In posterior draw s, the benchmark strength $B^{(s)}$ is the twelfth largest of the FBS strengths $\{r_i^{(s)} : \ell(i) = 1\}$.

The benchmark is a strength, not a team: in different draws it is a different team, and its uncertainty is part of the posterior.

Wins Above Benchmark. Let team i have played games $g \in G_i$, with result $W_g \in \{0, 1\}$ (overtime wins count), opponent $o(g)$ and venue $v_g \in \{-1, 0, 1\}$ from i's side. The probability that the benchmark would have won the same fixture is

$$\pi_g^{(s)} = 1 - F^{(s)}\big(0 \mid B^{(s)} - r_{o(g)}^{(s)} + \eta^{(s)} v_g\big),$$

the win probability of Section 3.1 with the benchmark in team i's place, and

$$\mathrm{WAB}_i^{(s)} = \sum_{g \in G_i} \big(W_g - \pi_g^{(s)}\big). \tag{8}$$

The published value is the posterior mean $\mathrm{WAB}_i = S^{-1} \sum_s \mathrm{WAB}_i^{(s)}$ with an 80% interval.

Equation 8 uses the scoreboard only through W_g: margin enters the resume solely by having informed the strengths of the opponents and the benchmark. Both are taken at the week of the snapshot, not at the week of the game: an opponent is valued at what the posterior now believes it to be, so a win over a team that improves later in the season gains value as it does. With the random walk the two can differ, by about $\phi_r \, |w - w(g)|^{1/2}$ points in sd. The following properties are elementary but they are the reason for the choice.

WAB Properties. For every draw s:

(i) *Additivity.* $\mathrm{WAB}_i^{(s)}$ is a sum of per-game contributions $W_g - \pi_g^{(s)}$, each in $(-1, 1)$.

(ii) *Monotonicity.* A win contributes $1 - \pi_g^{(s)} > 0$ and a loss $-\pi_g^{(s)} < 0$; with the draw held fixed, changing any loss to a win raises $\mathrm{WAB}_i^{(s)}$ by exactly one, whatever the opponent.

(iii) *Schedule response.* The contribution of a win is increasing in the opponent's strength and that of a loss is increasing too (a loss to a strong team costs less), and both are decreasing in the venue v_g: road wins count for more than home wins.

(iv) *Calibration.* If the model is correct and team i has strength $r_i^{(s)} = B^{(s)}$, then $\mathbb{E}[\mathrm{WAB}_i^{(s)} \mid \text{parameters}] = 0$ for every schedule.

Table 8: Wins Above Benchmark, game by game, for Notre Dame in 2025. H, A, N: home, away, neutral site. $\bar{\pi}_g$: probability that the twelfth-best team would win the same fixture.

Week	Opponent	Venue	Score	W_g	$\bar{\pi}_g$	$W_g - \bar{\pi}_g$	Running WAB
Week 1	Miami	A	24–27	0	0.272	-0.272	-0.27
Week 3	Texas A&M	H	40–41	0	0.660	-0.660	-0.93
Week 4	Purdue	H	56–30	1	0.958	0.042	-0.89
Week 5	Arkansas	A	56–13	1	0.818	0.182	-0.71
Week 6	Boise State	H	28–7	1	0.892	0.108	-0.60
Week 7	NC State	H	36–7	1	0.826	0.174	-0.43
Week 8	USC	H	34–24	1	0.631	0.369	-0.06
Week 10	Boston College	A	25–10	1	0.932	0.068	0.01
Week 11	Navy	H	49–10	1	0.887	0.113	0.12
Week 12	Pittsburgh	A	37–15	1	0.770	0.230	0.35
Week 13	Syracuse	H	70–7	1	0.975	0.025	0.38
Week 14	Stanford	A	49–20	1	0.910	0.090	0.47

Proof. Parts (i) and (ii) are immediate from Equation 8, since $0 < \pi_g^{(s)} < 1$. For (iii), $1 - F(0 \mid \mu)$ is increasing in μ for every family of Table 3 (for the scale-t because $\mu/(\bar{\sigma}^2 + \gamma^2\mu^2)^{1/2}$ is increasing), so $\pi_g^{(s)}$ is decreasing in $r_{o(g)}$ and increasing in v_g, and both contributions have the form const $- \pi_g^{(s)}$. For (iv), under the model $\Pr(W_g = 1) = 1 - F(0 \mid r_i - r_{o(g)} + \eta v_g) = \pi_g^{(s)}$ when $r_i = B^{(s)}$, so each term has expectation zero. □

Property (ii) holds draw by draw. The *published* WAB_i changes by a slightly different amount when a result flips, because the result is also data for the posterior: a win moves the opponent's strength down, and the benchmark with it, so the new draws differ from the old ones. Section 5.5 measures a related effect of the margin. Property (ii) excludes what the old composite resume did, rewarding a team for *playing* strong opponents regardless of result; (iii) is the precise sense in which the schedule is counted once; and (iv) gives the scale its meaning: a positive WAB is evidence of having done better than a bubble team would have with the same games. Summing over games makes WAB a count, not a rate, so a team that plays a thirteenth game can gain from it. That is intended: a conference championship game is a real opportunity to earn or lose standing.

Example. Table 8 works through WAB for Notre Dame, 10–2 and ranked 12 in the final 2025 resume. For each game, $\bar{\pi}_g$ is the posterior mean probability that the benchmark would have won the same fixture at the same venue; the contribution is $W_g - \bar{\pi}_g$, and the published WAB is their sum, 0.47. (The mean of the per-draw sums equals the sum of the per-game means, so the table reproduces the published value exactly.) The strength of record of Section 5.2 is the probability that a benchmark team with these independent chances wins at least 10 of the 12 games; the Poisson-binomial recursion over the column $\bar{\pi}_g$ gives 0.535.

5.2 Strength of Record and Ranks

Two further quantities are reported with the resume. The **strength of record** is the probability that the benchmark would do at least as well as team i did, $\mathrm{SOR}_i = \Pr\big(\sum_{g\in G_i} B_g \geq \sum_{g\in G_i} W_g\big)$ with independent $B_g \sim \mathrm{Bernoulli}(\bar\pi_g)$, $\bar\pi_g$ the posterior mean of $\pi_g^{(s)}$. The sum of independent Bernoulli variables with unequal probabilities is Poisson-binomial, and its distribution is computed exactly by the recursion

$$P_0^{(0)} = 1, \qquad P_k^{(t)} = (1-\bar\pi_t)P_k^{(t-1)} + \bar\pi_t P_{k-1}^{(t-1)}, \quad k = 0,\dots,t,$$

over a team's $t = 1,\dots,|G_i|$ games, in $O(|G_i|^2)$ operations [30]. Smaller is more impressive. **Schedule strength** is the benchmark's expected loss rate, $1 - |G_i|^{-1}\sum_g \bar\pi_g$.

Two choices in this definition need defending. First, SOR plugs in the posterior mean $\bar\pi_g$ rather than averaging the per-draw probabilities $\Pr(\sum B_g^{(s)} \geq \sum W_g)$ as WAB does. The two differ because the tail probability is non-linear in π_g; refitting each complete season's final week with every draw kept, the largest difference over all FBS teams and seasons is 0.021 and the rank correlation of the two versions is at least 0.999, so the published, cheaper version loses nothing. Second, "at least as well" counts ties as the benchmark matching the team. The strict version, $\Pr(\sum B_g > \sum W_g)$, is smaller for every team, but orders the teams almost identically (rank correlation at least 0.987); the weak inequality is kept because it makes $\mathrm{SOR} = 1$ for a winless team rather than for no team, which reads naturally as "anyone would have done at least this well".

Uncertainty in ranks is reported by ranking within each draw.

Posterior rank summaries. Let $R_i^{(s)}$ be the rank of $\mathrm{WAB}_i^{(s)}$ among the teams of i's tier in draw s (1 is best). The published rank orders the posterior means WAB_i; the 80% rank interval is the 10% and 90% quantiles of $\{R_i^{(s)}\}_s$; and $\Pr(\text{top } K) = S^{-1}\sum_s \mathbf{1}\{R_i^{(s)} \leq K\}$ for $K = 1, 4, 12$. The power rank is defined in the same way from $\tilde r_i^{(s)}$.

The published rank orders posterior means rather than taking the median rank, because the median ranks of different teams need not form a permutation.

5.3 Record-only Reference: the Colley Matrix

The model's resume uses opponent strengths from a model that sees margins. As a reference that sees only wins and losses we compute the Colley rating [6, 8].

Colley rating. Let n_i be team i's games, w_i and l_i its wins and losses, and n_{ij} the number of games between i and j. The Colley ratings $\boldsymbol{c}$ solve $C\boldsymbol{c} = \boldsymbol{f}$ with

$$C_{ii} = 2 + n_i, \qquad C_{ij} = -n_{ij}\ (i \neq j), \qquad f_i = 1 + \tfrac{1}{2}(w_i - l_i).$$

Colley Lemma. C is symmetric positive definite, so $\boldsymbol{c}$ exists and is unique, and $\frac{1}{n}\sum_i c_i = \frac{1}{2}$.

Table 9: Sensitivity of the final resume to the benchmark rank, relative to $b = 12$, over complete seasons. Rank correlation of WAB among FBS teams; teams entering or leaving the top 4 and the top 12; the largest rank change within the $b = 12$ top 25. Median over seasons, with the range.

Benchmark	ρ with $b = 12$	Top-4 changes	Top-12 changes	Max move, top 25
$b = 4$	0.989 (0.983–0.994)	0 (0–2)	1 (0–2)	10 (6–14)
$b = 25$	0.993 (0.986–0.996)	0 (0–1)	1 (0–2)	6 (4–10)

Proof. $C = 2I + L$, where L is the Laplacian of the schedule multigraph, which is positive semidefinite; so C is positive definite. Every column of L sums to zero, so $\mathbf{1}^\top C = 2\mathbf{1}^\top$ and $2\,\mathbf{1}^\top \boldsymbol{c} = \mathbf{1}^\top C\boldsymbol{c} = \mathbf{1}^\top \boldsymbol{f} = n + \frac{1}{2}\sum_i (w_i - l_i) = n$, since every game produces one win and one loss. □

The rating is Laplace's rule of succession, $(1 + w_i)/(2 + n_i)$, adjusted for the average rating of the opponents. We compute it on the same games as the posterior and rank within tier.

Example. Four teams play four games: A beats B and C, B beats C, and D beats C; A and D do not meet. Then A is 2–0, B 1–1, C 0–3 and D 1–0, and

$$C = \begin{pmatrix} 4 & -1 & -1 & 0 \\ -1 & 4 & -1 & 0 \\ -1 & -1 & 5 & -1 \\ 0 & 0 & -1 & 3 \end{pmatrix}, \qquad \boldsymbol{f} = \begin{pmatrix} 2 \\ 1 \\ -0.5 \\ 1.5 \end{pmatrix}, \qquad \boldsymbol{c} = C^{-1}\boldsymbol{f} = \begin{pmatrix} 0.683 \\ 0.483 \\ 0.250 \\ 0.583 \end{pmatrix}.$$

The ratings average 0.50, as the lemma requires. D is unbeaten but rated 0.583, below A's 0.683: its one win came against the winless C, while A also beat B, who has a win of its own. Laplace's rule alone would put D at $2/3$ and A at $3/4$; the opponent adjustment widens the gap.

5.4 Choosing the Benchmark

The benchmark rank $b = 12$ is a choice about what the resume is for, not a parameter that data could estimate: a resume is retrodictive, and there is no future outcome against which a backtest could score it. It was set to the size of the playoff field. Two facts complicate that. From 2014 to 2023 the playoff took four teams, so for those seasons the playoff boundary was the fourth-best team; and the twelve-team format reserves five places for conference champions, so the twelfth-*best* team is not the twelfth team *selected*, and the effective boundary for an at-large team is somewhat above twelve. Table 9 shows how much the choice matters, by recomputing WAB with $b = 4$ and $b = 25$ from each season's final posterior.

Over the whole FBS the order barely depends on the benchmark (median rank correlation 0.989 for $b = 4$ and 0.993 for $b = 25$), because a benchmark changes the price of each result but not which results are good. The top of the order does move: a strong benchmark makes a loss cheap and a win over a weak team nearly worthless, which favours difficult schedules, while a weak benchmark makes every win count and every loss costly. A typical season changes 1 of its top 12 under $b = 4$ and 1 under $b = 25$. The published $b = 12$ answers "what would a bubble team in the

Table 10: Effect of a blowout margin on the resume, 2025 final posterior. For each game: the change in each team's WAB and power rating when the result is refitted as a one-point win for the same team, and the mean absolute change in WAB of every other FBS team.

Game	Margin	ΔWAB winner	ΔWAB loser	Δ power winner	Δ power los
USC over Missouri State	60	-0.020	-0.024	-1.9	
Oregon over Oklahoma State	66	-0.022	0.053	-1.4	
Nebraska over Akron	68	-0.066	-0.014	-2.7	
Alabama over UL Monroe	73	-0.034	-0.006	-1.4	
Michigan over Central Michigan	60	-0.032	-0.019	-2.0	
Notre Dame over Syracuse	63	0.008	-0.047	-3.6	

current format have done"; the benchmark is a setting (`benchmark_rank` in `model_config()`), so the four-team-era question can be asked of the earlier seasons directly.

5.5 How Margin Reaches the Resume

WAB does not use margin directly, but margin informs the posterior strengths of the opponents and the benchmark, so it reaches the resume indirectly. A blowout win makes the loser look weaker, which lowers the winner's credit for beating it. Table 10 measures this on the 6 most lopsided FBS–FBS games of 2025: each is refitted with its regulation margin replaced by a one-point win for the same team (and its efficiency margin dropped), and the two teams' WAB recomputed.

Turning a blowout of 60–73 points into a one-point win moves the winner's power rating by 1.4–3.6 points, but its WAB by -0.066 to 0.008 wins and the loser's by -0.047 to 0.053; every other team's moves by at most 0.034 on average. For comparison, the Monte Carlo standard error of a published WAB is about 0.012, 0.008 (Section 6.1). The margin's influence on the resume runs through the opponents' strengths, as designed, and a team gains little by running up a score. The effect has no consistent sign: a close result makes the beaten opponent look stronger, which raises the winner's credit for that game, but it also changes the strengths that set the benchmark, and the two pull in different directions.

6 Computation and Pipeline

6.1 Sampling and Gates

Each posterior is sampled with the No-U-Turn sampler [31] in Stan [32] via CmdStan: 4 chains of 500 warm-up and 1,000 sampling iterations, target acceptance 0.9. A fit is accepted only if it passes four gates on every team strength and total tendency at the snapshot week and every structural parameter, including the random-walk steps and the family's shape parameters: rank-normalised $\widehat{R} \leq 1.01$ and bulk and tail effective sample sizes of at least 400 [33], no divergent transitions, and E-BFMI above 0.2 in every chain [28]. A fit that fails is rerun once with 1,000 warm-up iterations, target acceptance 0.95 and 2,000 sampling iterations (a much higher target acceptance makes the step size so small that chains stall in the neck of the random-walk funnel); the posterior for the

Table 11: Monte Carlo standard errors of the published resume quantities, FBS teams at the final posterior of every complete season: median and largest over teams and seasons. Thinned: the draws the pipeline keeps; all: every post-warm-up draw.

Draws	WAB	$\Pr(\text{top } 12)$	Rank, 10% quantile	Rank, 90% quantile
1,000 (thinned)	0.012 / 0.025	0.008 / 0.019	0.00 / 1.00	0.00 / 1.00
2,000 (thinned)	0.008 / 0.013	0.008 / 0.011	0.00 / 0.50	0.00 / 0.50
4,000 (all)	0.005 / 0.010	0.004 / 0.008	0.00 / 0.50	0.00 / 0.50
8,000 (all)	0.004 / 0.006	0.003 / 0.005	0.00 / 0.50	0.00 / 0.50

Table 12: Convergence gates by season, over all weekly posteriors. Worst values are taken over snapshots and over every monitored parameter.

Season	Snapshots	Passed	Retried	Max $\widehat{R}$	Min ESS	Divergent	Min E-BFMI	Minutes
2014	18	18	0	1.008	592	0	0.89	25.2
2015	18	18	0	1.010	536	0	0.87	15.5
2016	18	18	0	1.009	639	0	0.83	15.7
2017	18	18	1	1.009	738	0	0.87	20.6
2018	18	18	0	1.010	680	0	0.90	16.1
2019	19	19	0	1.009	654	0	0.88	18.1
2020	31	31	0	1.009	699	0	0.86	14.2
2021	18	18	0	1.008	755	0	0.87	23.9
2022	18	18	1	1.008	643	0	0.89	30.8
2023	18	18	0	1.008	473	0	0.88	21.4
2024	19	19	1	1.010	661	0	0.89	38.4
2025	19	19	0	1.009	624	0	0.87	20.6
2026	10	10	0	1.008	863	0	0.87	5.5

current week halts the pipeline if it still fails, while a historical week is recorded with a warning. The resume and predictions are computed from 1,000 equally spaced draws, which keeps the per-game probability matrices of the resume (draws $\times$ team-games) small enough to store for every week. Thinning costs precision, and Table 11 measures how much, from the final-week refits of every complete season with all 4,000 draws kept: the Monte Carlo standard errors of the published quantities with the thinned draws, against all draws.

The largest Monte Carlo error of a published $\Pr(\text{top } 12)$ is 0.019, 0.011, and of a WAB 0.025, 0.013 wins: small next to the posterior uncertainty they summarise, so the second decimal of a published probability is meaningful and the third is not. Thinning by four inflates the median error of WAB by a factor of 2.2; for independent draws the factor would be two.

Of the 242 weekly posteriors, 242 pass the gates, 3 after the retry. Table 12 summarises them by season.

6.2 The Pipeline

The computation is a `targets` pipeline in R [34, 35], drawn in Figure 6. Its structure enforces the two properties the evaluation depends on. First, the inputs to the week-w posterior contain only games of concluded weeks $\leq w$ and are hashed, so a cached posterior is reused exactly when its inputs are unchanged, and new results or betting lines for later weeks never invalidate it. Second, each season's preseason information d_i is the final posterior of the previous season, a dependency the pipeline tracks, so refitting one season refits every later one. The margin family is fixed for the whole build by the `margin_comparison` target (Section 7.4), which fits every complete season without carry-over and therefore does not depend on the published fits it configures.

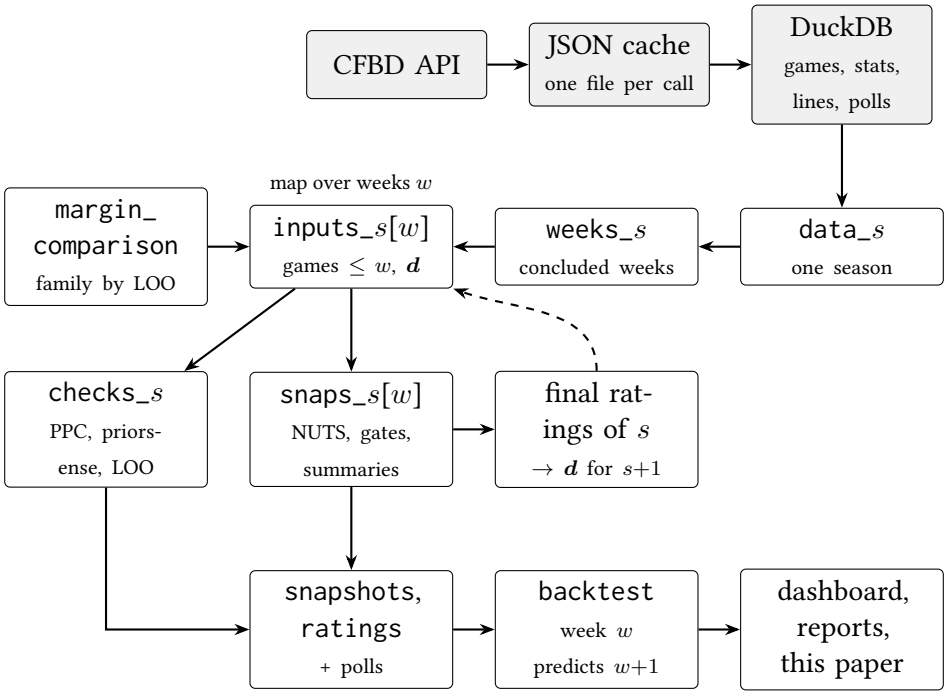


Figure 6: The computation as a `targets` pipeline. Grey boxes are external stores; white boxes are targets, with $[w]$ marking a dynamic branch per concluded week and s a season. The dashed edge is the only cross-season dependency: season s's final ratings become season $s+1$'s preseason information. The margin comparison, fitted once on every complete season, sets the family every fit uses. The checks feed the diagnostics report. A weekly update calls the API about six times and refits only the new week.

7 Model Checking

Each season's latest posterior is refitted with the pointwise log-likelihood stored, and three checks are run on it [24, 36]. This section summarises them across seasons; Section 11 examines two posteriors in full detail — sampler diagnostics, convergence, posterior geometry, prior-to-posterior learning, predictive distributions and influence.

7.1 Posterior Predictive Checks

Replicated regulation margins are drawn from Equation 2, each game at its expected margin in the week it was played, rounded to whole points so that they can land exactly on the thresholds of the statistics, and compared with five observed statistics: the standard deviation of the margin, the shares of games within seven and within three points, the share decided by 28 or more, and the home team's win rate in true home games. Replicated totals of the games without overtime are drawn from the marginal law of the total model and compared on their mean and standard deviation. Figure 7 shows the result for every season.

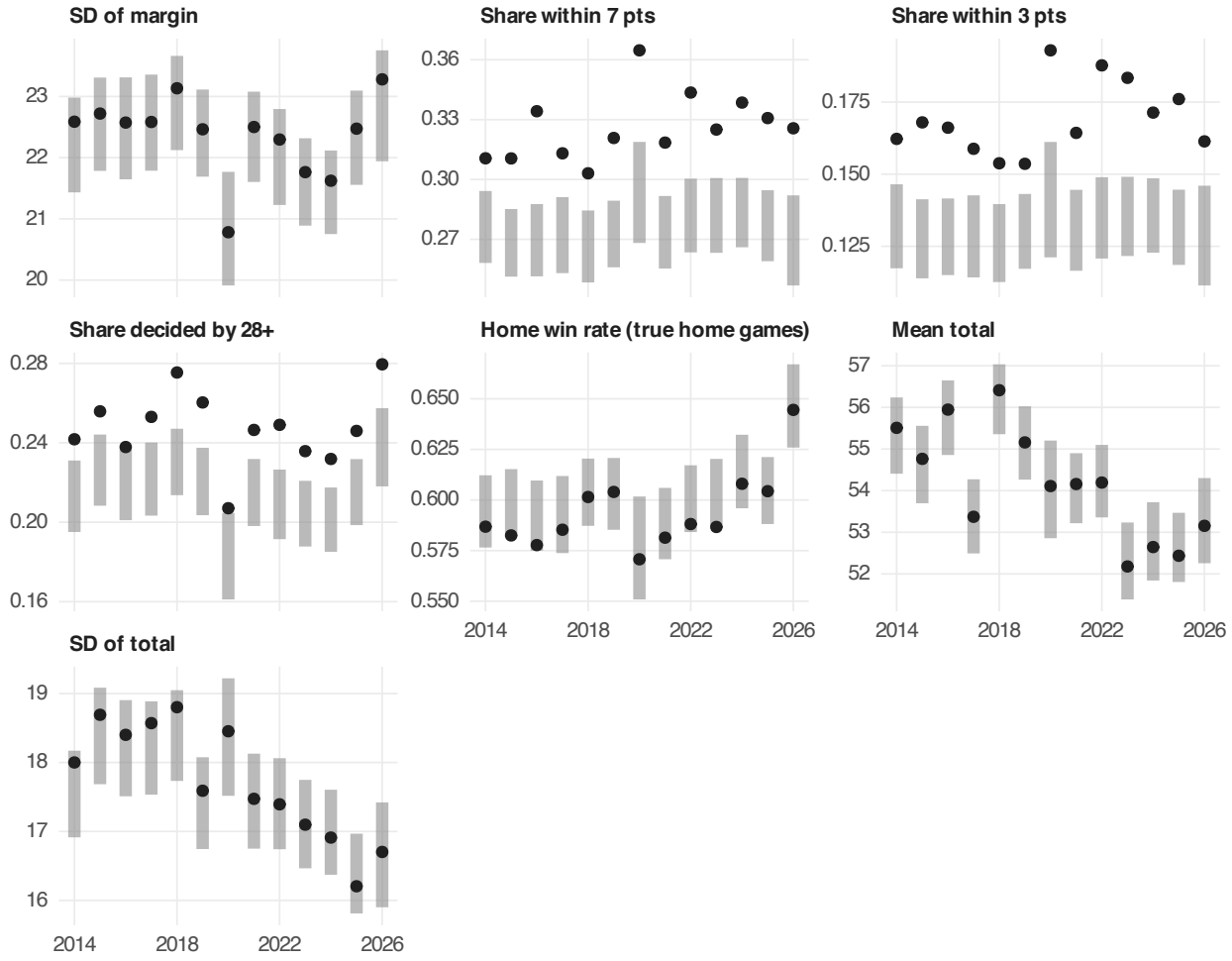


Figure 7: Posterior predictive checks at each season's latest posterior. Bars: central 90% interval of the statistic over 400 replicated seasons (as in Section 11.8); points: the observed value, filled when it falls outside the interval.

The model reproduces home win rate, mean total, sd of margin, sd of total (outside the interval in at most one season each). In 13 of 13 seasons the observed share of games decided by seven points or fewer (0.30–0.36) lies above the replicated interval, and in 13 the share within three points (0.15–0.19) does. Real margins bunch around one score more than the scale-t around the rating

difference allows: late-game behaviour (running out the clock when ahead, playing for a tying field goal when behind), key numbers, and the regulation ties that overtime games contribute all push mass towards zero. None of the families compared in Section 7.4 removes this misfit. It is in the *shape* of the margin distribution near zero; the win probabilities, which depend on the location and the tails, are well calibrated (Section 8).

7.2 Prior Sensitivity

Power-scaling sensitivity [37] perturbs the prior and the likelihood by raising each to a power near one, estimates the resulting change in each marginal posterior by importance sampling, and flags a parameter whose posterior moves under both (prior–data conflict) or under neither (weak likelihood). Table 13 counts the flags across seasons.

4 of the 31 parameters are flagged for a weak likelihood in any season, in at most 1 season(s) each. Prior–data conflict flags are the rule rather than the exception: 16 of the 31 parameters are flagged in all seasons or all but one. That is what the pooling of Section 3.4 produces, and not a sign that the seasons disagree with their history. Each prior carries the information of several seasons (Equation 6), so for 16 of the 31 pooled parameters its standard deviation is smaller than that of a single season's full-data posterior; power-scaling either the prior or the likelihood then moves the posterior, and priorsense labels a parameter that is sensitive to both a conflict. Whether a season actually departs from its prior is answered directly by the leave-one-season-out column of Table 28, which compares each season's own estimate with the prior pooled from the other seasons. The degrees of freedom ν are identified only weakly once they exceed about 15, where the t and normal laws are hard to tell apart (posterior means 29–32), so their posterior follows the pooled $\mathrm{Gamma}(56.2, 1.83)$ prior closely; Section 11.11 refits with it and the prior on γ widened fivefold. The sensitivity of ν matters for the tails of the predictive distribution, not for the ratings. Of the 12 parameters of the total model, 12 are flagged in some season; they affect the reported offence, defence and totals, not the power rating or the resume. How much the priors move the published products themselves is the subject of Section 11.11.

7.3 Leave-one-out Cross-validation

Pareto-smoothed importance-sampling leave-one-out cross-validation [38, 39] estimates the expected log predictive density of a new game, with the Pareto $\hat{k}$ diagnostic flagging games whose removal would change the posterior too much for the approximation.

The effective number of parameters is 68–109% of the number of teams, not always below it, as expected of a hierarchical model that shrinks the least-informed teams towards their conference. Because `log_lik` is the joint density of both signals where efficiency is observed, its $\widehat{\mathrm{elpd}}$ cannot be compared with that of a model without efficiency; the program therefore also stores the marginal density of the margin alone, on which Section 4 compares the two models. A few games per season have high $\hat{k}$; they are almost always the only link between a lower-division team and the rest of the graph, whose strength is then determined by that single game.

Table 13: Power-scaling sensitivity by parameter, at each season's latest posterior. Sensitivities are the cumulative Jensen–Shannon distance between the base and the power-scaled posteriors; the conventional threshold is 0.05. The parameters of the total model (Section 3.3) are listed last.

Parameter	Prior (median)	Likelihood (median)	Seasons flagged: conflict	weak likelihood
β	0.090	0.107	13/13	0/13
β_2	0.098	0.109	13/13	0/13
β_q	0.101	0.108	13/13	0/13
γ_a	0.091	0.098	13/13	0/13
γ_q	0.088	0.118	11/13	0/13
γ	0.103	0.149	13/13	0/13
η	0.069	0.134	12/13	0/13
κ	0.089	1.070	10/13	0/13
λ	0.074	0.159	13/13	0/13
m_2	0.058	0.106	10/13	0/13
m_3	0.068	0.113	9/13	0/13
ν	0.090	0.136	13/13	0/13
ω	0.103	0.104	13/13	0/13
σ_e	0.090	0.506	13/13	0/13
σ_m	0.069	0.927	10/13	0/13
τ_1	0.085	0.140	11/13	0/13
τ_2	0.099	0.096	13/13	0/13
τ_3	0.096	0.218	13/13	0/13
ϕ_r	0.093	1.514	13/13	0/13
α_s	0.077	0.092	11/13	0/13
β_t	0.058	0.091	10/13	0/13
θ	0.103	0.098	11/13	1/13
κ_t	0.072	0.092	8/13	0/13
λ_t	0.065	0.104	10/13	0/13
ω_t	0.088	0.097	12/13	1/13
σ_s	0.070	0.087	8/13	0/13
τ^t	0.106	0.101	13/13	0/13
σ_x	0.059	0.103	9/13	0/13
m_2^t	0.078	0.114	11/13	0/13
m_3^t	0.049	0.098	4/13	1/13
ϕ_t	0.117	0.095	12/13	1/13

Table 14: PSIS-LOO at each season's latest posterior. p_{loo} is the effective number of parameters; the last column counts games with $\hat{k} > 0.7$.

Season	Games	Teams	$\widehat{\text{elpd}}_{\text{loo}}$	SE	p_{loo}	$\hat{k} > 0.7$
2014	1,584	318	-9,621	84	325	13
2015	1,536	300	-9,446	79	319	13
2016	1,547	302	-9,507	81	314	8
2017	1,549	296	-9,509	79	290	3
2018	1,554	305	-9,654	79	308	6
2019	1,621	310	-9,850	80	306	11
2020	850	244	-5,520	56	183	6
2021	1,570	298	-9,688	79	303	4
2022	1,598	306	-11,764	63	316	17
2023	1,603	302	-11,858	56	314	13
2024	1,669	311	-12,578	53	339	16
2025	1,687	315	-12,892	49	304	7
2026	1,041	320	-8,042	34	219	19

Table 15: Margin families by PSIS-LOO, final week of every complete season, 2014–2025. $\Delta\widehat{\text{elpd}}$: pooled difference from the Student-t (positive is better) with its standard error; *Seasons better*: seasons in which the family beats the t, and by more than two of that season's standard errors; the last two columns count seasons whose observed share of games within 7 and within 3 points lies outside the 90% predictive interval.

Family	$\widehat{\text{elpd}}$	$\Delta\widehat{\text{elpd}}$	Seasons better	Within 7 missed	Within 3 missed
Scale-t (published)	-122,092	221.8 (30.3)	1, 1 of 12	12/12	12/12
Mixture	-122,295	18.4 (7.2)	1, 1 of 12	12/12	12/12
Student-t	-122,313	–	–	12/12	12/12

7.4 Choosing the Margin Family

The family $\mathcal{D}$ of Equation 1 is chosen by PSIS-LOO. Each family is fitted to the final week of every complete season, 2014–2025, under the vague priors of Section 11.6, without carry-over and without the total model. The vague priors and the absent carry-over make the comparison independent of the published fits, which depend on its result; the total model is left out because the family enters only the margin likelihood. Within a season the pointwise $\widehat{\text{elpd}}$ differences from the Student-t, the incumbent, give a difference and its standard error; seasons are independent, so the pooled difference is the sum and its standard error the root sum of squares. A family replaces the t only if its pooled difference exceeds two pooled standard errors. The normal is not refitted here, since the t contains it as the limit $\nu \to \infty$; it appears among the ablations of Section 8.4. Table 15 gives the result.

The Scale-t family has the best pooled $\widehat{\text{elpd}}$, 222 above the t (standard error 30), better in 1 of 12

seasons, and the published family is therefore the scale-t. The mixture also beats the t, by 18 (7), far less. The size of the gap between the two suggests that what the data want is less a sharper peak than a scale that grows with the mismatch: the margin of a lopsided game is less predictable than that of an even one, as a favourite may rest its starters at any lead, so one scale for all games is too wide for close matchups and too narrow for mismatches. The improvement does not extend to the one-score games. Figure 8 shows the shares of games within seven and within three points by season and family: for every family and every season the observed share within seven points lies above the replicated interval. The bunching of margins near zero is a property of the discrete score, with its key numbers and overtime ties, that a continuous location-scale law of any of these shapes does not reproduce (Section 10).

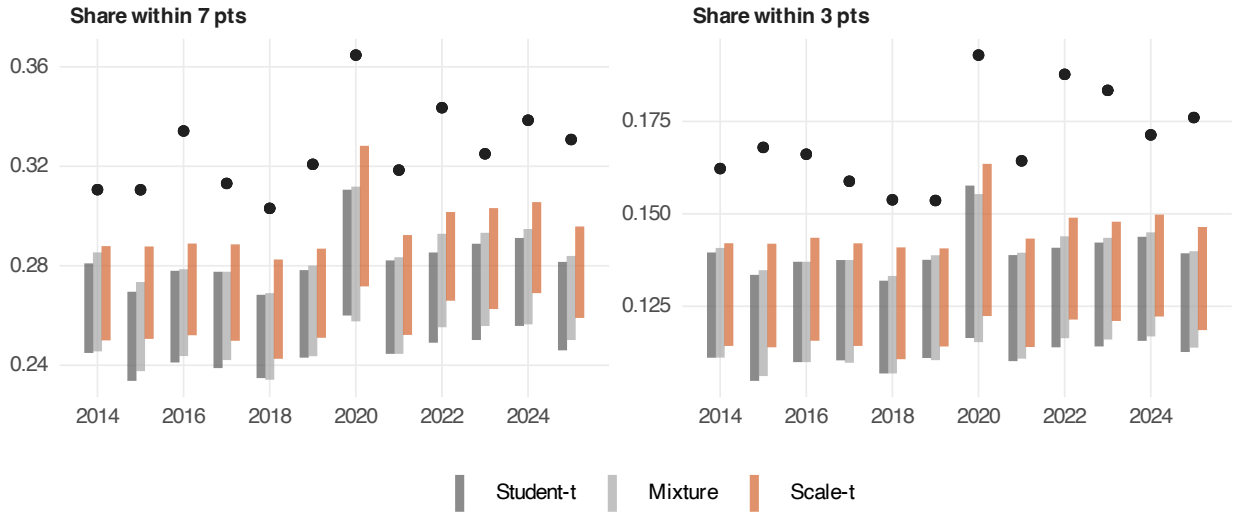


Figure 8: The share of games decided by seven or fewer points (left) and three or fewer (right) at each season's final week: observed (black points) and the 90% interval over replicates from each family's fit. The comparison fits use vague priors and no carry-over.

8 Out-of-sample Evaluation

8.1 Design

Every game of week $w + 1$ is predicted from the posterior fitted through week w, which by construction (Section 6.2) contains no information from week $w + 1$ or later. This is a rolling-origin evaluation [40] with one origin per week, and it covers every week of every season from 2015, when a preseason posterior first exists, including week 1 predicted from the preseason. Predictions are scored by the absolute error of the spread, the Brier score [41] and the log loss of the home win probability, both strictly proper [42], and against the closing betting line where one exists. A betting line is the sharpest public forecast available, since it aggregates information — injuries, weather, motivation — that no ratings model sees [43]. Beating it is not the aim; coming close to it with public results alone is evidence that the ratings are sound.

Table 16: Rolling out-of-sample accuracy, all predicted games in each scope. MAE and RMSE are errors of the predicted margin in points; CRPS scores the whole predictive distribution of the margin, in points; Brier score, log loss and accuracy score the home win probability. Week 5+ means the predicted week, so the posterior has at least four weeks of games.

Games	N	MAE	RMSE	CRPS	Brier	Log loss	Accuracy
All games	19,408	12.89	16.30	9.21	0.166	0.498	74.6%
FBS vs FBS	9,529	12.80	16.19	9.15	0.180	0.533	72.0%
FBS vs FBS, week 5+	7,762	12.73	16.08	9.09	0.185	0.545	71.1%

Table 17: The model against the closing betting line, games with a line only. The line's win probability applies the model's own predictive distribution to the line's spread. ATS is the share of games in which the model's side of the line covered, pushes excluded; the break-even rate against a standard -110 price is 52.4%.

Games	N with line	MAE, model	MAE, line	Brier, model	Brier, line	ATS
All games	13,640	12.74	12.23	0.166	0.160	50.0%
FBS vs FBS	9,529	12.80	12.28	0.180	0.172	49.6%
FBS vs FBS, week 5+	7,762	12.73	12.29	0.185	0.178	49.5%

8.2 Accuracy

Table 16 scores every predicted game, and Table 17 compares the model with the closing line on the games that have one. The two tables have different denominators: every FBS–FBS game has a line, but only 70% of all games do, since lines are rarely posted for games against lower-division teams.

Over all 19,408 predicted games the mean absolute error is 12.9 points (95% interval 0.16–0.17), the Brier score 0.166 (9.087–9.340) and the log loss 0.498 (0.486–0.509), and the favourite wins 74.6% of the time. Games predicted by the same posterior share its errors, so these intervals come from a bootstrap that resamples whole season-weeks, and the comparisons below use standard errors clustered the same way. On the 9,529 FBS–FBS games the model's error is 12.80 points against the line's 12.28. The difference, 0.52 points per game with standard error 0.05 ($z = 9.9$ in a Diebold–Mariano comparison of absolute errors [44]), is clearly distinguishable from zero: the market is better. Its win probabilities are better too (Brier difference 0.0079, $z = 7.8$). Against the line the model picks the covering side in 49.6% of 9,358 games (95% interval 48.6%–50.7%), consistent with the 50% of a forecast with no edge on the market and well short of the 52.4% needed to profit at standard prices.

Figure 9 shows how accuracy evolves through the season. The error is largest in week 1 (14.5 points), predicted from the preseason alone, and from week ∞ on it stays within 0.6 points of its week-5-and-later mean of 12.9. The gap to the line ranges from 0.1 to 1.1 points across regular-season weeks. The Brier score *rises* through the season, and the right panel shows why: the mean distance of the predicted probability from one half falls from 0.25 in the early weeks to 0.15 later on, as conference play pairs more evenly matched teams, and the line's own Brier score

rises in step.

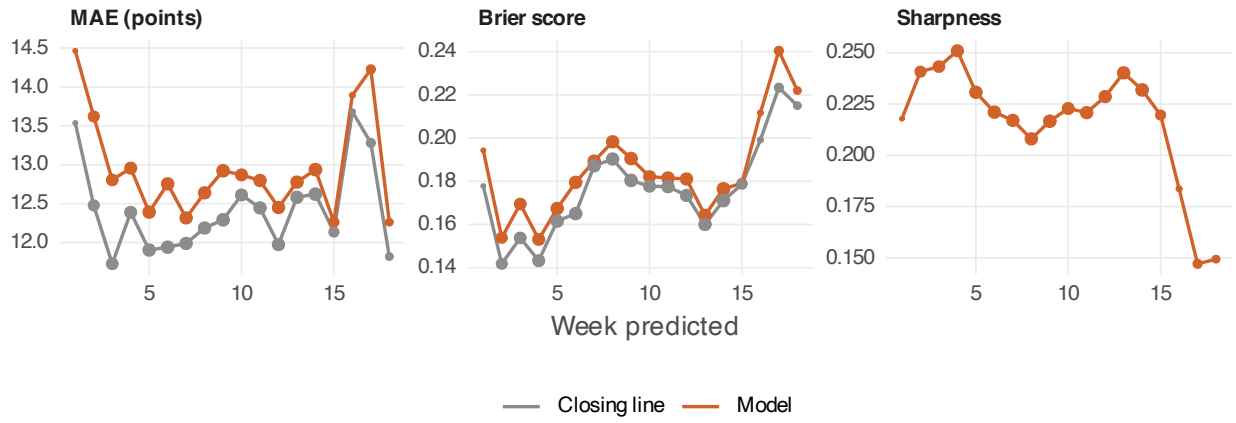


Figure 9: Rolling out-of-sample accuracy on FBS–FBS games by the week predicted. Left: mean absolute error of the model's spread and of the closing line on the same games. Centre: Brier score of the model's win probability and of the line's (the line's spread through the model's predictive distribution). Right: sharpness, the mean of $|p - 1/2|$ over the model's predicted probabilities. Weeks after the last regular-season week are the postseason and have few games.

The predictive distribution of the margin. A spread is a point forecast; the model also implies a full predictive distribution of the margin. For each game of the week ahead it is represented by quantiles of posterior predictive draws: for each posterior draw, the expected margin plus a week of random-walk drift for both teams, and then a margin from the scale-t law around it, so that the uncertainty in the ratings, the drift and the game noise all enter. Its central 50%, 80% and 95% intervals contain 49.9%, 79.2% and 94.5% of observed margins (FBS–FBS: 49.3%, 78.5%, 94.2%), and its continuous ranked probability score [42, 45], computed from the quantiles, is 9.21 points (12.71–13.07). The intervals are close to nominal in the aggregate, but they are the right width for the wrong reason near zero: Section 7.1 shows that real margins pile up within one score more than the model allows, so central intervals are slightly too wide for close games and slightly too narrow in the shoulders.

Totals. The total model of Section 3.3 predicts the final combined score of each game, and the closing over/under is the market's forecast of the same quantity. Both are scored against the final total, overtime included, since that is how the over/under is settled; the model's total, fitted on regulation-decided games only, therefore runs slightly low for games that go to overtime. On the 9,529 FBS–FBS games with an over/under the model's mean absolute error is 13.24 points against the over/under's 12.92 (all games: 13.16 against 12.85 on 13,246). The gap, 0.32 points, is smaller than the gap on the spread, 0.52: the market knows about weather, tempo changes and injuries to quarterbacks, which matter for totals as well as for margins.

8.3 Calibration

A win probability is useful to the resume only if it is calibrated, because WAB subtracts π_g from an observed result. Figure 10 groups all predicted games into ten fixed-width bins of their win probability and compares the mean prediction in each bin with the observed frequency, overall and split by kind of game and part of the season.

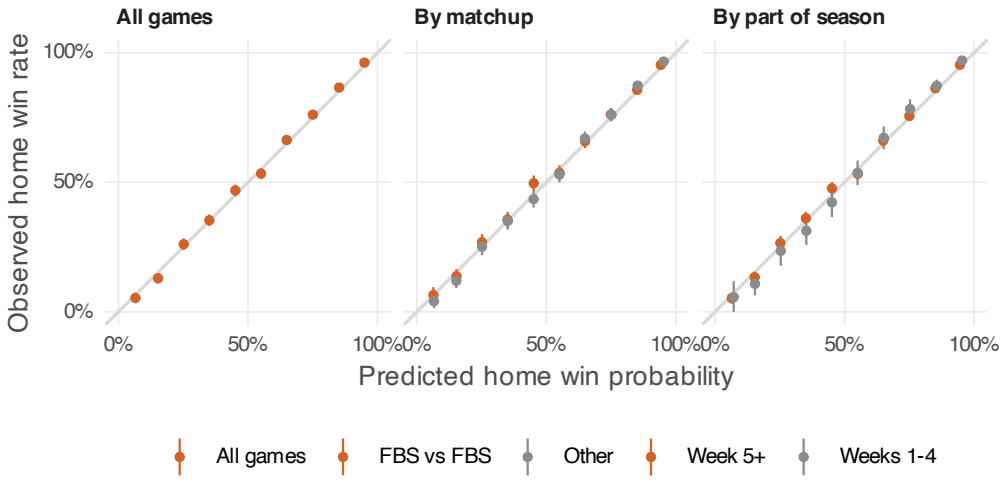


Figure 10: Calibration of out-of-sample home win probabilities. Games are grouped into fixed-width bins of predicted probability, $[0, 0.1], (0.1, 0.2], \ldots$; each point is a bin's mean prediction and observed home win rate, with a 95% binomial interval. The diagonal is perfect calibration. Left: all games. Centre: FBS–FBS games and the rest. Right: predicted weeks 1–4 and week 5 on.

Every bin lies within 2.4 percentage points of the diagonal, but with 19,408 games a gap of a few points can be real. 3 of the 10 bins are more than two standard errors from it: (0.1,0.2], predicted 0.153 against observed 0.130 with $n = 1{,}135$ ($z = -2.4$); (0.8,0.9], predicted 0.852 against observed 0.866 with $n = 2{,}568$ ($z = 2.1$); (0.9,1], predicted 0.949 against observed 0.962 with $n = 3{,}151$ ($z = 3.7$). The expected calibration error, the game-weighted mean gap between prediction and outcome within bins, is 0.013, and Spiegelhalter's test [46] gives $z = -3.5$ ($p = 0.00045$). The splits locate the misfit. In weeks 1–4, $z = -4.2$ (expected calibration error 0.021); from week 5, $z = -1.6$ (0.014). For FBS–FBS games $z = -0.9$, and for the rest $z = -4.0$. The early-season and cross-tier probabilities are thus less well calibrated than the rest: they lean on the preseason carry-over and on the tier offsets, both of which are least certain before any game.

The calibration of predicted games bears on the resume only indirectly. The benchmark probabilities π_g of Equation 8 use the same predictive distribution, but they describe a hypothetical fixture — the benchmark against the opponent — and are computed in sample, from the posterior that includes the game itself; no outcome can check them directly. That the model's out-of-sample probabilities for real fixtures are close to calibrated, especially in the second half of the season when resumes matter, is supporting evidence that the scale of π_g is right, not proof.

Figure 11 repeats the comparison with the line season by season. The gap is stable, which suggests it reflects information the market has and the model does not rather than a drifting model.

Table 18: Rolling out-of-sample accuracy of model variants, every game of 2022–2025, each predicted from the posterior through the previous week. The published model has the scale-t margin and the weekly random walk; each variant changes one thing: no efficiency likelihood, no carry-over, no conference level, ratings frozen within the season, or another margin family. Δ columns: difference from the published model on the same games, with a standard error clustered by season-week; positive is worse.

Variant	MAE	Brier	CRPS	Δ MAE	Δ Brier $\times 10^3$	Δ CRPS
Published model	12.53	0.1660	8.98			
Margin only	12.48	0.1650	8.95	-0.05 (0.01)	-0.9 (0.3)	-0.03 (0.01)
No carry-over	13.39	0.1756	9.61	+0.86 (0.15)	+9.6 (1.3)	+0.63 (0.11)
No conference level	12.54	0.1661	8.99	+0.01 (0.01)	+0.1 (0.1)	+0.01 (0.01)
Static ratings	12.54	0.1663	9.00	+0.02 (0.02)	+0.3 (0.3)	+0.02 (0.01)
Normal margin	12.49	0.1652	8.94	-0.03 (0.02)	-0.8 (0.3)	-0.04 (0.01)
Student-t margin	12.49	0.1652	8.94	-0.03 (0.01)	-0.7 (0.3)	-0.04 (0.01)
Mixture margin	12.50	0.1656	8.95	-0.03 (0.01)	-0.4 (0.3)	-0.03 (0.01)

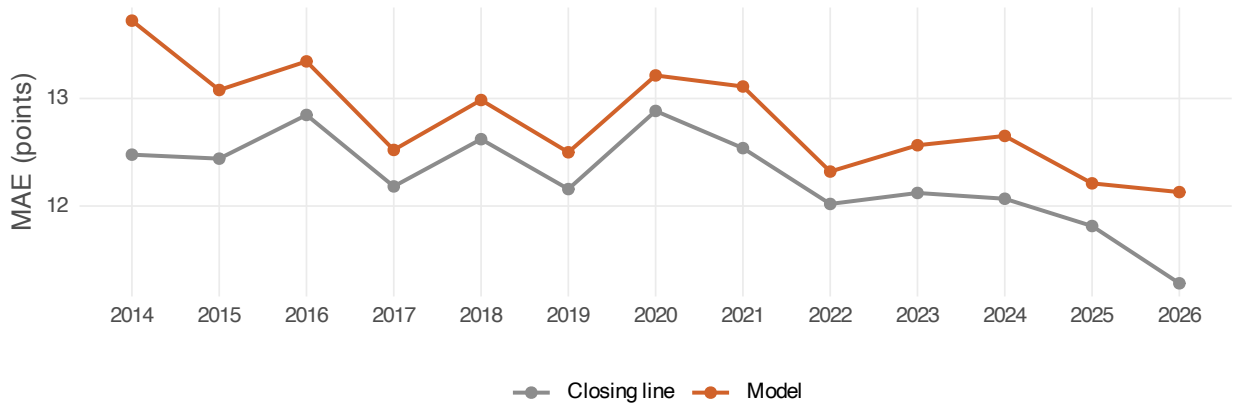


Figure 11: Out-of-sample mean absolute error by season on FBS–FBS games, model and closing line on the same games. The last season is in progress.

8.4 Ablations and Baselines

Each component of the model is tested by removing it and repeating the rolling backtest on the same games. Refitting every week of every season for each variant is expensive, so the ablations cover the 4 most recent complete seasons, 2022–2025 (6,557 games), with shorter chains than the published fits (two chains of 400 draws; the predictions are posterior means, for which that is ample). The published configuration is refitted the same way, so every row of Table 18 is comparable.

Removing a component makes the spreads measurably worse (by more than two standard errors) for: no carry-over. The published model is beaten by more than two standard errors by: margin only, normal margin, student-t margin. The margin-only model settles the question of Section 4

Table 19: Baselines against the published model on the same games (predicted weeks 2 and later, from the second season). Least squares covers every game; the CFBD Elo ratings exist only for games between rated teams.

Baseline	Games	N	MAE, baseline	MAE, model	Brier, baseline	Brier, model
CFBD Elo	All games	8,746	13.17	12.71	0.185	0.179
CFBD Elo	FBS vs FBS	8,728	13.16	12.72	0.185	0.179
Least squares (Massey)	All games	17,682	14.48	12.79	0.187	0.165
Least squares (Massey)	FBS vs FBS	8,728	13.93	12.72	0.195	0.179

empirically: the efficiency likelihood changes the mean absolute error by +0.05 points and the Brier score by +0.9 $\times 10^{-3}$ (published minus margin-only), in line with the few-percent information gain the proposition predicts. The carry-over matters most in the first weeks, where without it every team starts at its conference mean. Freezing ratings within the season (*static ratings*, $\phi_r = 0$) changes the mean absolute error by +0.02 points (se 0.02) and the CRPS by +0.02 (se 0.01) relative to the random walk: with drift of the size estimated in Figure 2 the two are hard to tell apart on these games, and the random walk is kept because it estimates how fast ratings should forget instead of fixing it at zero. The other margin families, fitted with the same published priors and carry-over, change the CRPS and the Brier score by -0.04 and -0.8 $\times 10^{-3}$ (normal); -0.04 and -0.7 $\times 10^{-3}$ (student-t); -0.03 and -0.4 $\times 10^{-3}$ (mixture); the family shapes the distribution of the margin, which the CRPS scores, more than its centre or the win probability.

The model can also be compared with rating systems built differently. Table 19 scores two baselines on the same games from 2015 (their win-probability scales are fitted on earlier seasons only): least-squares ratings on the margin, refitted weekly with a weak ridge penalty (the Massey method, unpooled), and the pregame Elo ratings that CFBD publishes, mapped to a spread and a win probability by regressions on earlier seasons. Published SP+ and FPI ratings are available only at the end of each season, after the games they would predict, so they cannot be backtested; at the end of each complete season the final power rating's rank correlation with SP+ is 0.90–0.96 and with FPI 0.96–0.99.

8.5 Priors and Preseason Terms Out of Sample

The pooling rule of Section 3.4 and the preseason terms of Equation 3 were chosen on predictions of the first weeks, where the priors matter most, with each season's priors set without that season. For each variant, every complete season's final week is refitted under the vague priors with the variant's model; for each season from 2015 the priors are then pooled from the *other* seasons only, weeks 0–3 are fitted with short chains, and each fit predicts the following week. Table 20 compares the published settings, on the same 3,868 games, with the priors in use before this version (no preseason terms, equal weights and twice the between-season sd, the random-walk step prior $\mathcal{N}^+(0.6, 0.25^2)$, pooled from every season including the one predicted), with the published model under the earlier pooling rule, and with the published model without each of its preseason terms.

Against the priors in use before, the published settings lower the mean absolute error of these predictions by 0.19 points and the CRPS by 0.14, and bring the coverage of the 80% interval from

Table 20: Predictions of weeks 1–4 from fits through weeks 0–3, 2015–2025, under priors pooled without the season predicted (the first row excepted). Differences are from the published settings (positive = worse), with paired bootstrap standard errors over games; the last two columns are the coverage of the central 80% and 95% predictive intervals.

Settings	MAE	Δ MAE	Δ log loss	Δ CRPS	80%	95%
Published	13.46	–	–	–	80%	96%
Before (2026-10-10 priors)	13.65	0.188 (0.048)	0.0085 (0.0018)	0.142 (0.028)	84%	97%
Published, equal weights and sd x 2	13.49	0.026 (0.029)	0.0039 (0.0011)	0.040 (0.017)	82%	97%
Without returning production	13.59	0.125 (0.030)	0.0017 (0.0011)	0.069 (0.016)	80%	96%
Without talent	13.56	0.100 (0.024)	0.0023 (0.0008)	0.061 (0.015)	80%	96%
Without two-season carry-over	13.50	0.032 (0.021)	0.0012 (0.0008)	0.037 (0.014)	81%	96%

84% to 80%, although the earlier priors had the advantage of being pooled from the season they predict. Each preseason term contributes: dropping returning production, talent and two-season carry-over raises the mean absolute error by 0.13, 0.10 and 0.03 points respectively. Two other candidates were tested in the same way and not adopted, since neither improved the log loss or the CRPS by more than its standard error: a separate carry-over for FCS and lower-division teams, and a wider preseason spread τ for teams with a new head coach. The pooling rule mostly affects calibration: the earlier rule's wider priors make the predictive intervals too wide in the first weeks.

The leakage of pooling itself can be measured on the published settings alone. Weeks 0–3 of every season from 2015 refitted with priors pooled from the *other* seasons predict weeks 1–4 with a mean absolute error of 13.46 points against 13.46 for the published predictions of the same 3,868 games, and a Brier score of 0.1270 against 0.1270: the leakage is negligible in the weeks where the priors matter most, and none at all later, when the data dominate them.

8.6 Preseason Ratings

Week-1 games are predicted from the preseason posterior, which rests on last season's final ratings through the carry-over β. The preseason power ranks agree with the preseason AP poll on the teams it ranks (rank correlation 0.49–0.89 by season), although the poll uses information the model lacks — coaching changes, the quality of individual returning starters and incoming transfers; the model sees returning production and the talent composite only as two team-level numbers. Against the season's final ratings, the preseason ratings have a rank correlation of 0.77–0.88. A comparison with preseason SP+ would be the natural next check, but CFBD publishes SP+ only as an end-of-season rating.

The transfer portal and the one-time transfer exception, from 2021, might be expected to weaken the link between seasons. Figure 12 shows β by season. The mean estimate is 0.60 for 2015–2020 and 0.59 for 2021–2025; with posterior standard deviations of about 0.07 per season, the data show

no clear change.

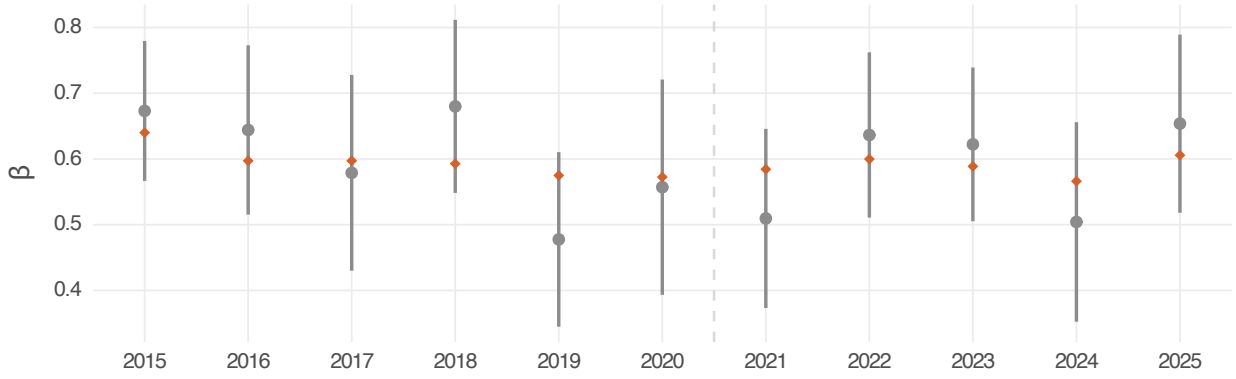


Figure 12: The carry-over β by season: full-season posterior mean ± 2 sd under vague priors (grey) and the published final posterior mean (orange). The first season has no previous ratings, so β is not identified there.

8.7 Tuned Choices

Every choice that was tuned or set from data is listed in Table 21, with the data it used. Several used the same seasons that the backtest reports, which makes the reported backtest slightly optimistic for those choices; Section 8.5 bounds the effect for the pooled priors, and the pooling rule and preseason terms were themselves chosen on leave-one-season-out predictions. The margin family was chosen on in-sample LOO at each season's final week, not on the backtest, but on the same seasons.

9 The 2025 Season

This section uses the 2025 season to show what the two products look like and how they relate. Most of it uses the final posterior, after the postseason.

Example. The 2025 national championship game, Indiana against Miami at a neutral site, was predicted by the posterior through the week before it. There the posterior mean ratings were $r = 30.9$ and 22.5 points, so the predicted spread is 8.4 points for Indiana (there is no home field, $\delta_g = 0$). With the posterior means $\bar{\sigma} = 12.26$, $\gamma = 0.287$ and $\nu = 29.4$ (so the scale at this spread is $(\bar{\sigma}^2 + \gamma^2\mu^2)^{1/2} = 12.50$), the plug-in win probability is $F_\nu(8.4/12.50) = 0.748$. The published probability averages $1 - F^{(s)}(0 \mid \cdot)$ over the posterior draws instead, 0.718, which is closer to one half because the uncertainty in the two ratings and a week of random-walk drift (together a posterior sd of the spread of 7.4 points) widen the predictive distribution. The total model projected a score of 30–21, a total of 51.3 against a closing over/under of 47.2. The closing line had Indiana favoured by 7.2, and the result was 27–21.

Table 22 lists the top of the final resume.

Table 21: Every tuned or data-driven setting, how it was set, and whether the data overlap the reported backtest.

Setting	Value	How set	Overlaps backtest
Structural priors	Tables 5, 6	Pooled full-season refits under vague priors, all complete seasons (Equation 6)	Yes (Section 8.5)
Pooling rule	half-life 3 seasons, predictive sd	Early-week predictions, leave-one-season-out priors (Section 8.5)	Yes
Preseason terms	returning production, talent, two seasons back	Early-week predictions, leave-one-season-out priors (Section 8.5)	Yes
Margin family	scale-t	Pooled PSIS-LOO, final week of every complete season (Section 7.4)	Yes
Seasons left out of the pooling	2014; 2020 for η	Structural reasons, not tuned	No
Benchmark rank b	12	Playoff size (Section 5.4)	No
Family shape priors	Table 6	ν pooled like the structural priors	Yes
Minimum plays for e_g	20	Judgement	No
Garbage-time filter	provider's	CFBD definition	No
Sampler and gates	4 chains, 500/1000	Conventional thresholds	No

Table 22: Final 2025 FBS resume. WAB with its 80% interval; the resume rank with its 80% rank interval; the power rating in points above an average FBS team with its rank; and the Colley and final AP ranks for reference.

	Team	Record	WAB	Rank 80%	$\Pr(\leq 12)$	Power	Colley	AP
1	Indiana	16–0	5.23 (4.6, 5.8)	1–1	1.00	35.6 (1)	1	1
2	Oregon	13–2	3.38 (2.8, 4.0)	2–2	1.00	27.1 (4)	2	4
3	Ohio State	12–2	2.18 (1.6, 2.7)	3–6	1.00	28.2 (3)	3	5
4	Miami	13–3	2.10 (1.4, 2.7)	3–6	1.00	25.3 (5)	4	2
5	Ole Miss	13–2	1.52 (1.0, 2.1)	4–8	1.00	20.8 (8)	5	3
6	Texas A&M	11–2	1.43 (0.8, 2.0)	5–9	1.00	16.5 (15)	12	8
7	Georgia	12–2	1.42 (0.7, 2.1)	4–10	0.99	21.8 (7)	6	6
8	BYU	12–2	1.38 (0.7, 2.1)	4–10	0.98	14.4 (20)	7	11
9	Texas Tech	12–2	0.99 (0.4, 1.6)	6–12	0.93	24.1 (6)	8	7
10	Alabama	11–4	0.86 (0.1, 1.6)	7–13	0.86	16.3 (16)	13	9
11	Texas	10–3	0.51 (-0.0, 1.0)	9–14	0.69	15.9 (17)	19	12
12	Notre Dame	10–2	0.47 (0.0, 0.9)	10–14	0.62	28.8 (2)	10	10
13	Oklahoma	10–3	0.26 (-0.4, 0.9)	10–17	0.39	14.6 (19)	9	13
14	Utah	11–2	0.14 (-0.4, 0.7)	11–18	0.27	20.3 (9)	14	14
15	Vanderbilt	10–3	-0.10 (-0.6, 0.4)	13–20	0.07	16.7 (14)	22	15
16	Iowa	9–4	-0.11 (-0.7, 0.5)	12–20	0.13	19.6 (10)	26	17
17	Navy	11–2	-0.22 (-0.6, 0.2)	14–20	0.02	4.6 (52)	18	23
18	Illinois	9–4	-0.34 (-0.9, 0.2)	14–22	0.03	10.4 (28)	21	–
19	James Madison	12–2	-0.34 (-0.6, -0.0)	15–21	0.01	12.1 (22)	16	19
20	USC	9–4	-0.56 (-1.1, -0.1)	16–24	0.01	17.4 (13)	20	20
21	Tulane	11–3	-0.63 (-1.1, -0.1)	17–25	0.00	4.4 (54)	15	18
22	Michigan	9–4	-0.68 (-1.2, -0.1)	17–25	0.00	14.6 (18)	23	21
23	North Texas	12–2	-0.76 (-1.1, -0.4)	19–25	0.00	8.3 (35)	11	24
24	Virginia	11–3	-1.13 (-1.6, -0.6)	21–28	0.00	8.1 (37)	17	16
25	Washington	9–4	-1.13 (-1.7, -0.6)	22–30	0.00	19.0 (11)	27	–

The rank intervals are the point of reporting a posterior. Figure 13 shows them for the top 25 in both products. Among the resume top 12 the median width of the 80% rank interval is 4 places for the resume and 13 for power. Resume ranks are the sharper, because once the results are fixed only the valuation of the opponents is uncertain, while power ranks describe strength itself.

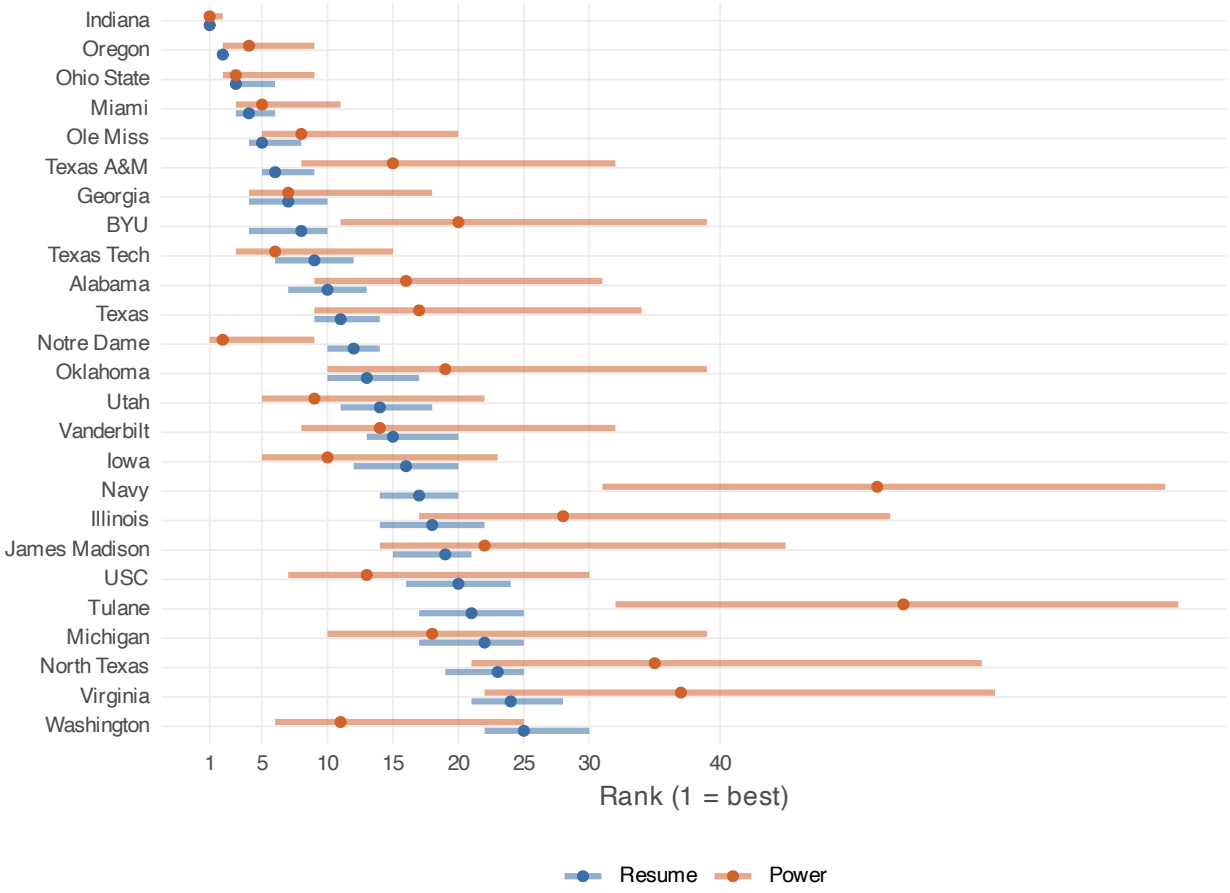


Figure 13: Posterior rank intervals at the end of 2025 for the top 25 of the resume. Points are the published ranks; bars the central 80% of the rank across posterior draws. Teams are ordered by resume rank.

Figure 14 plots the two products against each other for every FBS team. The relation is monotone but loose. The right panel shows what the scatter around the curve is: the vertical distance of each team from the curve against its *win luck*, actual wins minus the wins its own rating would predict against its schedule (the sum of its win probabilities in each game under the final posterior). The two are correlated at 0.90: teams above the curve have won more games than their strength predicts, often close ones, and teams below it have lost them.

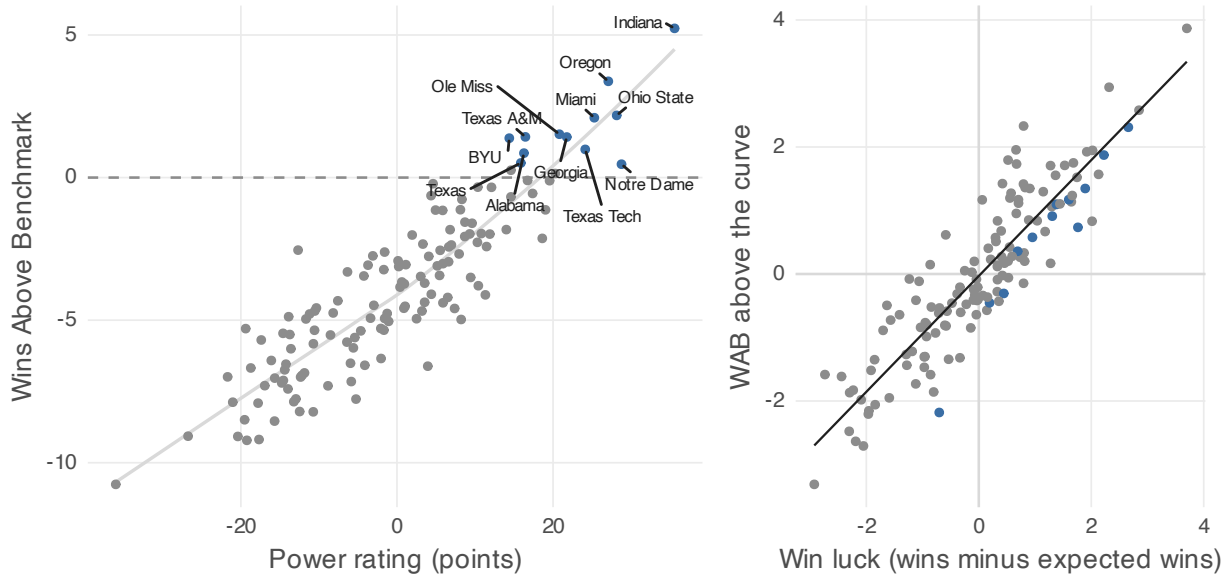


Figure 14: Resume against power at the end of 2025, all FBS teams. Left: the dashed line is zero Wins Above Benchmark, teams above it did better than the twelfth-best team would have with their schedule; the grey curve is a LOESS fit; labelled, the resume top 12. Right: each team's distance above or below that curve against its win luck, actual minus expected wins.

Figure 15 follows the power rating of the resume top six through the season. Preseason ratings come from the carry-over alone and are wide: the median 80% interval is 16.2 points in week 0, 13.3 in week 9, and 11.7 at the end of the season.

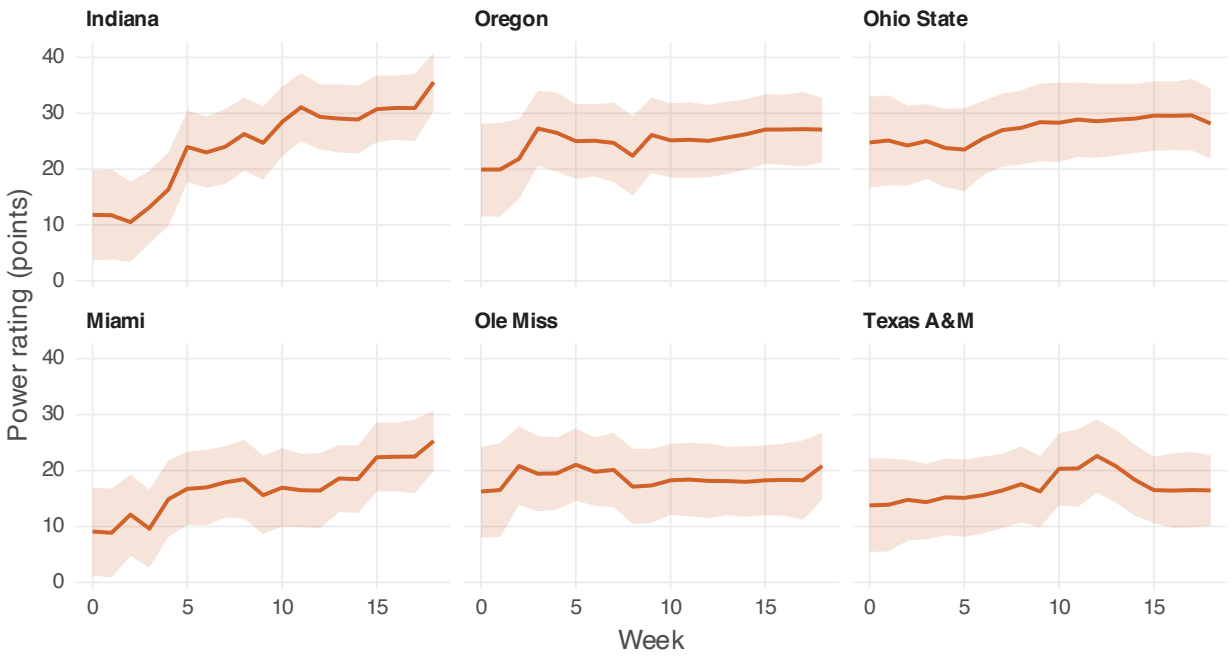


Figure 15: Weekly power rating in 2025 for the final resume top six, with 80% posterior intervals. Week 0 is the preseason posterior; the last point is the final posterior after the postseason.

With the random walk a rating can fall as well as rise late in the season, and the bands do not narrow to zero: the walk keeps adding uncertainty of ϕ_r per week while each game removes some.

Figure 16 splits the final 2025 power ratings into offence and defence (Section 3.3). The best offence by this measure is Indiana's, 44.4 points against an average FBS team, and the best defence Indiana's, 8.8 points allowed. Across FBS teams good offences and good defences tend to go together (correlation of Off with $-$Def 0.80), and teams with the same power rating can get there in very different ways: every point on one diagonal of the figure has the same Off $-$ Def.

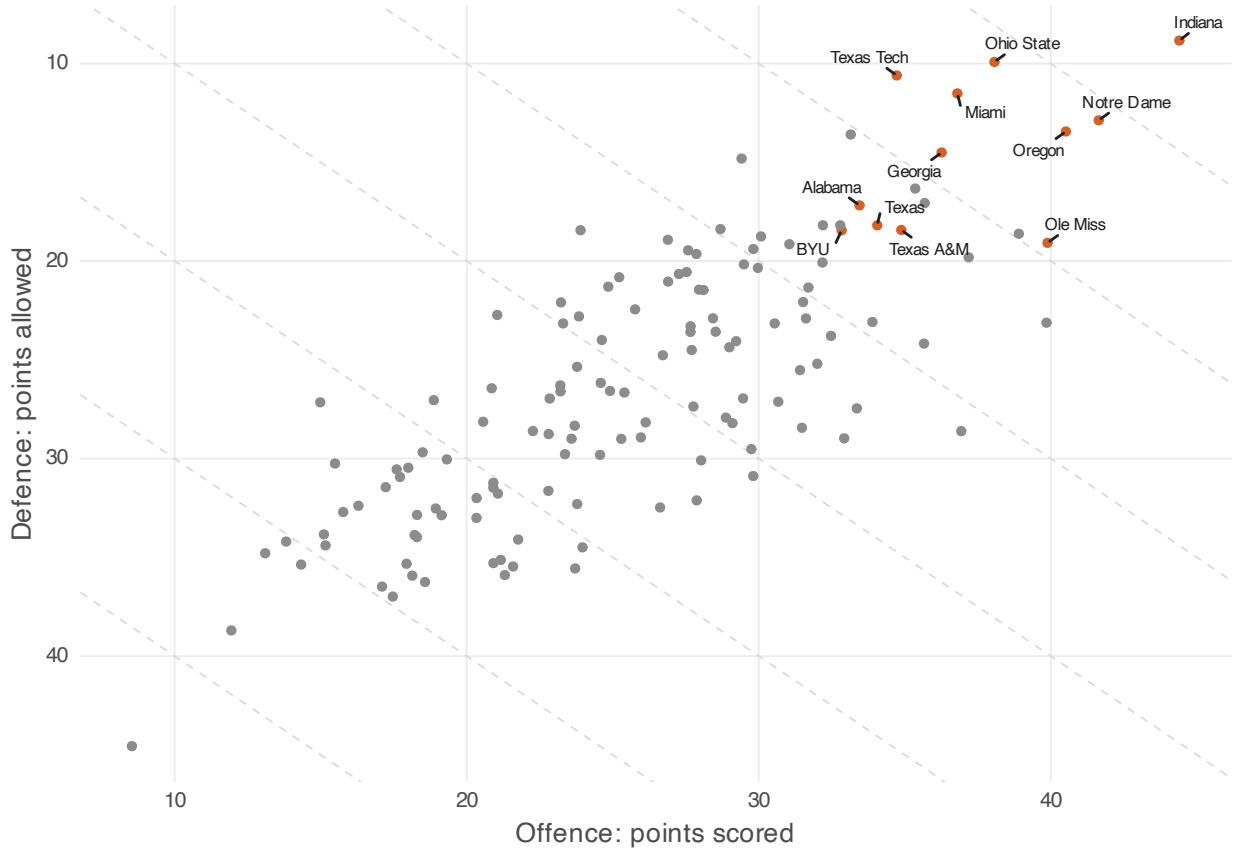


Figure 16: Offence and defence at the end of 2025, all FBS teams: points scored and allowed against an average FBS team on a neutral field. The defence axis is reversed, so better teams are up and to the right; dashed diagonals are lines of equal power rating, 10 points apart. Labelled, the resume top 12.

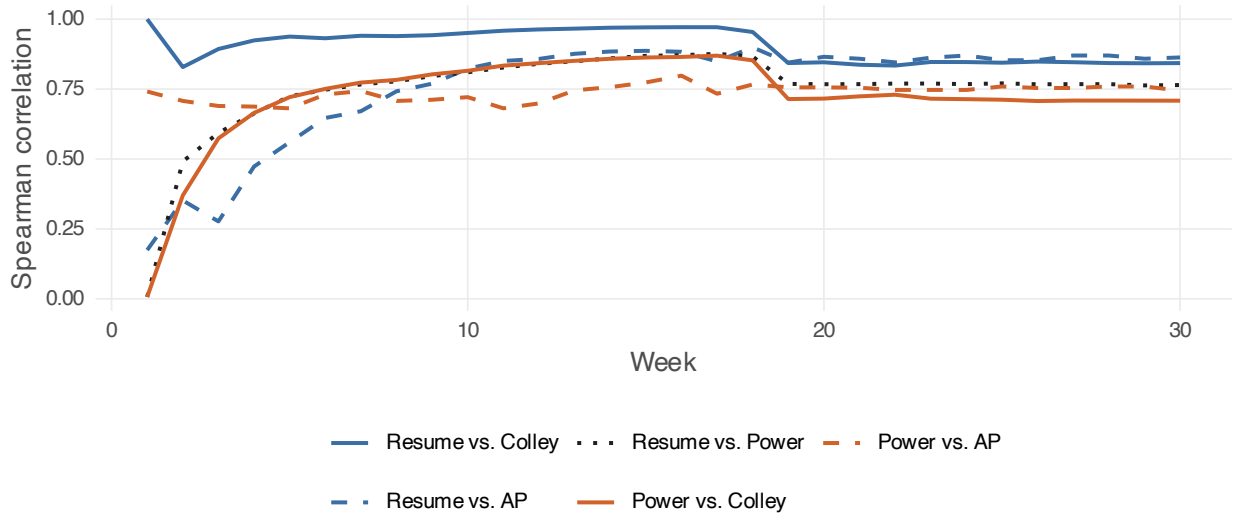


Figure 17: Spearman rank correlation between rankings of FBS teams, by week, averaged over complete seasons. Correlations with the AP poll use the teams it ranks.

Figure 17 measures agreement between the products and the references by Spearman rank correlation among FBS teams, week by week, averaged over the complete seasons. The resume tracks the record-only Colley ranking closely from the first weeks (mean correlation 0.91 in weeks 1–3 and 0.96 in weeks 10–14) — both are functions of the record — and diverges from it only in how opponents are valued. The power rating is less correlated with Colley (0.32 early, 0.84 late), as it should be, because it uses margin and efficiency. Agreement with the AP poll, which ranks only 25 teams, is computed on those 25.

Example. The products disagree most for teams that are strong but lost games. In 2025 the highest-rated team outside the resume top 25 is Penn State: power rank 12 (18.6 points) but resume rank 37, at 7–6. Table 23 shows why, game by game. Its own posterior gave it an expected 9.0 wins against this schedule, so its win luck is -2.0; 6 losses, 5 of them by one score, cost it 2.90 wins of WAB, while its 7 wins earned 0.77. The power rating reads those losses as close games that a strong team loses sometimes; the resume counts them as losses.

9.1 The 2025 Playoff Field

The first question a reader will ask of a resume is whether it picks the teams the committee picked. Table 24 compares the 2025 playoff field with the model at the committee's final ranking (the posterior through the conference championships, week 16). The resume's top 12 contains 10 of the 12 teams in the field. In the field but outside the resume top 12: Tulane (resume 20, CFP 20); James Madison (resume 18, CFP 24). In the resume top 12 but not in the field: BYU (resume 6, CFP 12); Vanderbilt (resume 12, CFP 14). The field is not simply the committee's top 12: the five highest-ranked conference champions qualify automatically, which is how a team outside the top 12 of any ranking can be selected.

Table 23: Resume against power, game by game, for Penn State in 2025. p_g: the team's own win probability under the final posterior; $\bar{\pi}_g$: the benchmark's.

Week	Opponent	Score	p_g	$\bar{\pi}_g$	$W_g - \bar{\pi}_g$
Week 1	Nevada	46–11	0.98	0.98	0.02
Week 2	Florida International	34–0	0.97	0.98	0.02
Week 3	Villanova	52–6	0.97	0.98	0.02
Week 5	Oregon	24–30	0.35	0.36	-0.36
Week 6	UCLA	37–42	0.87	0.89	-0.89
Week 7	Northwestern	21–22	0.83	0.86	-0.86
Week 8	Iowa	24–25	0.41	0.42	-0.42
Week 10	Ohio State	14–38	0.21	0.21	-0.21
Week 11	Indiana	24–27	0.17	0.16	-0.16
Week 12	Michigan State	28–10	0.81	0.83	0.17
Week 13	Nebraska	37–10	0.86	0.88	0.12
Week 14	Rutgers	40–36	0.81	0.83	0.17
Postseason	Clemson	22–10	0.72	0.75	0.25

Table 24: The 2025 playoff field and the model at the committee's final ranking. Teams in the field, in the resume top 12, or in the committee's top 12. Field: selected for the playoff.

Team	Record	Field	CFP	Resume	WAB	SOR	Power	AP
Indiana	13–0	yes	1	1	3.36	0.0053	1	1
Ohio State	12–1	yes	2	3	2.30	0.0810	2	3
Georgia	12–1	yes	3	5	1.86	0.1700	6	2
Texas Tech	12–1	yes	4	7	1.40	0.2400	4	4
Oregon	11–1	yes	5	2	2.43	0.0660	5	5
Ole Miss	11–1	yes	6	8	1.32	0.2600	11	6
Texas A&M	11–1	yes	7	4	1.88	0.1400	17	7
Oklahoma	10–2	yes	8	9	0.94	0.3900	16	8
Alabama	10–3	yes	9	10	0.75	0.4500	15	11
Miami	10–2	yes	10	11	0.46	0.5400	7	10
Notre Dame	10–2		11	13	0.39	0.5600	3	9
BYU	11–2		12	6	1.42	0.2400	18	12
Vanderbilt	10–2		14	12	0.46	0.5300	13	13
Tulane	11–2	yes	20	20	-0.42	0.8100	45	17
James Madison	12–1	yes	24	18	-0.17	0.8100	27	19

Table 25: Rankings against the committee, 2014–2025 (mean over seasons), at the snapshot before the committee's final ranking, and retrodictive accuracy at the end of each season. SOR ranks teams by strength of record.

Ranking	Field overlap	Top 4 (of 4)	Top 12 (of 12)	ρ, CFP top 25	Retrodictive accuracy
Resume	84%	3.33	10.2	0.88	80.7%
SOR	86%	3.33	10.2	0.89	81.4%
Colley	83%	3.25	10.2	0.83	81.6%
Power	67%	2.67	9.1	0.77	78.1%
AP	94%	3.75	11.1	0.96	–

9.2 The Resume Against the Committee

A retrodictive ranking has no future outcome to score it against, so the evidence for the resume has to be external. Table 25 compares each ranking with the committee's final ranking in every season since the playoff began: overlap with the playoff field (the top four through 2023, the twelve selected teams since 2024), with the committee's top 4 and top 12, and the rank correlation over the committee's 25. It also reports retrodictive accuracy: the share of the season's FBS–FBS games whose winner the final ranking places higher. The legacy composite that this model replaced is not included, because its code was retired with the defects listed in the project plan and cannot be rerun on the same data.

The AP poll agrees with the committee most closely, which is unsurprising: both are made by people watching the same games, and the poll is known to the committee. Among the computer rankings the resume and SOR match the committee's field in 84% and 86% of places, Colley in 83% and the power rating in 67%. The committee selects on results, as the resume does, more than on strength. Retrodictive accuracy is 80.7% for the resume, 81.4% for SOR, 81.6% for Colley and 78.1% for power: a ranking built from the record explains the record better than one built to predict.

10 Discussion

The design principle of this paper is that a ranking should say which question it answers, and that different questions can share a model without sharing a formula. The power rating and the resume come from one posterior and are never mixed: the resume borrows the model's estimate of how hard each game was, and nothing else. That separation has a cost — two numbers where the public expects one — but each number can then be checked against something observable: the power rating against future games (Section 8), and the resume, less directly, against the committee's selections and the season's own results (Section 9.2).

Limitations. Several choices deserve more work.

1. *The margin distribution near zero.* The model underpredicts one-score games in 13 of 13 seasons (Section 7.1), and neither a two-normal scale mixture nor a scale that grows with the mismatch removes the misfit, although the latter improves the LOO density considerably (Section 7.4). The remaining reason is that margins are integers with spikes at key numbers

(3 and 7) and at zero (overtime games, whose regulation margin is zero by construction), while the likelihood is continuous. This affects spread and cover probabilities rather than win probabilities. Alternatives that respect the discreteness are models for the two scores themselves, such as the bivariate Poisson of Karlis and Ntzoufras [47] or the hierarchical score model of Baio and Blangiardo [48], or a rounded (interval-censored) version of the present likelihood. They should be compared on the margin's LOO density.

2. *One step size for every team and week.* The random walk of Equation 4 moves every team by Gaussian steps of the same size, so a team that loses its quarterback in October is pulled down only gradually, at the rate the whole sport's drift allows, and a slow decline and a sudden one look alike. Heavier-tailed steps, or a step size that varies by team or by part of the season, would let a rating jump; the data identify even one step size only weakly (Section 3.2), so such extensions would need informative structure, such as injury reports, rather than more parameters alone.
3. *Asymmetric garbage time.* The efficiency margin excludes garbage time (the provider's definition), but the scoreboard margin cannot: a team that leads by 30 and rests its starters concedes late points that enter y_g but not e_g. That shrinks the scoreboard margin of lopsided games relative to their efficiency margin, which the model can only absorb in λ (the efficiency scale per point of margin) and κ; both are therefore averages over close and lopsided games. A margin-dependent λ, or a garbage-time-filtered scoreboard margin, would separate the two.
4. *Priors from the whole history.* The structural priors use every complete season (Section 3.4), weighted towards the recent ones. The leave-one-season-out comparisons of Section 8.5 bound what that borrows; a strictly prospective version would set each season's priors from earlier seasons only. The preseason terms use only two team-level roster numbers; player-level returning production, incoming transfers and coaching staffs would sharpen the first weeks further, where the model still trails the market most (Section 8.2).
5. *A simple total model.* Offence and defence come from a total model that is Gaussian, independent of the margin given the ratings, and blind to the score of overtime games (Section 3.3). It trails the closing over/under (Section 8.2), and the priors of the total model and of the random-walk steps are provisional: they were set from preliminary fits rather than pooled from the vague-prior refits as the other structural priors were, and the twelve pooled priors themselves date from the static Student-t model (Section 3.4). Re-pooling all of them under the published model is the next step.
6. *Games are exchangeable.* Motivation, rest, travel, injuries and weather are all absent, and home field is one number per season for every venue (Figure 3); the gap to the closing line in Section 8 is a measure of how much they matter.

11 Appendix: Supplementary Posterior Checks

The weekly posteriors of Section 6 are checked by gates and summaries only, because keeping every draw of 242 fits is not needed for publication. This appendix examines two posteriors in full, following the checklist of Kahle et al. [49] and Miyakawa and Kahle [50] for applied Bayesian computation — sampler behaviour; convergence (smooth, chain-consistent marginal densities, well-mixed traces, quickly damping autocorrelation, a Gelman–Rubin statistic approaching one);

posterior geometry; the justification of informative priors and the priors they induce on derived quantities; sensitivity of the posterior to changes in the prior; predictive adequacy; and influence — together with the visual workflow of Gabry et al. [36]. The two posteriors bracket the range the pipeline fits:

- **2025, week 1**: 8 games, about one per team. The data barely constrain a team's strength, the hierarchy and the carry-over do most of the work, and this is where the hierarchical geometry is hardest for the sampler.
- **2025, final**: all 1,687 games of the season, including the postseason.

Both are refitted with the production settings of Section 6.1 (4 chains, 1,000 post-warm-up draws each, 4,000 in all) by the `supplement` target in `R/06_supplement.R`.

11.1 Sampler Behaviour Across All Snapshots

Before the two detailed fits, Figure 18 shows the gate statistics of every weekly posterior. The retry of Section 6.1 is visible as the fits with doubled sampling.

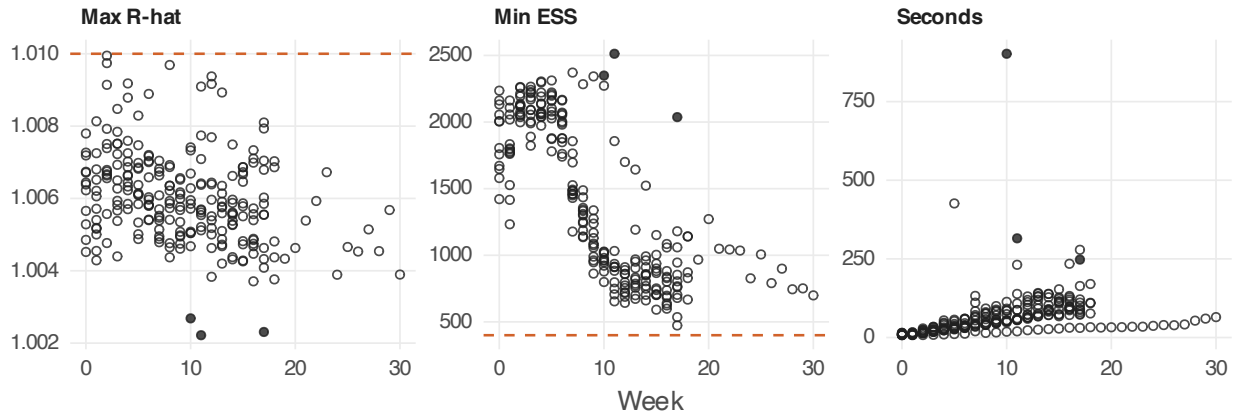


Figure 18: Gate statistics of all weekly posteriors by week, every season overlaid. Left: the largest rank-normalised $\widehat{R}$ over all monitored parameters, with the gate at 1.01. Centre: the smallest bulk or tail effective sample size, with the gate at 400. Right: wall-clock time of the fit, four chains in parallel. Filled points are fits that needed the retry.

11.2 Sampler Diagnostics

Table 26 reports the Hamiltonian Monte Carlo diagnostics of each chain [28]: the adapted step size, the mean number of leapfrog steps per iteration, how often the trajectory hit the maximum tree depth (which would indicate a truncated, inefficient exploration), divergent transitions (which indicate curvature the integrator cannot follow and a potentially biased posterior), and the energy Bayesian fraction of missing information, E-BFMI [51].

Neither fit has a single divergent transition, no iteration reaches the maximum tree depth, and the smallest E-BFMI is 0.91, far above 0.2. The week-1 fit, with few data, needs no longer trajectories than the final one: the non-centred hierarchy keeps its geometry close to a product of normals.

Table 26: Sampler diagnostics by chain. E-BFMI below 0.2 would indicate that the momentum resampling cannot move the chain across energy levels fast enough; the maximum tree depth is 10.

Fit	Chain	Step size	Leapfrog steps	Max depth	At max depth	Divergent	E-BFMI	Warm
2025, week 1	1	0.222	15.0	4	0	0	1.02	
	2	0.245	15.0	4	0	0	1.16	
	3	0.209	23.1	5	0	0	0.95	
	4	0.209	24.1	5	0	0	1.03	
2025, final	1	0.100	63.0	6	0	0	0.97	
	2	0.114	31.0	5	0	0	0.96	
	3	0.103	59.4	6	0	0	0.91	
	4	0.111	31.0	5	0	0	0.96	

Figure 19 compares the marginal energy distribution with the distribution of energy changes between iterations. When the second is much narrower than the first, the sampler explores the energy levels of the posterior slowly and tails may be under-sampled; overlapping histograms are what a healthy fit looks like.

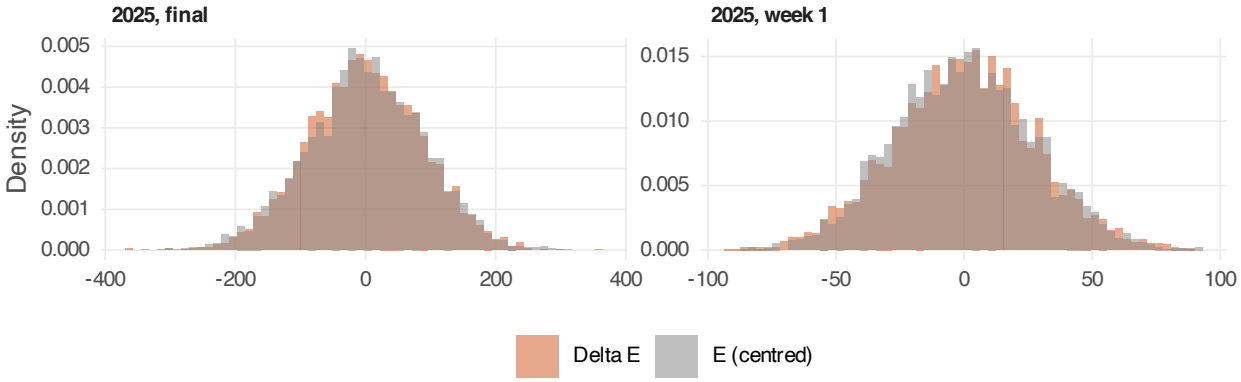


Figure 19: Energy diagnostic. Grey: marginal energy E, centred; orange: the change in energy between successive iterations, ΔE. Pooled over chains.

11.3 Convergence

Table 27 gives posterior summaries and convergence diagnostics of the structural parameters: the rank-normalised split-$\widehat{R}$, which compares between- and within-chain variation after rank-normalising and folding the draws [33, 52]; bulk and tail effective sample sizes, which account for autocorrelation [53]; and the Monte Carlo standard error of the posterior mean.

Every structural $\widehat{R}$, including those of the 12 total-model parameters, is at most 1.004, and the smallest bulk effective sample size is 667, for ϕ_r in the final fit (smallest among the total model's: 2,081). The random-walk step ϕ_r mixes most slowly: the data inform it only weakly, and its

Table 27: Posterior summaries and convergence diagnostics for the structural parameters of the margin model, the random-walk step and the family's shape parameters. *MCSE* is the Monte Carlo standard error of the posterior mean. The total model's parameters are summarised in the text.

Fit	Parameter	Mean	SD	5%	95%	$\widehat{R}$	ESS bulk	ESS tail	MCSE
2025, week 1	η (home field)	2.44	0.37	1.83	3.05	1.001	6,027	2,900	0.005
	β (carry-over)	0.59	0.04	0.53	0.65	1.000	4,898	2,834	0.001
	λ	0.79	0.02	0.76	0.82	1.002	5,014	3,056	0.000
	κ	0.75	0.02	0.71	0.79	1.000	5,482	2,877	0.000
	σ_e	14.00	0.17	13.73	14.28	1.002	5,423	2,729	0.002
	σ_m	6.59	0.40	5.94	7.24	1.003	5,499	2,896	0.005
	ν	30.70	4.02	24.44	37.57	1.002	5,640	2,565	0.053
	γ	0.28	0.01	0.26	0.31	1.000	5,674	2,749	0.000
	ω	2.43	0.33	1.89	2.98	1.000	5,634	2,850	0.004
	τ_1 (FBS)	4.84	1.23	2.80	6.88	1.001	3,994	1,761	0.019
	τ_2 (FCS)	5.02	0.47	4.25	5.81	1.002	5,144	2,880	0.007
	τ_3 (lower)	11.40	1.73	8.57	14.31	0.999	4,333	2,545	0.026
	m_2 (FCS)	-28.29	2.11	-31.72	-24.81	1.001	3,845	3,088	0.034
	m_3 (lower)	-60.24	4.84	-68.30	-52.22	1.001	5,668	2,734	0.064
	ϕ_r	1.57	0.28	1.10	2.03	1.001	5,102	2,271	0.004
2025, final	η (home field)	2.52	0.26	2.10	2.93	1.001	5,993	3,068	0.003
	β (carry-over)	0.61	0.03	0.55	0.66	1.002	3,375	3,039	0.001
	λ	0.78	0.01	0.76	0.80	1.000	5,476	2,674	0.000
	κ	0.73	0.01	0.70	0.75	1.002	4,079	2,810	0.000
	σ_e	13.94	0.15	13.71	14.18	1.001	5,821	3,084	0.002
	σ_m	6.90	0.18	6.60	7.20	1.001	3,932	2,977	0.003
	ν	29.45	3.90	23.25	36.16	1.001	5,449	3,037	0.053
	γ	0.29	0.01	0.27	0.31	1.001	5,250	3,082	0.000
	ω	2.31	0.32	1.80	2.84	1.001	3,339	3,064	0.005
	τ_1 (FBS)	5.45	0.69	4.28	6.55	1.001	1,953	2,268	0.016
	τ_2 (FCS)	5.21	0.42	4.53	5.88	1.000	3,630	2,947	0.007
	τ_3 (lower)	10.51	1.48	8.10	12.96	1.000	3,183	2,431	0.026
	m_2 (FCS)	-29.15	1.37	-31.37	-26.94	1.001	4,079	2,895	0.021
	m_3 (lower)	-61.57	2.96	-66.38	-56.79	1.000	3,818	2,355	0.048
	ϕ_r	1.46	0.19	1.14	1.77	1.003	667	1,686	0.007

posterior is correlated with every step of every team (Section 3.2). Monte Carlo standard errors are at most 3.9% of the posterior standard deviations.

The team strengths are too many to tabulate. Figure 20 shows the distribution of their $\widehat{R}$ and of their bulk effective sample size as a fraction of the number of draws.

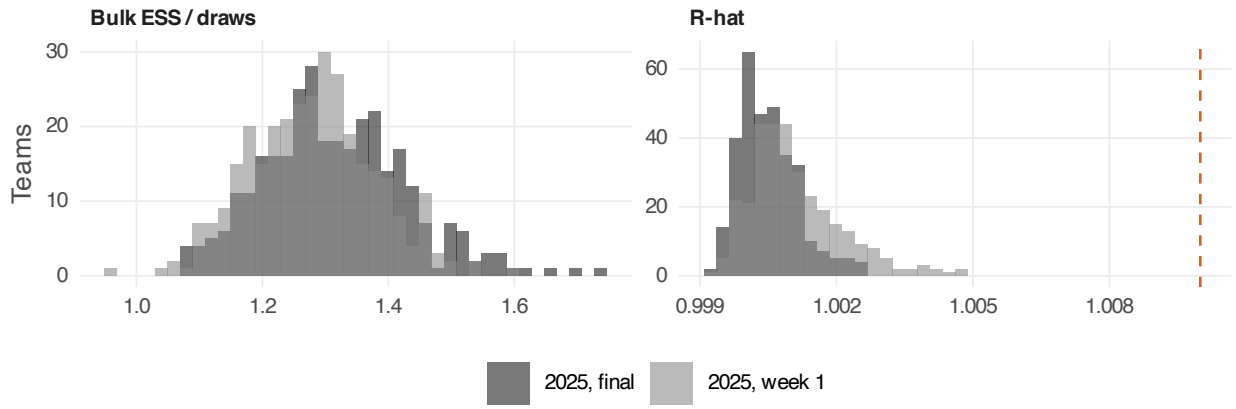


Figure 20: Convergence of all team strengths r_i. Left: rank-normalised $\widehat{R}$, with the gate at 1.01. Right: bulk effective sample size per draw; values above one are possible for anti-correlated chains, which NUTS often produces for well-identified parameters.

Trace plots (Figure 21, Figure 22) show the chains iteration by iteration; well-mixed chains overlap and look like stationary noise, with no trends, no stuck stretches and no chain wandering off on its own. Rank plots (Figure 23) are more sensitive: the draws of all chains are ranked together, and if the chains sample the same distribution each chain's ranks are uniform [33]. Kernel density estimates of each chain's marginal (Figure 24) should be smooth, unimodal where the model implies it, and indistinguishable between chains. Autocorrelation (Figure 25) should damp quickly with the lag, since it determines the effective sample size. Finally, the Gelman–Rubin shrink factor (Figure 26) should fall to 1 as the chains lengthen and stay there [52].

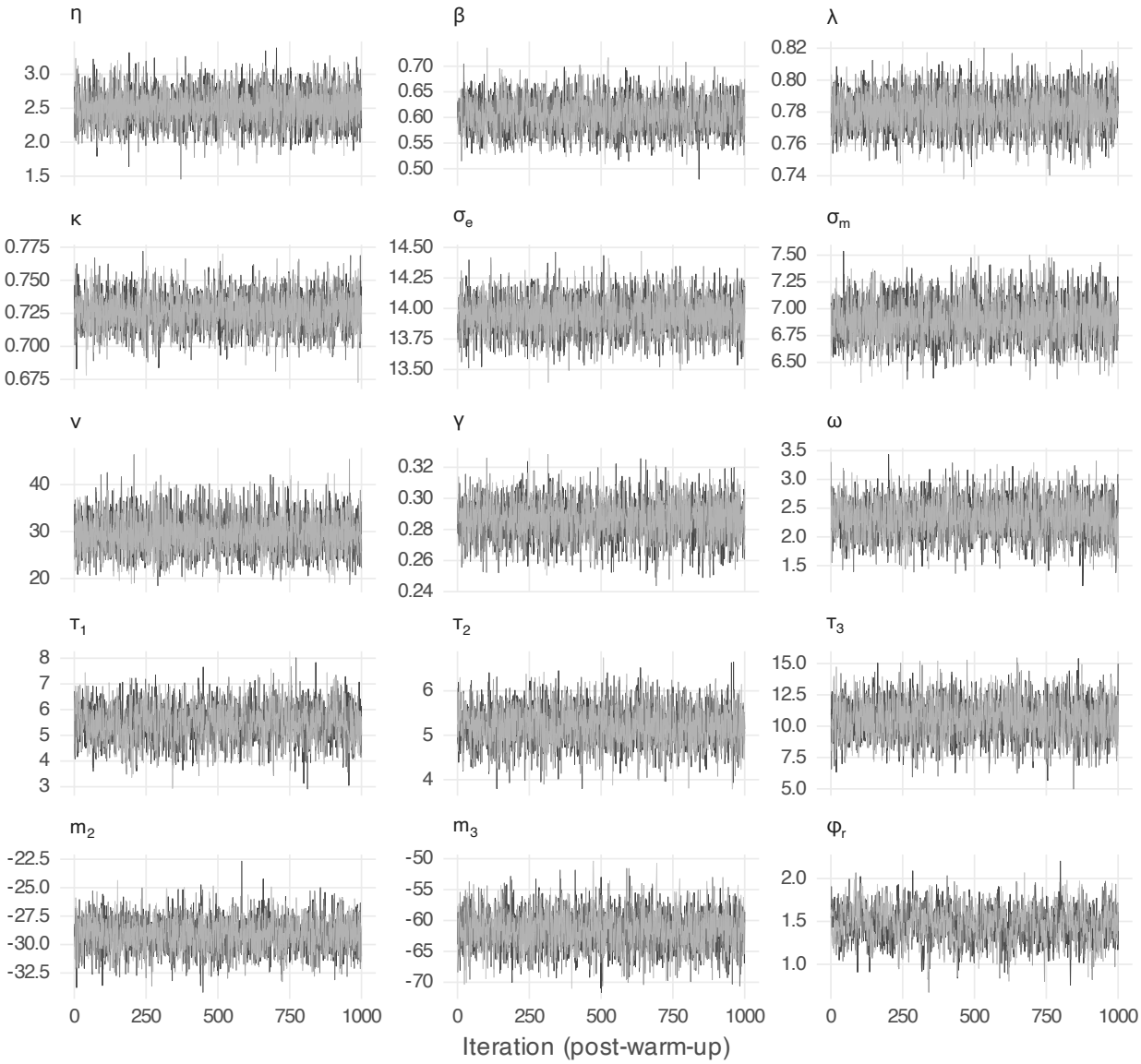


Figure 21: Trace plots of the structural parameters, 2025, final. One shade of grey per chain.

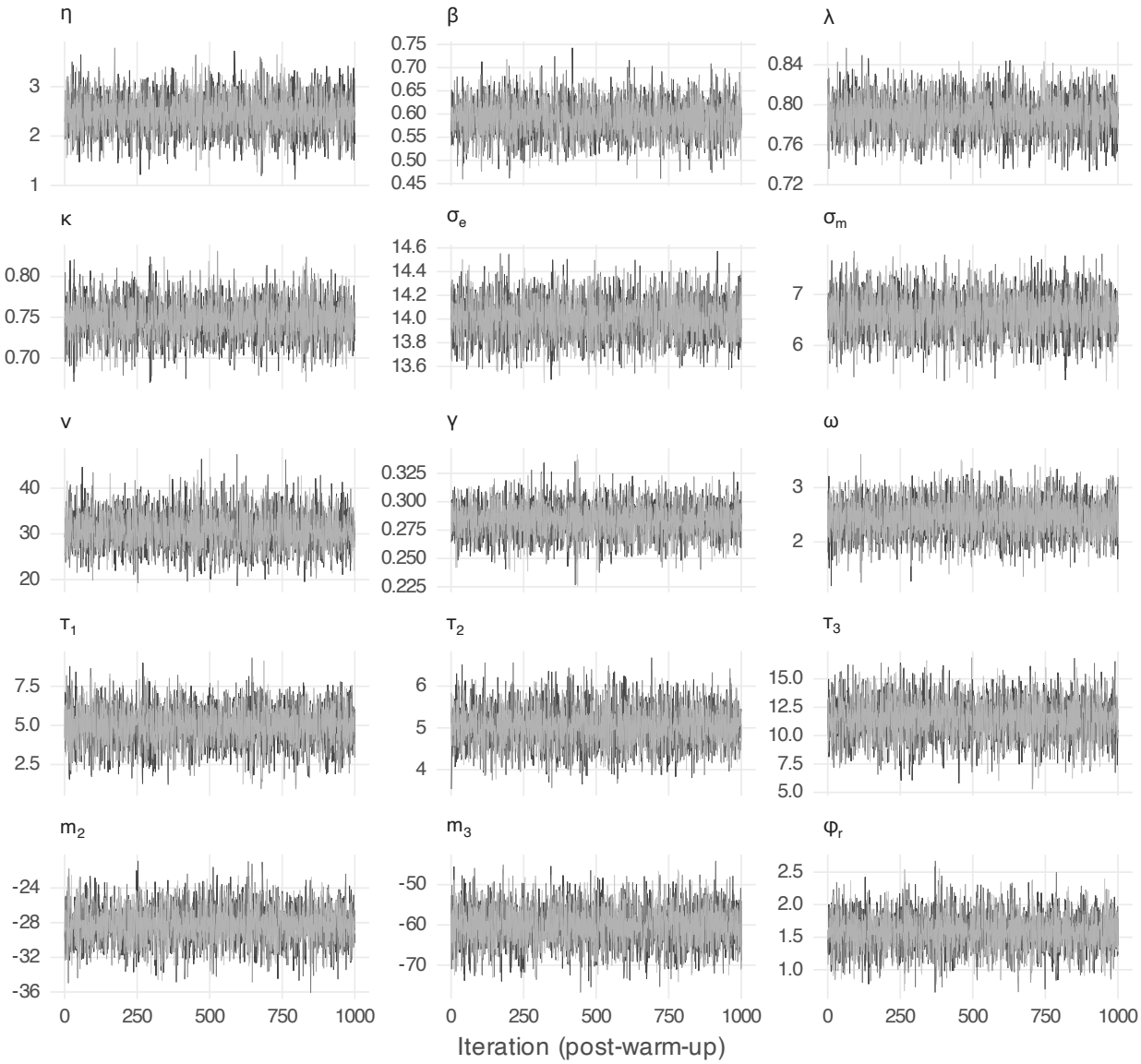


Figure 22: Trace plots of the structural parameters, 2025, week 1. With about one game per team several parameters are close to their priors, and the chains explore those priors freely.

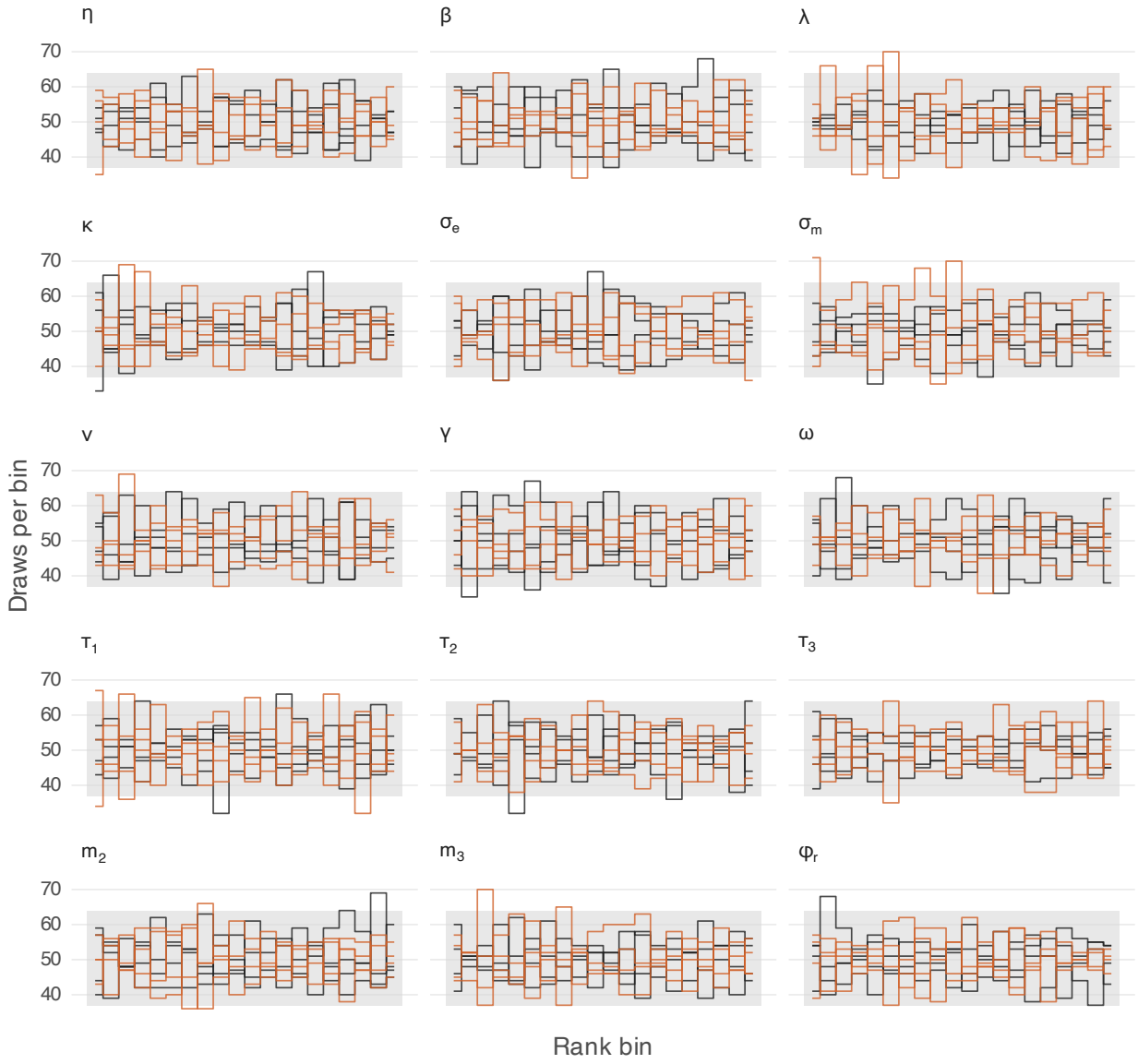


Figure 23: Rank plots of the structural parameters for both fits. For each parameter the draws of all chains are ranked jointly and each chain's ranks are binned into 20 bins; the band is the central 95% range of a bin count under uniformity. Lines are chains; 2025, week 1 in orange, 2025, final in grey.

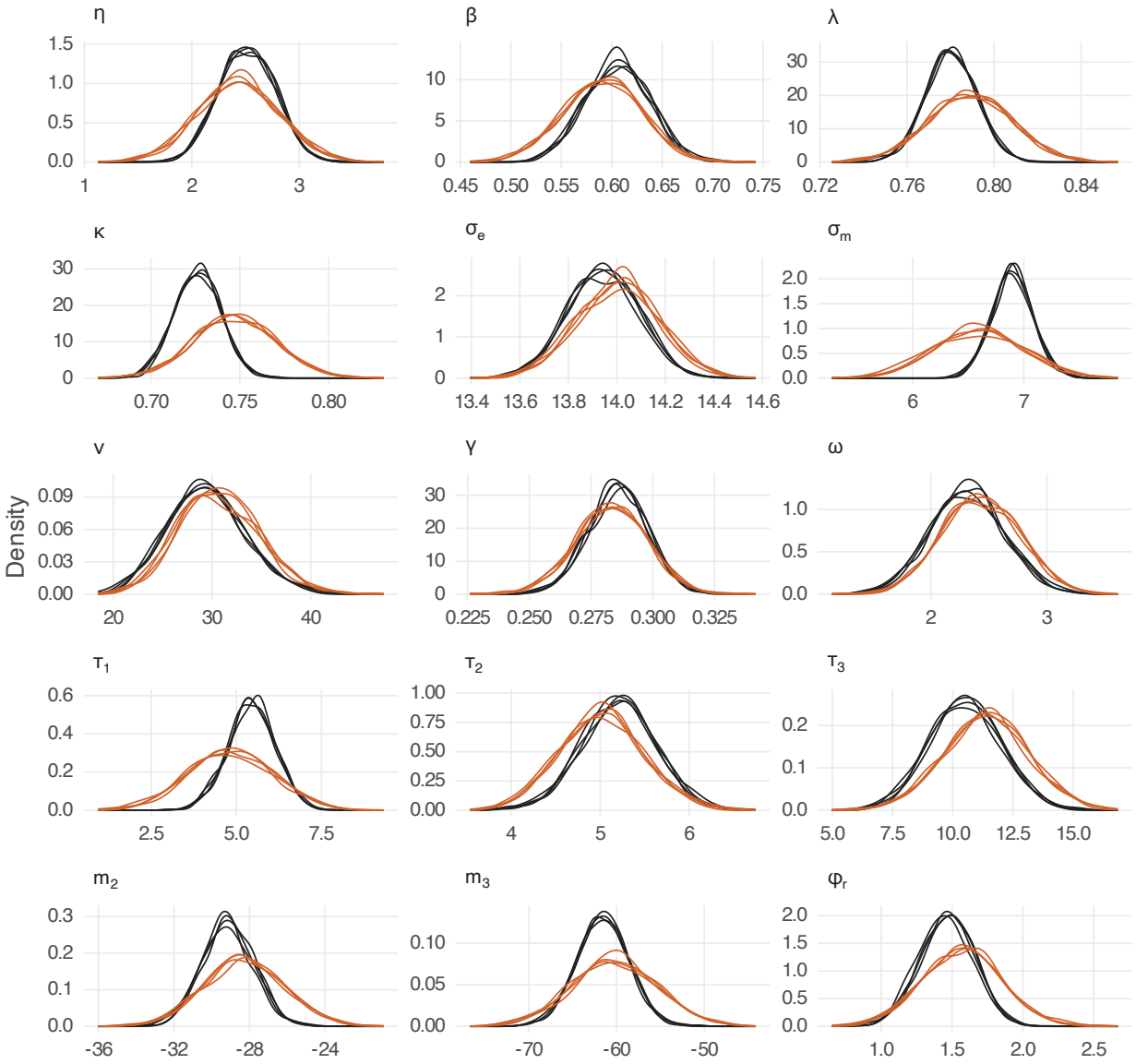


Figure 24: Kernel density estimates of each structural parameter's marginal posterior, one line per chain. Orange: week 1; grey: final. Chains that agree give overlapping curves; a bumpy or multimodal curve, or one chain apart from the others, would indicate poor exploration.

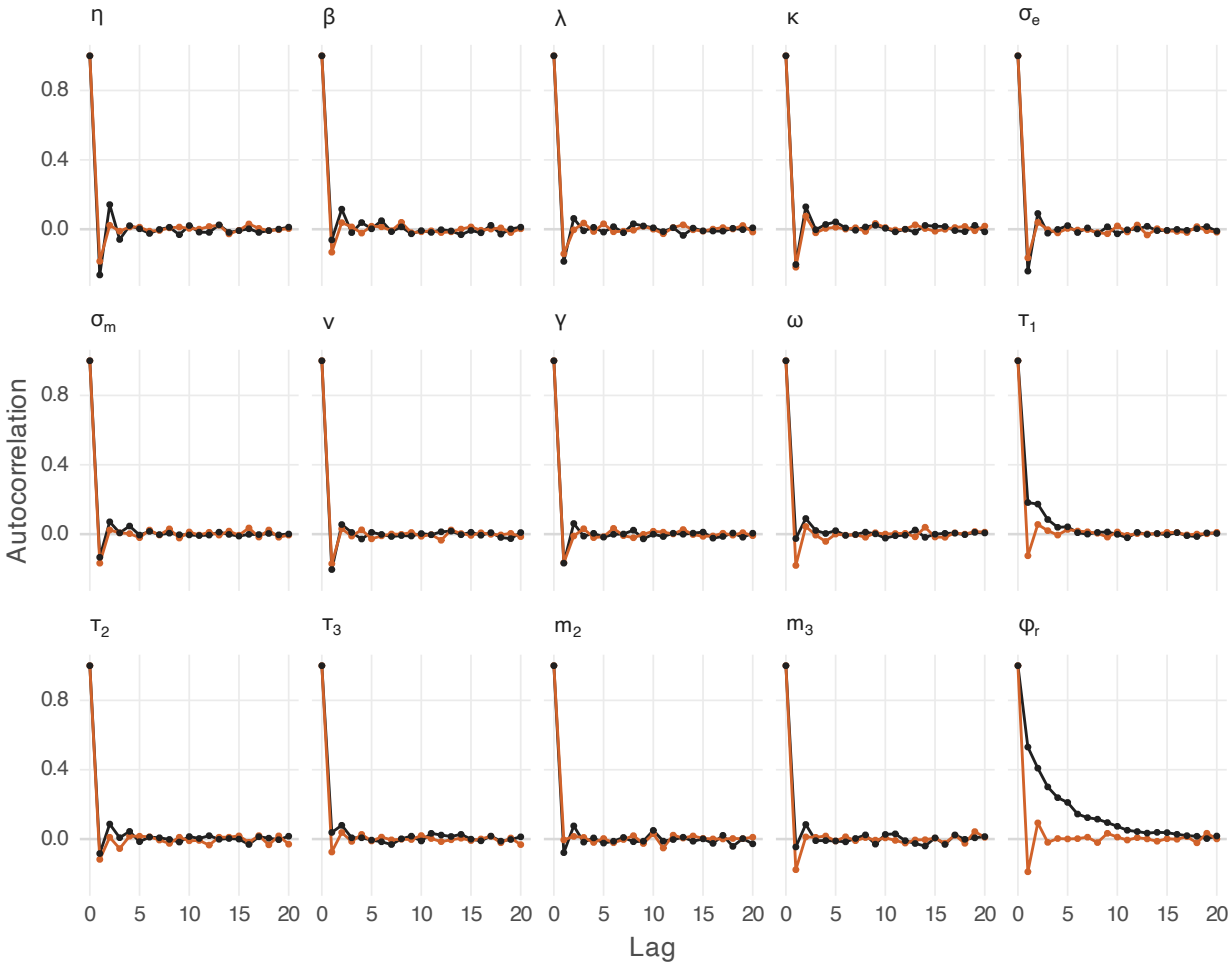


Figure 25: Autocorrelation of the structural parameters up to lag 20, averaged over chains. Orange: week 1; grey: final.

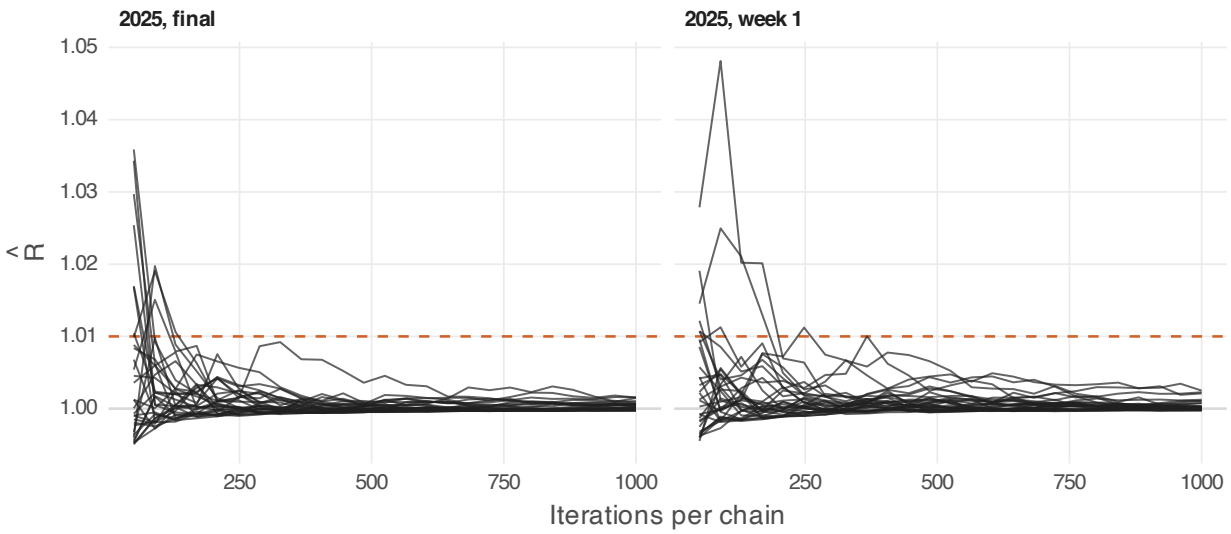


Figure 26: Gelman–Rubin shrink factor (rank-normalised split-$\widehat{R}$) of each structural parameter computed on the first n post-warm-up iterations of every chain, as n grows. The dashed line is the gate at 1.01.

The autocorrelation of 29 of the 30 parameter–fit pairs drops below 0.1 within three lags; the slowest is ϕ_r in the final fit, which needs 9 lags, and even there the effective sample size is well above the gate. The shrink factors of every parameter are below the gate after 208 iterations per chain in both fits and stay there, so the 1,000 iterations used are more than enough.

11.4 Posterior Geometry

Hierarchical models have funnels: when a scale parameter such as ω or τ_1 is small, the group effects it governs must be close to their mean, and the posterior narrows into a region the sampler can struggle to enter. The non-centred parameterisation of Equation 3 removes the funnel from the sampled coordinates when the data are weak [27]. Figure 27 plots each scale against a standardised effect it governs, in the sampled coordinates; divergent transitions, if any, would be drawn in orange and would typically concentrate at the neck of a funnel.

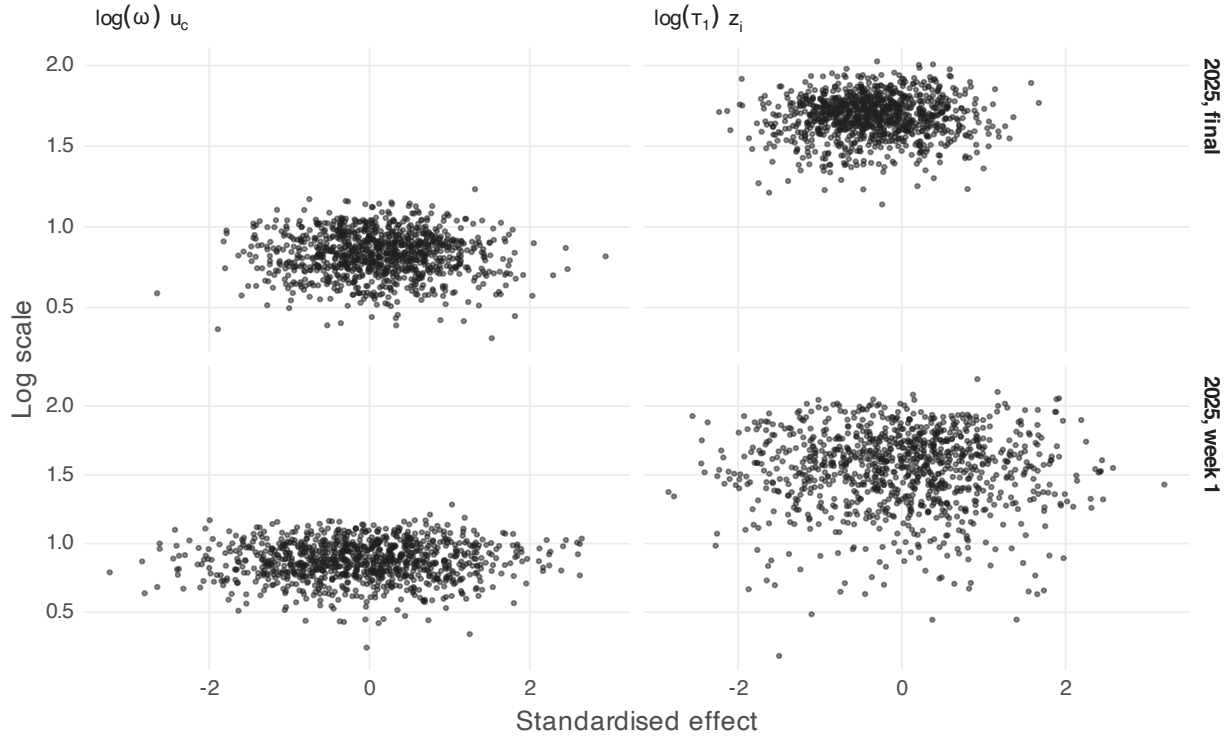


Figure 27: Posterior geometry in the sampled coordinates, 1,000 draws per panel. Each panel plots the log of a scale parameter against a standardised effect it governs: the conference effect u_c of the first FBS conference against $\log \omega$, and the team effect z_i of the FBS team with the fewest games against $\log \tau_1$. Divergent transitions would be orange.

Neither fit shows a funnel in the sampled coordinates: the standardised effects have the same spread at every value of the scale, which is the purpose of the non-centred parameterisation. In the final fit the team effect is nearly uncorrelated with τ_1 and concentrated, because a full season pins the team down; in week 1 it spreads over its $\mathcal{N}(0, 1)$ prior, as it should with one game. The random walk has a funnel of the same kind, between the step ϕ_r and the standardised steps $\varepsilon_{i,k}$ of Equation 4, which is not drawn because the steps are not kept; the evidence that the sampler enters its neck is the absence of divergent transitions, which the gates require of every published fit (Section 6.1), together with the informative prior that keeps ϕ_r away from zero.

11.5 Prior and Posterior

Figure 28 overlays the prior density of each structural parameter on its posterior in both fits. The comparison shows what the data teach: in week 1 many posteriors stay close to the prior, which is why the priors of Section 3.4 were set empirically, while by the end of the season most have contracted well inside it.

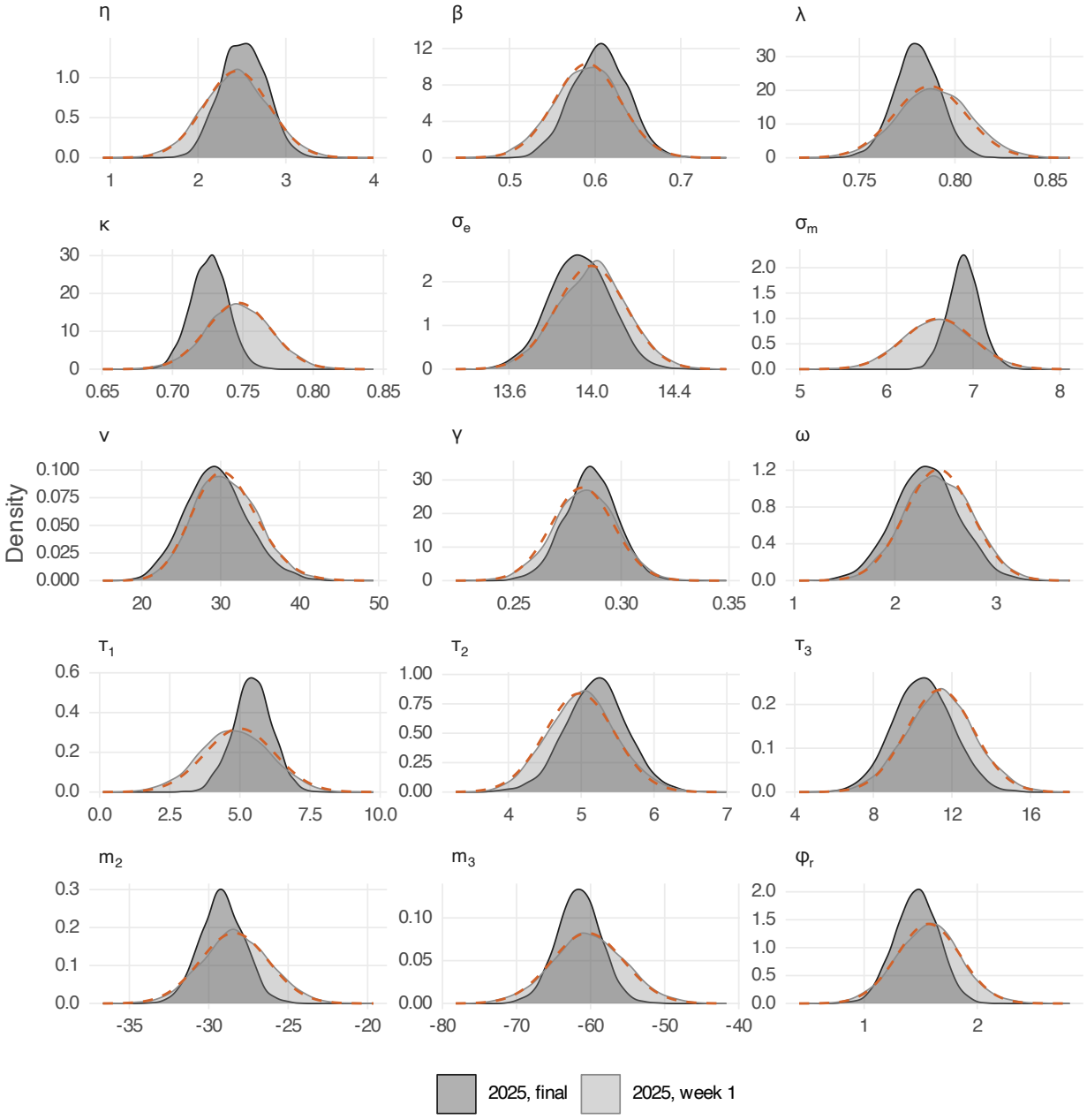


Figure 28: Prior (dashed) and posterior densities of the structural parameters. Priors are normal, truncated to the parameter's support, except $\nu \sim \mathrm{Gamma}(56.2, 1.83)$ on $\nu > 1$.

Figure 29 condenses the same comparison into two numbers per parameter [54]: the posterior contraction $1 - \mathrm{Var}_{\mathrm{post}} / \mathrm{Var}_{\mathrm{prior}}$, near one when the data dominate the prior, and the posterior z-score $(\mathbb{E}_{\mathrm{post}} - \mathbb{E}_{\mathrm{prior}})/\mathrm{sd}_{\mathrm{post}}$, large in magnitude when the posterior sits far from where the prior expected it. Contraction near one with small $|z|$ is the ideal; high contraction with large $|z|$ indicates a prior–data conflict; low contraction indicates that the prior, not the data, is determining the answer.

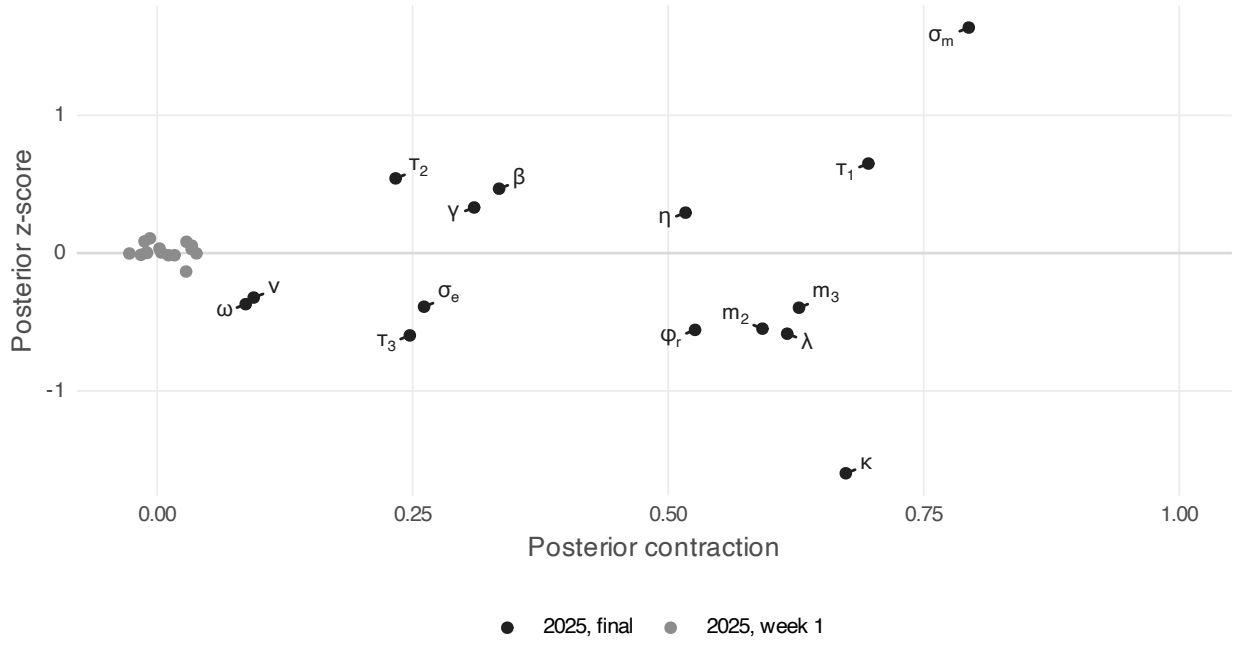


Figure 29: Posterior contraction against posterior z-score for the structural parameters, for both fits. Prior moments are those of the truncated priors.

In week 1, 15 of 15 structural parameters contract by less than half — η, σ_m, λ, σ_e, κ, β, m_2, m_3, τ_1, τ_2, τ_3, ω, ν, γ and ϕ_r — so their priors carry the fit, as intended in a week with one game per team. By the end of the season every parameter except σ_e, β, τ_2, τ_3, ω, ν and γ contracts by more than half. No posterior lies more than two of its own standard deviations from its prior mean.

11.6 Justification of the Informative Priors

Informative priors need a justification that does not come from the data they are applied to. Ours come from the history of the sport. The structural parameters describe the game rather than any team — how much a home field is worth, how noisy a margin is, how far FCS teams sit below FBS teams, how much of a team's strength carries over a summer — and they should change little from year to year. The `prior_refits` target checks this directly: it refits the full-season posterior of every complete season, each with about 1,550 games, under deliberately vague priors — $\eta \sim \mathcal{N}(0, 10^2)$, tier offsets $\mathcal{N}(0, 100^2)$, β nearly uniform on $(0, 1)$, half-normal priors with scale 25 on τ_ℓ and ω, scale 2 on λ and κ, and scale 50 on σ_e and σ_m, and comparably wide priors on the total model and the family's shape parameters — so that a season's estimate owes nothing to the published priors. The random-walk steps get wide priors centred away from zero, $\mathcal{N}^+(0.8, 0.8^2)$ for ϕ_r and $\mathcal{N}^+(1, 0.8^2)$ for ϕ_t, because under one with its mode at zero the sampler stalls in the funnel at small ϕ_r (Section 3.2); ν gets $\mathrm{Gamma}(2, 0.05)$, with mean 40, and the preseason coefficients normal priors several times wider than their estimates. Figure 30 plots the resulting per-season posteriors against the priors now in use, and Table 28 gives the numbers.

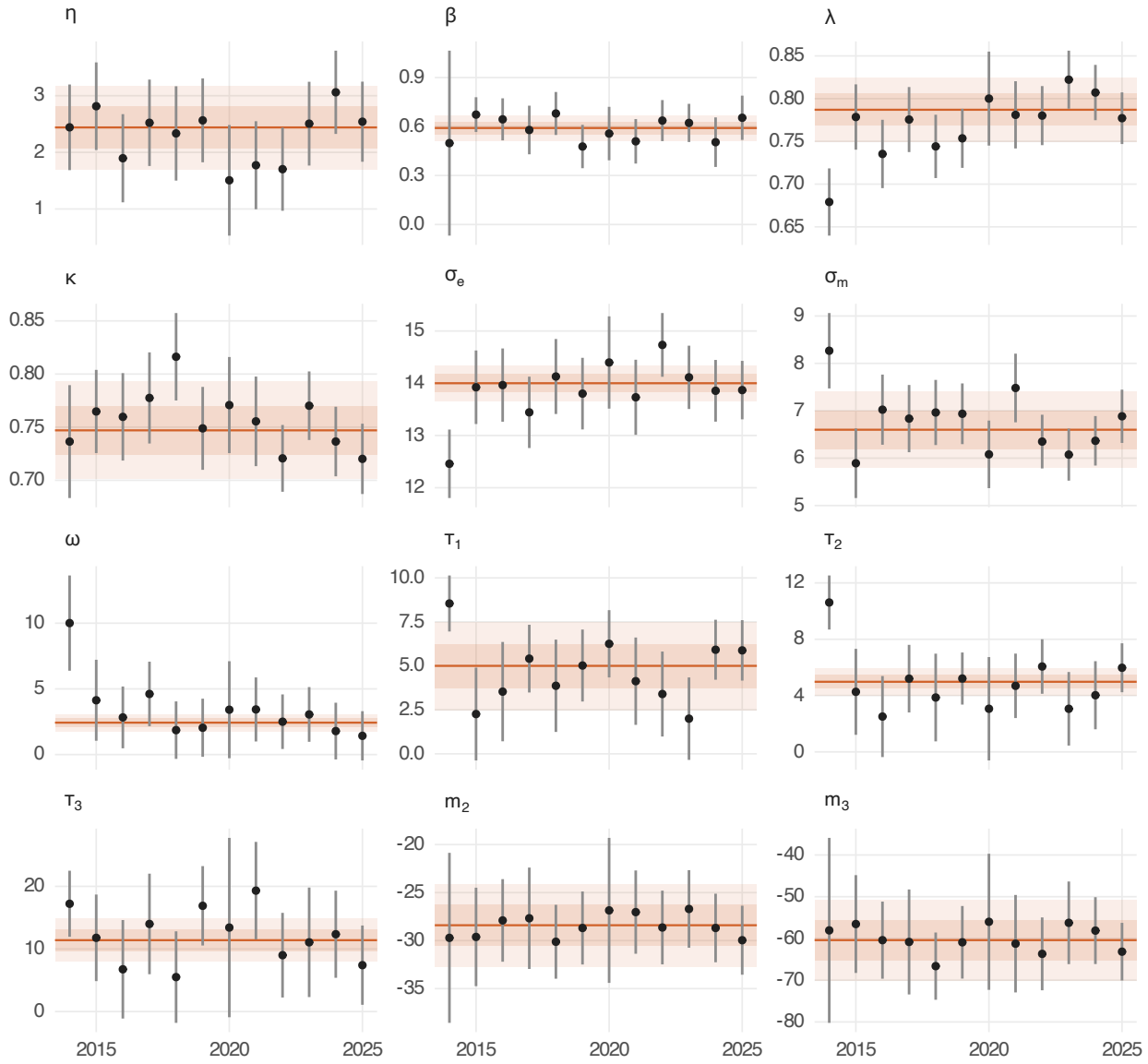


Figure 30: Per-season full-season posteriors under vague priors (points: mean; bars: ± 2 posterior sd) against the prior now in use (line: prior mean; bands: ± 1 and ± 2 prior sd). The priors are centred on the recency-weighted mean of the seasons after the first, with the predictive standard deviation of Equation 6.

Four points qualify this justification. First, pooling every complete season is empirical Bayes at the level of the sport, and saying that a season's estimate lies inside a prior partly built from it would be circular. The last column of Table 28 avoids that: for each season it pools the *other* seasons only and asks whether the season's own estimate lies within two standard deviations of that leave-one-season-out prior, where the standard deviation combines the prior's with the estimate's own posterior standard deviation. For every parameter it does in 82% or more of seasons, so a new season should be expected to fall inside its prior. Second, 2014 is left out of the pooling for the structural reason given in Section 3.4, and 2020 for η. Third, the prior standard deviation removes

Table 28: Empirical basis of the structural priors: the pooled mean of the per-season full-season posterior means under vague priors, their between-season standard deviation, the typical within-season posterior sd (root mean square), and the prior now in use. *Ratio* is the prior sd over the between-season sd; *LOSO* is the share of seasons whose own estimate lies within two standard deviations of the prior pooled from the other seasons, counting the estimate's own posterior sd.

Parameter	Pooled mean	Between-season sd	Posterior sd	Prior mean	Prior sd	Ratio	LOSO
η (home field)	2.44	0.48	0.37	2.44	0.37	0.8	100%
β (carry-over)	0.59	0.07	0.07	0.59	0.04	0.5	100%
λ	0.79	0.02	0.02	0.79	0.02	0.8	91%
κ	0.75	0.03	0.02	0.75	0.02	0.8	91%
σ_e	14.02	0.34	0.32	14.00	0.17	0.5	91%
σ_m	6.60	0.47	0.30	6.60	0.40	0.9	82%
ω	2.43	0.94	1.15	2.43	0.33	0.3	100%
τ_1 (FBS)	4.63	1.60	1.06	5.00	1.25	0.8	91%
τ_2 (FCS)	4.61	1.27	1.20	4.98	0.48	0.4	100%
τ_3 (lower)	11.22	4.06	3.97	11.40	1.71	0.4	91%
m_2 (FCS)	-28.44	1.32	2.14	-28.40	2.15	1.6	100%
m_3 (lower)	-60.40	3.36	4.83	-60.40	4.85	1.4	100%

estimation noise from the between-season spread (Equation 6). For β, m_2, m_3, ω, σ_e, τ_2 and τ_3 that spread is about the size of the within-season posterior sd, so their priors are narrow, as narrow as the uncertainty of a mean over about 7 effective seasons. That is the right width if the parameter is the same every season, and the leave-one-season-out column checks it. Fourth, the priors in use are copied into the configuration from one run of the refits and lag them by a rebuild: 1 of the 12 prior means is more than half a prior sd from the current pooled mean (τ_2).

11.7 Induced Priors

A prior on parameters induces a prior on everything computed from them, and a set of individually reasonable priors can combine into an unreasonable one. The quantities that matter here are not ω or τ_1 but the spread of team strengths, the probability that the home team wins an even game, and the distribution of margins the model expects before seeing any game. Table 29 simulates them from the priors alone, with the 2025 schedule and carry-over, each game played at the random-walk states of its week and its margin drawn from the scale-t law, and Figure 31 compares the prior predictive distribution of three margin statistics with the values observed that season.

Table 29: Induced priors, by simulation from the priors alone (4,000 draws) on the 2025 schedule, with the carry-over d_i from 2024. The last column gives the value observed in 2025 where one exists.

Quantity	5%	Median	95%	Observed
Marginal margin scale $\bar{\sigma}$ (pts)	11.76	12.37	12.97	
Share of margin variance from efficiency, R^2	0.67	0.72	0.76	
Degrees of freedom ν	24.18	30.42	37.67	
Weekly random-walk step ϕ_r (pts)	1.10	1.56	2.04	
SD of FBS strengths (pts)	10.75	12.55	14.42	
Best minus 12th-best FBS team (pts)	7.42	12.84	20.82	
P(home win), evenly matched teams	0.56	0.58	0.60	
P(best FBS team beats an average one), neutral	0.95	0.97	0.99	
SD of regulation margin, all games (pts)	20.95	22.36	23.88	22.47
Share of games within 7 points	0.25	0.28	0.30	0.33
Home win rate, true home games	0.59	0.61	0.63	0.60

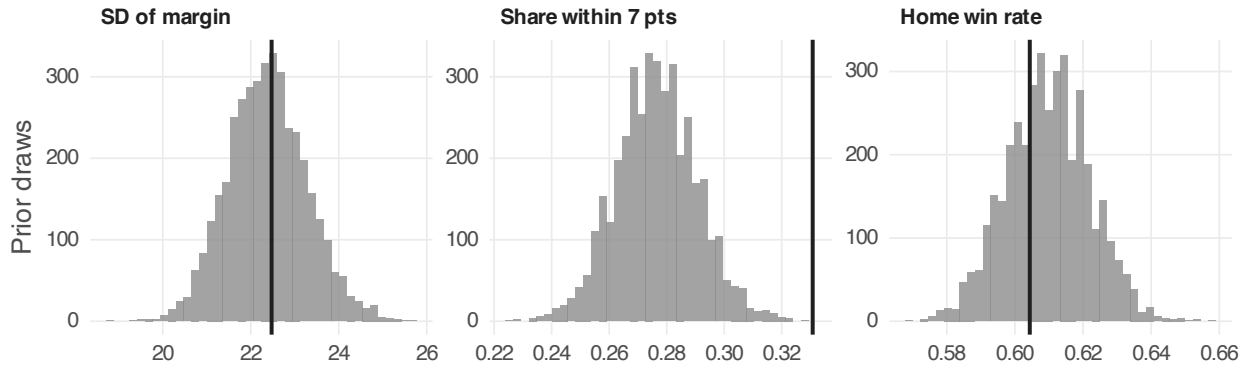


Figure 31: Prior predictive distributions of three statistics of the 2025 regulation margins, simulated from the priors alone; the vertical line is the observed value.

The induced priors are on the scale of the sport: before any game, the model expects the best FBS team to be about 13 points better than the twelfth (90% prior interval 7–21), an even game to be won at home 58% of the time, and margins with a standard deviation of about 22 points. 2 of the three observed statistics lie inside the central 95% of their prior predictive distribution; the exception is the share within 7 pts, the statistic the fitted model also misses (Section 7.1). Three induced quantities deserve care. The first is the tail of the margin law. A vague prior on ν such as $\mathrm{Gamma}(2, 0.1)$ puts 3.7% of its mass below 3, where the t law has little or no variance, and such draws produce prior-predictive seasons with a margin standard deviation of 50 points or more: an implausible implication of a prior on one parameter for an observable. The pooled prior avoids it, because every season's vague-prior estimate of ν lies between 16 and 56 (Table 6): 0.0% of its draws fall below 3, and the 99th percentile of the prior-predictive margin standard deviation is 25 points. The spread of FBS strengths combines ω, τ_1 and βd_i, so a prior that looks modest for

each piece can imply a wide or narrow league; the table shows it lands at a plausible 13 points. And the prior on $\bar{\sigma} = (\sigma_m^2 + \kappa^2 \sigma_e^2)^{1/2}$, which with the family's shape parameters sets every win probability and therefore every π_g in the resume, is induced by three separate priors; its 90% prior interval, 11.8–13.0 points, is narrow, which is appropriate only because each of the three is well estimated in every season (Figure 30).

11.8 Posterior Predictive Checks

Section 7.1 compared four summary statistics with their replicates. Here the whole distributions are compared (Figure 32), and then the location and spread of the margin within each kind of matchup (Figure 33), since a model that fits the pooled distribution can still misfit a subgroup, in particular the games across tiers.

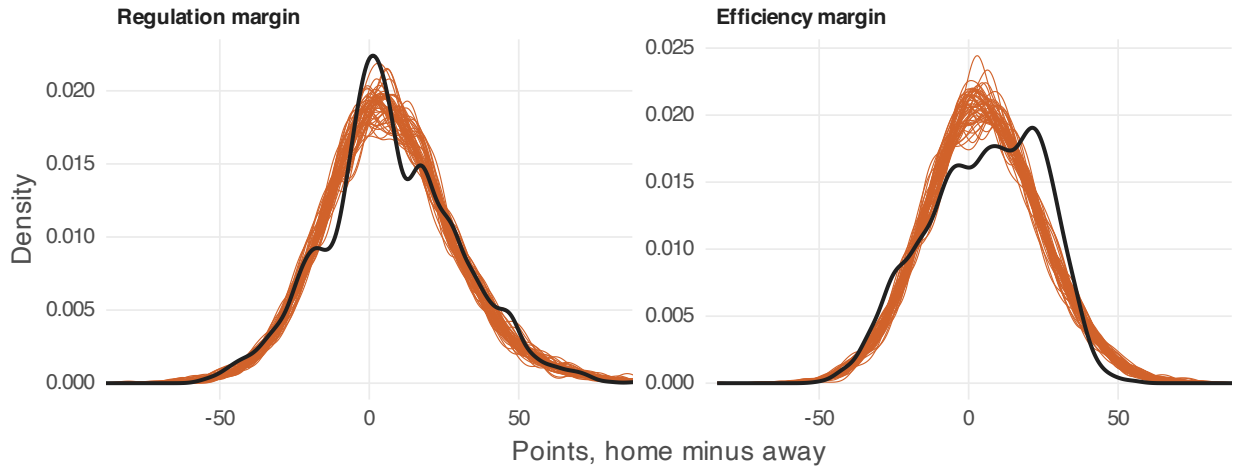


Figure 32: Observed distribution (black) against 50 replicated data sets (orange) for the 2025, final fit. Left: regulation margin, home minus away, replicated from the marginal law of Equation 2 and rounded. Right: efficiency margin, for games that have one.

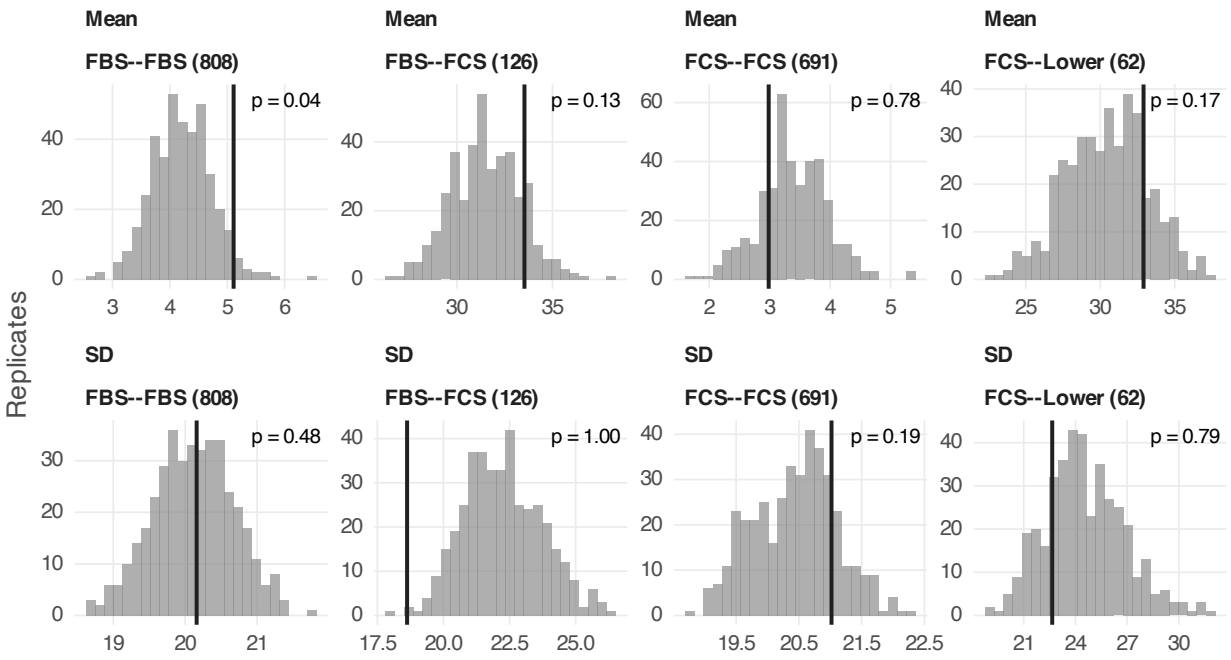


Figure 33: Grouped test statistics, 2025, final. Histograms: the mean and standard deviation of the regulation margin over 400 replicated data sets (as in Figure 7), within each kind of matchup; vertical line: the observed value. The label gives the posterior predictive p-value, the share of replicates at least as large as the observed statistic.

Across the 8 grouped statistics the posterior predictive p-values run from 0.04 to 1.00, and 1 falls outside $(0.025, 0.975)$: the sd in FBS–FCS games, where the observed value is smaller than almost every replicate. The tier offsets therefore place cross-tier games about right in location, and the misfit of Section 7.1 is mainly one of shape. Games across tiers are, if anything, less variable than a single margin scale for all games allows: a tier-specific scale is a natural refinement. The density overlay shows the same shape misfit directly: the observed margin has a sharper peak near zero than any replicate.

The leave-one-out probability integral transform (LOO-PIT) checks calibration game by game without using a game to predict itself: for each game, $\Pr(y_g^{\text{rep}} \le y_g \mid \text{all other games})$, estimated with the Pareto-smoothed importance weights of the LOO computation. If the model is calibrated these values are uniform; a U shape indicates predictive distributions that are too narrow, a hump that they are too wide [36, 55]. Figure 34 shows them for both signals and both fits.

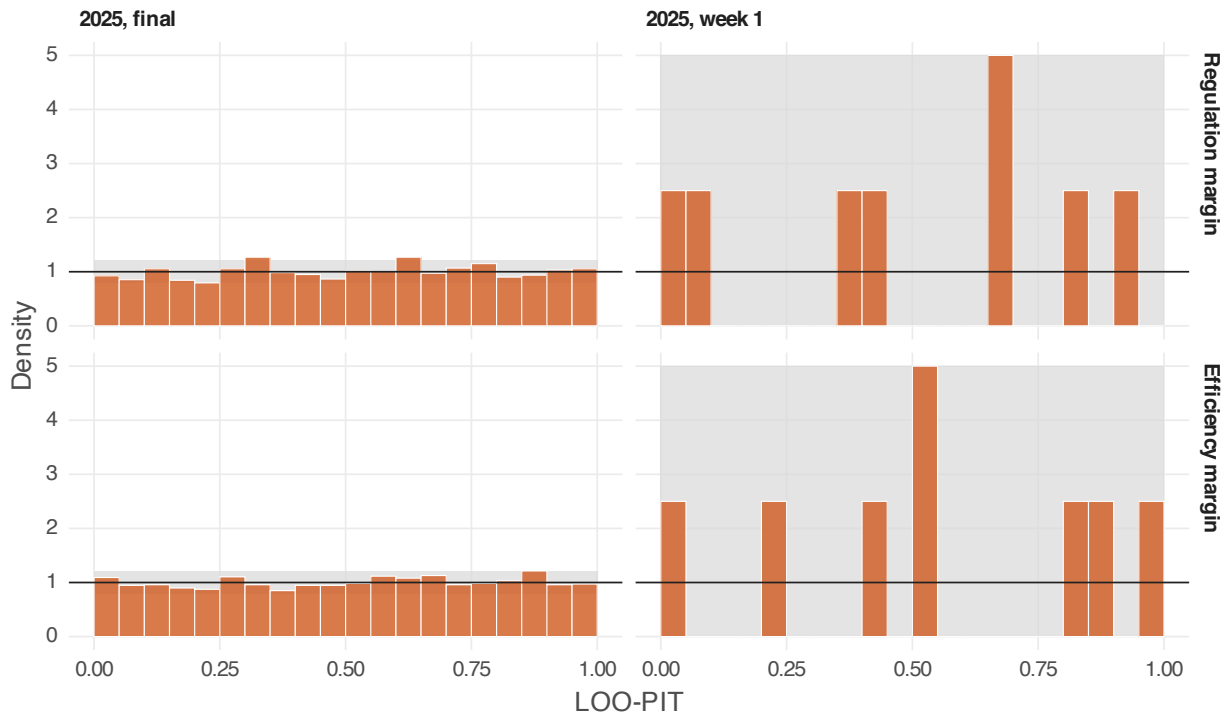


Figure 34: LOO-PIT histograms, 20 bins; the band is the central 95% range of a bin count under uniformity. Margins are whole numbers while the model is continuous, which spreads each PIT value over an interval of width about $1/\bar{\sigma} \approx 0.08$ and does not by itself cause non-uniformity at this bin width.

Of the 54 histogram bins across the four panels, 2 fall outside their 95% band, against about 3 expected by chance. There is no U shape or hump, so the predictive distributions are neither systematically too narrow nor too wide, in week 1 or at the end of the season.

Figure 35 plots the observed margin minus its posterior mean against that mean. A systematic trend would mean that the model's spreads are biased for lopsided or for close matchups — for example, if strong favourites won by less than predicted.

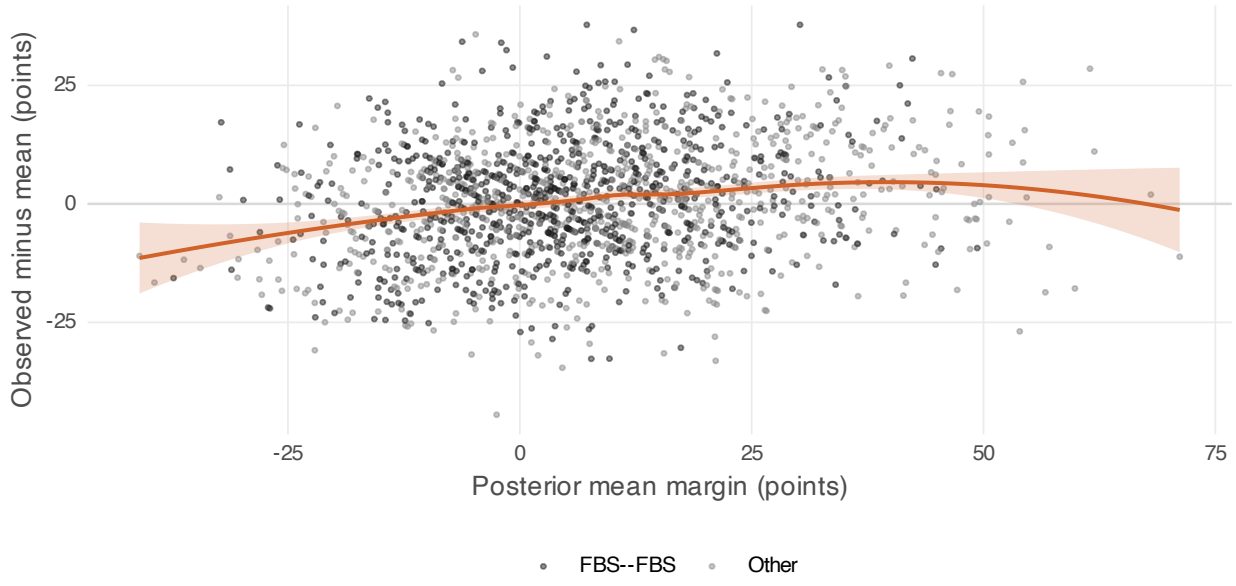


Figure 35: Residual regulation margin against the posterior mean margin, 2025, final. The line is a LOESS smooth with a 95% band; points are games, by kind of matchup.

11.9 Influence and PSIS Diagnostics

The Pareto $\hat{k}$ of each game estimates the tail shape of its importance ratios in PSIS-LOO; values above 0.7 mean the leave-one-out approximation is unreliable for that game, and in practice that the game is highly influential on the posterior [39]. Figure 36 plots every game's $\hat{k}$ in both fits, and Table 30 lists the games above 0.7 in the final fit.

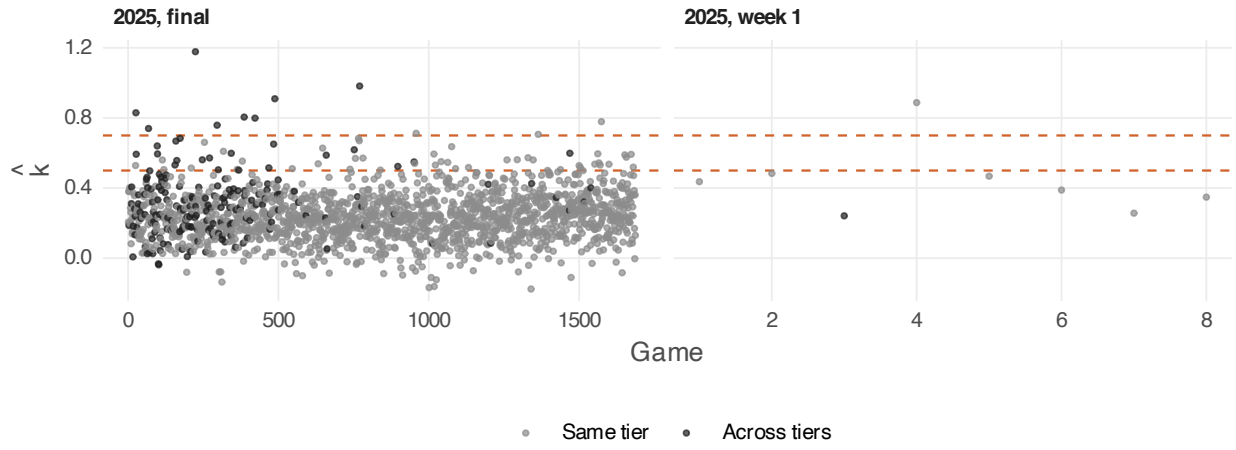


Figure 36: Pareto $\hat{k}$ for each game, in schedule order. Dashed lines at 0.5 and 0.7.

11 of 1,687 games exceed 0.7 in the final fit and 1 of 8 in week 1. In the final fit 8 of them involve a

Table 30: Games with Pareto $\hat{k} > 0.7$ in the 2025, final fit, largest first. *Games* counts each team's games in the fit.

Week	Home (tier, games)	Away (tier, games)	Score	$\hat{k}$
3	Stetson (FCS, 12)	Warner (Lower, 1)	59–13	1.18
8	New Haven (FCS, 10)	Western Connecticut State (Lower, 1)	69–0	0.98
5	Davidson (FCS, 12)	Greensboro (Lower, 1)	48–0	0.91
2	Houston Christian (FCS, 12)	Arkansas Baptist (Lower, 1)	69–0	0.83
4	Cal Poly (FCS, 12)	Western Oregon (Lower, 1)	56–7	0.80
5	Presbyterian (FCS, 12)	BLUEFIELD (Lower, 1)	76–3	0.80
15	South Dakota State (FCS, 14)	New Hampshire (FCS, 13)	41–3	0.78
4	Howard (FCS, 12)	Morehouse (Lower, 1)	38–10	0.76
2	Valparaiso (FCS, 12)	Virginia Lynchburg (Lower, 4)	67–10	0.74
10	Dartmouth (FCS, 10)	Columbia (FCS, 10)	49–3	0.71
13	Mercer (FCS, 12)	Chattanooga (FCS, 12)	63–17	0.71

lower-division team and 7 are the only game of one of the two teams: such a game is the only evidence about that team, so removing it changes the team's posterior completely. None involves an FBS team, and the estimated expected log predictive density is not materially affected. A K-fold refit would remove the approximation for these games if it were needed.

11.10 Power-scaling Sensitivity

Table 13 summarised power-scaling sensitivity by a single distance per parameter. Figure 37 shows what lies behind it for the parameters flagged most often: the posterior density as the prior (top row) or the likelihood (bottom row) is raised to powers between 0.8 and 1.25 [37]. A density that shifts with the prior power is prior-sensitive; one that shifts with the likelihood power is informed by the data.

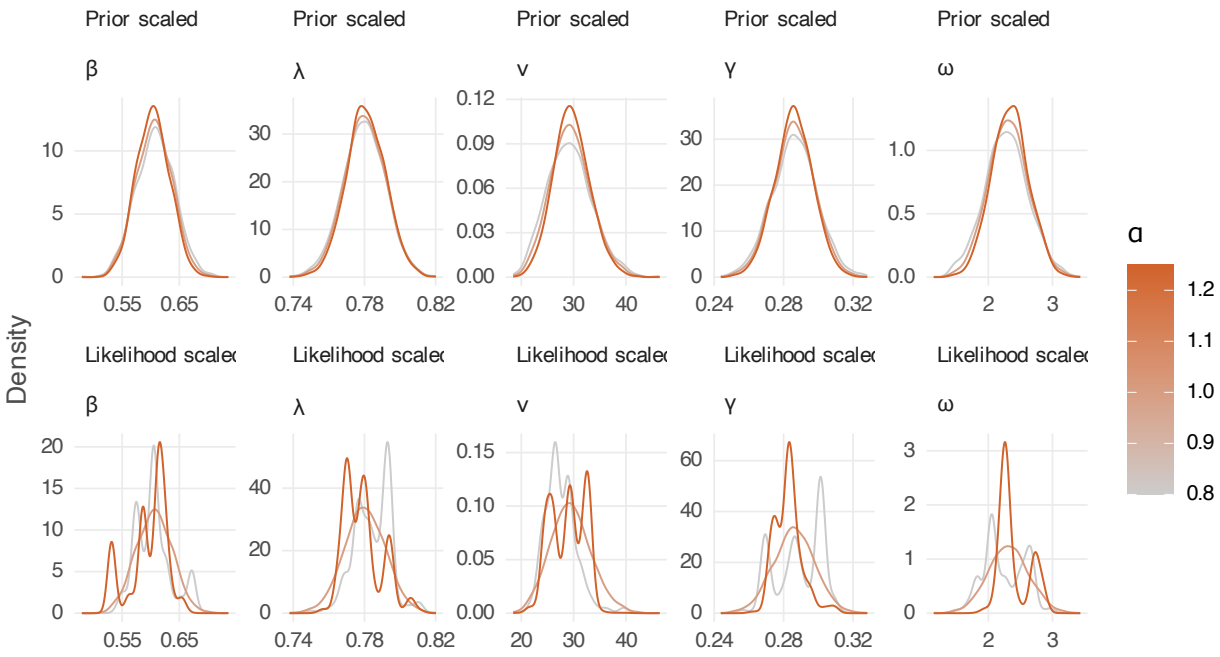


Figure 37: Power-scaled posterior densities, 2025, final. Darker lines are larger powers α; the base posterior is $\alpha = 1$.

11.11 Sensitivity to the Prior: Perturbation Refits

Power-scaling is a local check: it asks what happens under small multiplicative changes to the prior. The requirement that matters in practice is global — that a reasonable analyst with a different prior would publish essentially the same ratings. We therefore refit both posteriors under four joint alternatives for every structural prior: all prior standard deviations doubled, all multiplied by five (close to vague on the scale of these parameters), and all prior means moved up, or down, by one prior standard deviation. The last two move every mean at once in the same direction, a deliberately harsh test. The family's shape parameters (ν and γ) are pooled like the others and move with them; the mean and standard deviation of the Gamma prior on ν are moved and matched back to a Gamma. Joint changes can cancel, so each prior group is also widened fivefold on its own. Table 31 gives the structural posteriors and Table 32 compares what is published: the FBS power ratings and the resume. Table 33 gives the one-at-a-time results, and Table 34 repeats the joint alternatives for 2020, the most atypical season.

Table 31: Posterior means (posterior sd) of selected structural parameters under each prior. *Gates* reports whether the refit passed the convergence gates of Section 6.1.

Fit	Prior	η	β	σ_m	κ	ω	m_2	Gates
week 1	Baseline	2.44 (0.37)	0.59 (0.04)	6.59 (0.40)	0.75 (0.02)	2.43 (0.33)	-28.29 (2.11)	pass
	SD $\times$ 2	2.44 (0.72)	0.59 (0.08)	6.61 (0.77)	0.75 (0.04)	2.45 (0.67)	-27.99 (4.28)	pass
	SD $\times$ 5	2.48 (1.87)	0.58 (0.18)	6.71 (1.65)	0.74 (0.10)	2.78 (1.44)	-26.03 (9.92)	pass
	Means $+1$ SD	2.80 (0.37)	0.63 (0.04)	7.00 (0.39)	0.77 (0.02)	2.76 (0.33)	-26.14 (2.16)	pass
	Means -1 SD	2.07 (0.36)	0.55 (0.04)	6.21 (0.41)	0.72 (0.02)	2.11 (0.33)	-30.40 (2.17)	pass
final	Baseline	2.52 (0.26)	0.61 (0.03)	6.90 (0.18)	0.73 (0.01)	2.31 (0.32)	-29.15 (1.37)	pass
	SD $\times$ 2	2.54 (0.31)	0.62 (0.05)	6.91 (0.23)	0.72 (0.02)	2.05 (0.56)	-29.56 (1.67)	pass
	SD $\times$ 5	2.54 (0.35)	0.64 (0.06)	6.84 (0.28)	0.72 (0.02)	1.61 (0.85)	-29.90 (1.76)	pass
	Means $+1$ SD	2.71 (0.26)	0.62 (0.03)	6.96 (0.18)	0.73 (0.01)	2.58 (0.32)	-27.81 (1.37)	pass
	Means -1 SD	2.33 (0.25)	0.60 (0.03)	6.83 (0.18)	0.72 (0.01)	2.05 (0.31)	-30.42 (1.32)	pass

Table 32: Effect of the prior on the published FBS products, relative to the baseline prior. Power: Spearman correlation of the ratings, and the mean and largest absolute change in a team's rating (points). Resume: Spearman correlation of WAB, the largest change in resume rank among the baseline top 25, and the number of teams whose membership of the top 12 changes.

Fit	Prior	ρ power	Mean $\lvert\Delta\rvert$	Max $\lvert\Delta\rvert$	ρ WAB	Max rank Δ (top 25)	Top-12 changes
week 1	SD $\times$ 2	1.000	0.13	0.41	1.000	0	0
	SD $\times$ 5	0.999	0.38	1.18	1.000	0	0
	Means $+1$ SD	0.998	1.33	4.06	1.000	0	0
	Means -1 SD	0.996	1.33	3.71	1.000	0	0
final	SD $\times$ 2	1.000	0.15	0.61	1.000	1	0
	SD $\times$ 5	0.999	0.29	1.04	1.000	1	0
	Means $+1$ SD	1.000	0.14	0.45	1.000	1	0
	Means -1 SD	1.000	0.15	0.57	1.000	1	0

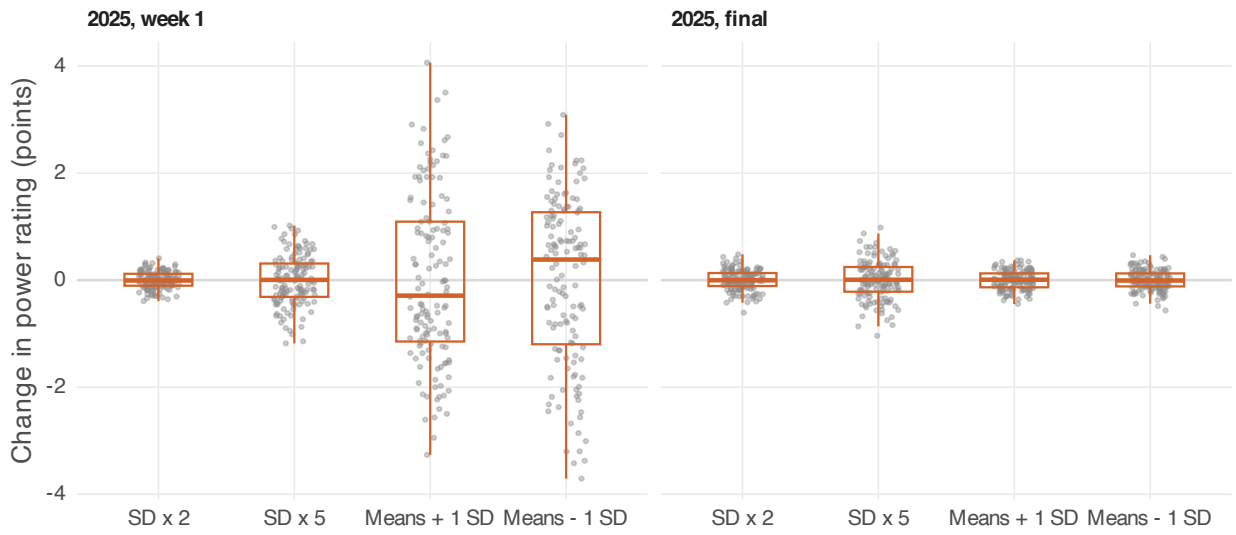


Figure 38: Change in each FBS team's power rating relative to the baseline prior, by alternative prior. Each point is a team; the box spans the middle half. The week-1 ratings depend on the prior by design, the final ratings hardly at all.

At the end of the season the prior is immaterial to what is published: under every alternative the FBS power ratings correlate with the baseline at $\rho \geq 0.999$, no team's rating moves by more than 1.0 points (mean 0.29), and the resume top 12 is unchanged. More strikingly, the week-1 ratings are also robust, although several structural posteriors there stay close to their priors (Section 11.5): widening every prior fivefold moves no FBS rating by more than 1.2 points, shifting every prior mean by one prior standard deviation moves none by more than 4.1, and the ratings keep their order ($\rho \geq 0.996$). The reason is that a week-1 rating rests mainly on the carry-over βd_i from last season's final ratings, and β is estimated from the week-1 games together with last season's ratings, whatever its prior; the priors on the scales and offsets change how much each team is shrunk towards its conference, which moves ratings by a point or two, not their order. The structural priors therefore stabilise the sampler in data-poor weeks (Section 3.4) without determining what is published. The priors on lower-division offsets and spreads, which the diagnostics flag most, affect FBS ratings only through the handful of cross-tier games.

One prior group at a time (Table 33), the largest effect in the final fit comes from widening the prior on preseason terms, which moves no FBS rating by more than 0.6 points; in week 1 it is the prior on preseason terms, at 1.3 points. For 2020 (Table 34), a season with atypical home field, few early games and a spring FCS season, the final ratings under the joint alternatives keep a rank correlation of at least 0.998 and no rating moved by more than 1.6 points.

11.12 Simulation-based Calibration

The checks so far ask whether the model fits the data. Simulation-based calibration asks whether the *computation* recovers the model: if parameters are drawn from the prior, data simulated from them, and the posterior computed, then the rank of each true value among the posterior draws is

Table 33: One prior group at a time: each group's prior standard deviation multiplied by five, all others at baseline. Columns as in Table 32.

Fit	Prior	ρ power	Mean $\lvert\Delta\rvert$	Max $\lvert\Delta\rvert$	ρ WAB	Max rank Δ (top 25)	Top-12 c
week 1	η sd $\times$ 5	1.000	0.11	0.37	1.000	0	
	m_2, m_3 sd $\times$ 5	1.000	0.12	0.37	1.000	0	
	β sd $\times$ 5	1.000	0.13	0.58	1.000	0	
	τ_ℓ sd $\times$ 5	1.000	0.10	0.36	1.000	0	
	ω sd $\times$ 5	1.000	0.11	0.34	1.000	0	
	λ sd $\times$ 5	1.000	0.11	0.34	1.000	0	
	κ sd $\times$ 5	1.000	0.09	0.33	1.000	0	
	σ_e sd $\times$ 5	1.000	0.11	0.37	1.000	0	
	σ_m sd $\times$ 5	1.000	0.11	0.34	1.000	0	
	ϕ_r, ϕ_t sd $\times$ 5	1.000	0.10	0.35	1.000	0	
	ν, γ sd $\times$ 5	1.000	0.11	0.37	1.000	0	
	total-model sd $\times$ 5	1.000	0.11	0.39	1.000	0	
	preseason terms sd $\times$ 5	0.999	0.40	1.29	1.000	0	
final	η sd $\times$ 5	1.000	0.07	0.31	1.000	1	
	m_2, m_3 sd $\times$ 5	1.000	0.09	0.29	1.000	1	
	β sd $\times$ 5	1.000	0.09	0.42	1.000	3	
	τ_ℓ sd $\times$ 5	1.000	0.10	0.34	1.000	1	
	ω sd $\times$ 5	1.000	0.18	0.55	1.000	2	
	λ sd $\times$ 5	1.000	0.08	0.33	1.000	1	
	κ sd $\times$ 5	1.000	0.09	0.31	1.000	2	
	σ_e sd $\times$ 5	1.000	0.10	0.29	1.000	2	
	σ_m sd $\times$ 5	1.000	0.09	0.32	1.000	1	
	ϕ_r, ϕ_t sd $\times$ 5	1.000	0.17	0.48	1.000	2	
	ν, γ sd $\times$ 5	1.000	0.10	0.41	1.000	2	
	total-model sd $\times$ 5	1.000	0.08	0.24	1.000	2	
	preseason terms sd $\times$ 5	1.000	0.14	0.64	1.000	0	

Table 34: The joint prior alternatives for the 2020 season (week 1 and the final posterior). Columns as in Table 32.

Fit	Prior	ρ power	Mean $\lvert\Delta\rvert$	Max $\lvert\Delta\rvert$	ρ WAB	Max rank Δ (top 25)	Top-12 chan
2020, week 1	SD $\times$ 2	1.000	0.11	0.38	NA	0	
	SD $\times$ 5	1.000	0.27	1.50	NA	0	
	Means $+1$ SD	0.997	1.13	4.46	NA	0	
	Means -1 SD	0.996	1.12	4.17	NA	0	
2020, final	SD $\times$ 2	0.999	0.30	1.05	1.000	1	
	SD $\times$ 5	0.998	0.43	1.62	0.999	2	
	Means $+1$ SD	0.999	0.37	1.38	1.000	2	
	Means -1 SD	0.999	0.36	1.10	1.000	1	

uniform when the sampler and the code are correct [56, 57]. We simulate 100 seasons on the real 2025 schedule through week 8 (840 games, with the same pattern of missing efficiency margins and the same carry-over d_i and s_i), each with its own random walk and margins from the scale-t law, refit each with four chains of 250 draws, and rank the true values among 99 thinned draws. The simulation covers the margin model only; the total model, which shares no parameter with it, is switched off. Besides structural parameters the test quantities include what is published: every FBS team's centred power rating, its WAB (the true value computed from the true strengths and the simulated results), and both ranks.

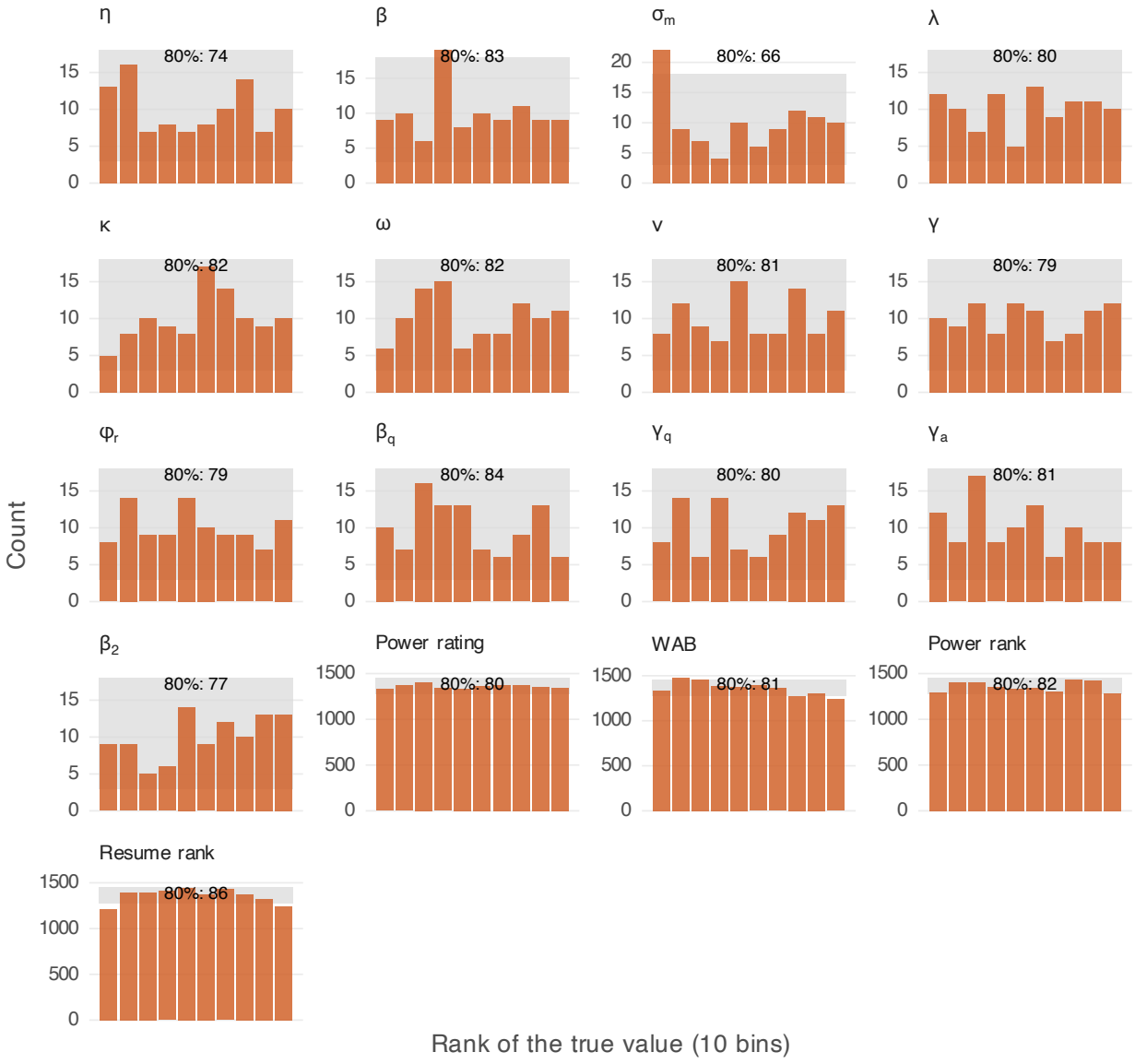


Figure 39: Simulation-based calibration, 100 simulated seasons. Histograms of the rank of the true value among 99 posterior draws, in 10 bins; the band is the central 99% range of a bin count under uniformity (for the team-level quantities, pooled over all FBS teams, so the band is narrow). The label gives the coverage of the central 80% posterior interval.

The structural parameters are recovered with coverage of the 80% interval between 66% and 84%, and for the published quantities it is power rank 82%, power rating 80%, resume rank 86%, wab 81%. A chi-square test of uniformity of the rank histograms rejects at the 1% level for Resume rank, WAB; with 544 team-level values per simulated season the team-level tests are very sensitive, and the rank statistics of different teams in one simulation are not independent, so the size of the departure, visible in the figure, matters more than its significance. Ranks are discrete, so ties between a true rank and the posterior draws are broken at random.

12 Appendix: The Stan Program

The program below is `stan/power.stan`, reproduced verbatim at render time.

```
// ============================================================================
// Power model: each team has a net strength r (points, neutral field) and a total tendency
// t (how many points its games produce), both moving week to week as random walks.
// Offence and defence follow: off = (r + t) / 2, def = (r - t) / 2, so a game's expected
// home score is base + (mu + T) / 2 and away score base + (T - mu) / 2, with
//   mu = r[home] - r[away] + hfa * home_adv      (expected margin)
//   T  = t[home] + t[away]                       (expected total, relative to 2 * base)
//
// Margin, observed through two per-game signals that share r:
//   eff    : net garbage-time-filtered PPA (home offence total - away offence total)
//            eff ~ Normal(lambda * mu, sigma_eff)
//   margin : regulation margin (OT games -> 0), conditioned on the eff residual so the
//            two signals are not double-counted as independent evidence
//            margin ~ F(mu + kappa * (eff - lambda * mu); sigma_m, ...)
// Games without eff use the margin's marginal law: the same family with the scales widened by
// kappa^2 sigma_eff^2 (exact for the normal and the mixture, an approximation for the t).
// F is one of four families (margin_family):
//   1 normal(loc, sigma)
//   2 Student-t(nu, loc, sigma)
//   3 two-normal scale mixture: (1 - p_wide) normal(loc, sigma) + p_wide normal(loc, sigma_w),
//     sigma_w = sigma + sigma_gap > sigma (ordered, so the labels are identified)
//  4 Student-t whose scale grows with the expected margin: sigma_g = sqrt(sigma^2 + gamma_m^2 mu^2)
//
// Total (final score, games without overtime), observed the same way through the summed
// offensive PPA of both teams:
//   eff_sum ~ Normal(alpha_es + lambda_t * T, sigma_es)
//   total   ~ Normal(2 base + T + kappa_t * (eff_sum - alpha_es - lambda_t * T), sigma_tot)
//
// Preseason level (week 1): r0 = mu_conf[conf] + b_prev * prev_dev + s_team * z,
//         mu_conf ~ Normal(mu_tier[tier], sigma_conf), tiers FBS (anchored at 0) / FCS / non-D1;
//         t0 has the same structure with its own conference spread, tier offsets and carry-over.
// prev_dev is last season's final rating relative to its tier's mean (0 when unknown); prev_sd
// is its posterior sd, integrated out: s_team = sqrt(sigma_team^2 + (b_prev prev_sd)^2).
// Optional preseason terms (each behind a switch, 0 for teams without the data):
//   b_prev_tier  FCS and non-D1 teams get their own carry-over b_prev_low
//   use_ret     carry-over b_prev + b_ret * ret and a level shift c_ret * ret, where ret is the
//                share of last season's production that returns (centred within FBS)
//   use_tal      level shift b_tal * tal (roster talent composite, standardised within FBS)
//   use_prev2   level shift b_prev2 * prev2_dev (the rating two seasons back vs. its tier mean)
```

```
//   use_coach    sigma_team scaled by exp(delta_coach) for teams with a new head coach
//
// Dynamics: a team's state at each week it plays ("states", in week order per team) is its
// previous state plus tau * sqrt(weeks elapsed) * N(0, 1); the first state steps from r0 (the
// week-1 level). r in generated quantities is the rating at the snapshot week: the last state
// plus the drift since. dynamic = 0 freezes ratings within the season (tau = 0).
//
// Switches for the ablations: conf_level = 0 drops the conference effects; use_total = 0 drops
// the total model (its parameters are then absent).
// ============================================================================

functions {
  // Margin density for family fam; m is the expected margin from the ratings (family 4 only)
 real margin_lpdf(real y, int fam, real loc, real s, real s_w, real p_w, real nu, real gam, real m) {
    if (fam == 1) return normal_lpdf(y | loc, s);
    if (fam == 2) return student_t_lpdf(y | nu, loc, s);
    if (fam == 3) return log_mix(p_w, normal_lpdf(y | loc, s_w), normal_lpdf(y | loc, s));
    return student_t_lpdf(y | nu, loc, sqrt(square(s) + square(gam * m)));
  }
  // The same, summed over a vector of games (vectorised except for the mixture)
 real margins_lpdf(vector y, int fam, vector loc, real s, real s_w, real p_w, real nu, real gam, vector m) {
    if (fam == 1) return normal_lpdf(y | loc, s);
    if (fam == 2) return student_t_lpdf(y | nu, loc, s);
    if (fam == 3) {
      real lp = 0;
    for (i in 1:rows(y)) lp += log_mix(p_w, normal_lpdf(y[i] | loc[i], s_w), normal_lpdf(y[i] | loc[i], s));
      return lp;
    }
    return student_t_lpdf(y | nu, loc, sqrt(square(s) + square(gam * m)));
  }
}

data {
  int<lower=1> N_teams;
  int<lower=1> N_conf;
  array[N_teams] int<lower=1, upper=N_conf> team_conf;
  array[N_conf] int<lower=1, upper=3> conf_tier;
  array[N_teams] int<lower=1, upper=3> team_tier;
  vector[N_teams] prev_dev;
  vector<lower=0>[N_teams] prev_sd;
  vector[N_teams] prev_dev_t;                 // last season's total tendency vs. its tier mean
  vector<lower=0>[N_teams] prev_sd_t;

  // Weekly states: one per (team, week with a game), grouped by team and in week order within
  // a team, so a team's walk is a cumulative sum over its block
  int<lower=0> N_states;
  array[N_states] int<lower=1, upper=N_teams> state_team;
  array[N_states] int<lower=1, upper=N_states> state_first;  // first state of the same team
 vector<lower=0>[N_states] state_gap;               // weeks since the previous state (r0 for the first)
  array[N_teams] int<lower=0, upper=N_states> now_state;     // latest state (0 = no game yet)
  vector<lower=0>[N_teams] now_gap;                          // weeks from it to the snapshot

  int<lower=0> N_games;
  array[N_games] int<lower=1, upper=N_states> home_state;
```

```
 array[N_games] int<lower=1, upper=N_states> away_state;
 vector<lower=0, upper=1>[N_games] home_adv;   // 1 = true home game, 0 = neutral site
 vector[N_games] margin;                       // regulation margin, home - away
 array[N_games] int<lower=0, upper=1> has_eff;
 vector[N_games] eff;                          // 0 where has_eff == 0
vector[N_games] eff_sum;               // home + away offensive PPA totals; 0 where has_eff == 0
 array[N_games] int<lower=0, upper=1> has_total;
 vector[N_games] total;                        // final total points; 0 where has_total == 0

 real hfa_mean;
 real<lower=0> hfa_sd;
 vector[2] tier_mean;                          // prior means for FCS, non-D1 tier offsets
 vector<lower=0>[2] tier_sd;
 real b_prev_mean;
 real<lower=0> b_prev_sd;
 // Informative priors (mean, sd) for the structural noise/scale parameters: they barely move
 // between seasons, and without them week-1 fits (~1 game per team) funnel
 vector[3] sigma_team_mean;
 vector<lower=0>[3] sigma_team_sd;
 vector[2] sigma_conf_prior;
 vector[2] lambda_prior;
 vector[2] kappa_prior;
 vector[2] sigma_eff_prior;
 vector[2] sigma_m_prior;
 vector<lower=0>[2] nu_prior;                // Gamma(shape, rate), truncated to nu > nu_lower
 real<lower=0> nu_lower;
 vector[2] tau_r_prior;                       // weekly step sd of r (normal, truncated at 0)
 vector[2] sigma_gap_prior;                   // mixture: sigma_w - sigma_m
vector<lower=0>[2] p_wide_prior;            // mixture: Beta(a, b) weight of the wide component
 vector[2] gamma_prior;                       // scale-t: growth of the scale with |mu|
 // Total model
 vector[2] base_prior;                        // points per team, average FBS matchup
 vector[2] t_tier_mean;
 vector<lower=0>[2] t_tier_sd;
 vector[2] sigma_t_team_prior;
 vector[2] sigma_conf_t_prior;
 vector[2] b_prev_t_prior;
 vector[2] tau_t_prior;
 vector[2] lambda_t_prior;
 vector[2] alpha_es_prior;
 vector[2] kappa_t_prior;
 vector[2] sigma_es_prior;
 vector[2] sigma_tot_prior;

 // Preseason covariates (0 where unknown) and their priors
 vector[N_teams] ret;
 vector[N_teams] tal;
 vector[N_teams] prev2_dev;
 vector<lower=0, upper=1>[N_teams] coach_new;
 int<lower=0, upper=1> b_prev_tier;
 int<lower=0, upper=1> use_ret;
 int<lower=0, upper=1> use_tal;
 int<lower=0, upper=1> use_prev2;
 int<lower=0, upper=1> use_coach;
```

```
  vector[2] b_prev_low_prior;
  vector[2] b_ret_prior;
  vector[2] c_ret_prior;
  vector[2] b_tal_prior;
  vector[2] b_prev2_prior;
  vector[2] delta_coach_prior;

  int<lower=1, upper=4> margin_family;
  int<lower=0, upper=1> dynamic;
  int<lower=0, upper=1> use_total;
  int<lower=0, upper=1> conf_level;
  int<lower=0, upper=1> compute_ll;
}

transformed data {
  int N_eff = sum(has_eff);
  array[N_eff] int idx_eff;
  array[N_games - N_eff] int idx_noeff;
  int is_mix = margin_family == 3;
  int is_sct = margin_family == 4;
  int dyn_t = dynamic * use_total;
  vector[N_states] sqrt_gap = sqrt(state_gap);
  // Totals: games with eff_sum; with eff_sum and a total; with a total but no eff_sum
  int N_es_tot = 0;
  int N_tot_only = 0;
  for (k in 1:N_games) {
    if (has_eff[k] == 1 && has_total[k] == 1) N_es_tot += 1;
    if (has_eff[k] == 0 && has_total[k] == 1) N_tot_only += 1;
  }
  array[N_es_tot] int idx_es_tot;
  array[N_tot_only] int idx_tot_only;
  {
    int i = 1;
    int j = 1;
    for (k in 1:N_games) {
      if (has_eff[k] == 1 && has_total[k] == 1) { idx_es_tot[i] = k; i += 1; }
      if (has_eff[k] == 0 && has_total[k] == 1) { idx_tot_only[j] = k; j += 1; }
    }
  }
  {
    int i = 1;
    int j = 1;
    for (k in 1:N_games) {
      if (has_eff[k] == 1) { idx_eff[i] = k; i += 1; }
      else { idx_noeff[j] = k; j += 1; }
    }
  }
}

parameters {
  vector[2] mu_tier_raw;
  real<lower=0> sigma_conf;
  vector[N_conf] conf_raw;
  vector<lower=0>[3] sigma_team;
```

```
  real<lower=0, upper=1> b_prev;
  vector[N_teams] z;
  vector<lower=0>[dynamic] tau_r_raw;
  vector[dynamic ? N_states : 0] eps_r;

  real hfa;
  real<lower=0> lambda;
  real<lower=0> sigma_eff;
  real kappa;
  real<lower=0> sigma_m;
  real<lower=nu_lower> nu;                      // used by families 2 and 4 only
  vector<lower=0>[is_mix] sigma_gap_raw;
  vector<lower=0, upper=1>[is_mix] p_wide_raw;
  vector<lower=0>[is_sct] gamma_raw;

  // Preseason terms
  vector<lower=0, upper=1>[b_prev_tier] b_prev_low_raw;
  vector[use_ret] b_ret_raw;
  vector[use_ret] c_ret_raw;
  vector[use_tal] b_tal_raw;
  vector[use_prev2] b_prev2_raw;
  vector[use_coach] delta_coach_raw;

  // Total model
  vector[use_total] base_raw;
  vector[use_total ? 2 : 0] t_tier_raw;
  vector<lower=0>[use_total] sigma_conf_t_raw;
  vector[use_total ? N_conf : 0] conf_t_raw;
  vector<lower=0>[use_total] sigma_t_team_raw;
  vector<lower=0, upper=1>[use_total] b_prev_t_raw;
  vector[use_total ? N_teams : 0] z_t;
  vector<lower=0>[dyn_t] tau_t_raw;
  vector[dyn_t ? N_states : 0] eps_t;
  vector<lower=0>[use_total] lambda_t_raw;
  vector[use_total] alpha_es_raw;
  vector<lower=0>[use_total] sigma_es_raw;
  vector[use_total] kappa_t_raw;
  vector<lower=0>[use_total] sigma_tot_raw;
}

transformed parameters {
  vector[3] mu_tier = append_row(0, mu_tier_raw);
  vector[N_conf] mu_conf = mu_tier[conf_tier] + conf_level * sigma_conf * conf_raw;
  vector[N_teams] r0;
  {
    vector[N_teams] carry = rep_vector(b_prev, N_teams);
    vector[N_teams] s_team = sigma_team[team_tier];
    if (b_prev_tier) for (i in 1:N_teams) if (team_tier[i] > 1) carry[i] = b_prev_low_raw[1];
    if (use_ret) carry += b_ret_raw[1] * ret;
    if (use_coach) s_team = s_team .* exp(delta_coach_raw[1] * coach_new);
   r0 = mu_conf[team_conf] + carry .* prev_dev + sqrt(square(s_team) + square(carry .* prev_sd)) .* z;
    if (use_ret) r0 += c_ret_raw[1] * ret;
    if (use_tal) r0 += b_tal_raw[1] * tal;
    if (use_prev2) r0 += b_prev2_raw[1] * prev2_dev;
```

```
}
real sigma_marg = sqrt(square(sigma_m) + square(kappa * sigma_eff));
// Family parameters as scalars (defaults where the family does not use them)
real tau_r = 0;
real sigma_w = sigma_m;
real p_wide = 0;
real gamma_m = 0;
if (dynamic) tau_r = tau_r_raw[1];
if (is_mix) { sigma_w = sigma_m + sigma_gap_raw[1]; p_wide = p_wide_raw[1]; }
if (is_sct) gamma_m = gamma_raw[1];
real sigma_marg_w = sqrt(square(sigma_w) + square(kappa * sigma_eff));

// Random walk as a cumulative sum of steps within each team's block of states
vector[N_states] r_state = r0[state_team];
if (dynamic && N_states > 0) {
  vector[N_states] step = sqrt_gap .* eps_r;
  vector[N_states] cs = cumulative_sum(step);
  r_state += tau_r * (cs - cs[state_first] + step[state_first]);
}

// Total model (empty when use_total = 0)
vector[use_total ? N_teams : 0] t0;
vector[use_total ? N_states : 0] t_state;
if (use_total) {
  vector[3] t_tier = append_row(0, t_tier_raw);
  vector[N_conf] t_conf = t_tier[conf_tier] + conf_level * sigma_conf_t_raw[1] * conf_t_raw;
  t0 = t_conf[team_conf] + b_prev_t_raw[1] * prev_dev_t
       + sqrt(square(sigma_t_team_raw[1]) + square(b_prev_t_raw[1] * prev_sd_t)) .* z_t;
  t_state = t0[state_team];
  if (dyn_t && N_states > 0) {
    vector[N_states] step = sqrt_gap .* eps_t;
    vector[N_states] cs = cumulative_sum(step);
    t_state += tau_t_raw[1] * (cs - cs[state_first] + step[state_first]);
  }
}

// Log prior of the population-level parameters (used by priorsense power-scaling)
real lprior = normal_lpdf(mu_tier_raw | tier_mean, tier_sd)
            + normal_lpdf(sigma_conf | sigma_conf_prior[1], sigma_conf_prior[2])
            + normal_lpdf(sigma_team | sigma_team_mean, sigma_team_sd)
            + normal_lpdf(b_prev | b_prev_mean, b_prev_sd)
            + normal_lpdf(hfa | hfa_mean, hfa_sd)
            + normal_lpdf(lambda | lambda_prior[1], lambda_prior[2])
            + normal_lpdf(sigma_eff | sigma_eff_prior[1], sigma_eff_prior[2])
            + normal_lpdf(kappa | kappa_prior[1], kappa_prior[2])
            + normal_lpdf(sigma_m | sigma_m_prior[1], sigma_m_prior[2])
       + gamma_lpdf(nu | nu_prior[1], nu_prior[2]) - gamma_lccdf(nu_lower | nu_prior[1], nu_prior[2]);
if (dynamic) lprior += normal_lpdf(tau_r_raw | tau_r_prior[1], tau_r_prior[2]);
if (is_mix) lprior += normal_lpdf(sigma_gap_raw | sigma_gap_prior[1], sigma_gap_prior[2])
                    + beta_lpdf(p_wide_raw | p_wide_prior[1], p_wide_prior[2]);
if (is_sct) lprior += normal_lpdf(gamma_raw | gamma_prior[1], gamma_prior[2]);
if (use_total) {
  lprior += normal_lpdf(base_raw | base_prior[1], base_prior[2])
          + normal_lpdf(t_tier_raw | t_tier_mean, t_tier_sd)
```

```
            + normal_lpdf(sigma_conf_t_raw | sigma_conf_t_prior[1], sigma_conf_t_prior[2])
            + normal_lpdf(sigma_t_team_raw | sigma_t_team_prior[1], sigma_t_team_prior[2])
            + normal_lpdf(b_prev_t_raw | b_prev_t_prior[1], b_prev_t_prior[2])
            + normal_lpdf(lambda_t_raw | lambda_t_prior[1], lambda_t_prior[2])
            + normal_lpdf(alpha_es_raw | alpha_es_prior[1], alpha_es_prior[2])
            + normal_lpdf(kappa_t_raw | kappa_t_prior[1], kappa_t_prior[2])
            + normal_lpdf(sigma_es_raw | sigma_es_prior[1], sigma_es_prior[2])
            + normal_lpdf(sigma_tot_raw | sigma_tot_prior[1], sigma_tot_prior[2]);
    if (dyn_t) lprior += normal_lpdf(tau_t_raw | tau_t_prior[1], tau_t_prior[2]);
  }
  if (b_prev_tier) lprior += normal_lpdf(b_prev_low_raw | b_prev_low_prior[1], b_prev_low_prior[2]);
  if (use_ret) lprior += normal_lpdf(b_ret_raw | b_ret_prior[1], b_ret_prior[2])
                       + normal_lpdf(c_ret_raw | c_ret_prior[1], c_ret_prior[2]);
  if (use_tal) lprior += normal_lpdf(b_tal_raw | b_tal_prior[1], b_tal_prior[2]);
  if (use_prev2) lprior += normal_lpdf(b_prev2_raw | b_prev2_prior[1], b_prev2_prior[2]);
  if (use_coach) lprior += normal_lpdf(delta_coach_raw | delta_coach_prior[1], delta_coach_prior[2]);
}

model {
  target += lprior;
  conf_raw ~ std_normal();
  z ~ std_normal();
  eps_r ~ std_normal();
  conf_t_raw ~ std_normal();
  z_t ~ std_normal();
  eps_t ~ std_normal();

  if (N_games > 0) {
    vector[N_games] mu = r_state[home_state] - r_state[away_state] + hfa * home_adv;
    vector[N_eff] mu_e = mu[idx_eff];
    vector[N_games - N_eff] mu_n = mu[idx_noeff];
    vector[N_eff] res = eff[idx_eff] - lambda * mu_e;
    target += normal_lpdf(eff[idx_eff] | lambda * mu_e, sigma_eff);
   target += margins_lpdf(margin[idx_eff] | margin_family, mu_e + kappa * res, sigma_m, sigma_w, p_wide, nu, gamma
   target += margins_lpdf(margin[idx_noeff] | margin_family, mu_n, sigma_marg, sigma_marg_w, p_wide, nu, gamma_m

    if (use_total) {
      vector[N_games] tt = t_state[home_state] + t_state[away_state];
      vector[N_games] es_mu = alpha_es_raw[1] + lambda_t_raw[1] * tt;
     real tot_marg = sqrt(square(sigma_tot_raw[1]) + square(kappa_t_raw[1] * sigma_es_raw[1]));
      target += normal_lpdf(eff_sum[idx_eff] | es_mu[idx_eff], sigma_es_raw[1]);
      target += normal_lpdf(total[idx_es_tot] | 2 * base_raw[1] + tt[idx_es_tot]
                  + kappa_t_raw[1] * (eff_sum[idx_es_tot] - es_mu[idx_es_tot]), sigma_tot_raw[1]);
     target += normal_lpdf(total[idx_tot_only] | 2 * base_raw[1] + tt[idx_tot_only], tot_marg);
    }
  }
}

generated quantities {
  // Ratings at the snapshot week: the latest state plus the random-walk drift since
  vector[N_teams] r;
  vector[use_total ? N_teams : 0] t;
  for (i in 1:N_teams) {
    if (now_state[i] == 0) r[i] = r0[i];
```

```
    else r[i] = r_state[now_state[i]];
    if (dynamic) r[i] += tau_r * sqrt(now_gap[i]) * normal_rng(0, 1);
  }
  if (use_total) {
    for (i in 1:N_teams) {
      if (now_state[i] == 0) t[i] = t0[i];
      else t[i] = t_state[now_state[i]];
      if (dyn_t) t[i] += tau_t_raw[1] * sqrt(now_gap[i]) * normal_rng(0, 1);
    }
  }

  // log_lik: joint p(e, y) where e is observed, p(y) otherwise (the fitted margin likelihood).
  // log_lik_margin: p(y) alone for every game, the marginal law of the margin; comparable
  // across models whether or not they use the efficiency margin.
  // log_lik_total: the total's contribution (joint with eff_sum where observed).
  vector[compute_ll ? N_games : 0] log_lik;
  vector[compute_ll ? N_games : 0] log_lik_margin;
  vector[compute_ll * use_total ? N_games : 0] log_lik_total;
  if (compute_ll) {
    vector[N_games] mu = r_state[home_state] - r_state[away_state] + hfa * home_adv;
    for (k in 1:N_games) {
    log_lik_margin[k] = margin_lpdf(margin[k] | margin_family, mu[k], sigma_marg, sigma_marg_w, p_wide, nu, gamm
      if (has_eff[k] == 1) {
        real res = eff[k] - lambda * mu[k];
        log_lik[k] = normal_lpdf(eff[k] | lambda * mu[k], sigma_eff)
             + margin_lpdf(margin[k] | margin_family, mu[k] + kappa * res, sigma_m, sigma_w, p_wide, nu, gamma_m,
      } else {
        log_lik[k] = log_lik_margin[k];
      }
    }
    if (use_total) {
      vector[N_games] tt = t_state[home_state] + t_state[away_state];
     real tot_marg = sqrt(square(sigma_tot_raw[1]) + square(kappa_t_raw[1] * sigma_es_raw[1]));
      for (k in 1:N_games) {
        log_lik_total[k] = 0;
        if (has_eff[k] == 1) {
          real es_mu = alpha_es_raw[1] + lambda_t_raw[1] * tt[k];
          log_lik_total[k] = normal_lpdf(eff_sum[k] | es_mu, sigma_es_raw[1]);
          if (has_total[k] == 1)
        log_lik_total[k] += normal_lpdf(total[k] | 2 * base_raw[1] + tt[k] + kappa_t_raw[1] * (eff_sum[k] - es_mu
        } else if (has_total[k] == 1) {
          log_lik_total[k] = normal_lpdf(total[k] | 2 * base_raw[1] + tt[k], tot_marg);
        }
      }
    }
  }
}
```

13 Appendix: Reproducibility

The pipeline is run with `targets::tar_make()` from the project root, after setting an API key for CollegeFootballData.com; the first full build fits all 242 weekly posteriors and the analyses

Table 35: Software versions used for this render.

Component	Version
R	4.6.1
CmdStan	2.37.0
targets	1.12.0
tarchetypes	0.14.1
cmdstanr	0.9.0.9000
posterior	1.7.0
loo	2.10.1
priorsense	1.4.0
dplyr	1.2.1
ggplot2	4.0.3
igraph	2.3.3

of this paper and takes several hours on eight cores, and later runs fit only new weeks. `Rscript check_build.R` then verifies the store. This paper is itself a target, rendered by `tar_make()` whenever a target it reads changes; every number and figure is read from the store and nothing is refitted at render time. The analyses beyond the weekly fits (the margin-family comparison, vague-prior refits, leave-one-season-out priors, ablations, baselines, benchmark and Monte Carlo checks, the margin refits and simulation-based calibration) are targets defined in `R/07_analysis.R`, so they are rerun whenever their inputs change.

Randomness. Every fit uses a fixed seed (2026 for the published fits and the analyses, $1000 + i$ for the i-th simulation-based calibration season, and a seed per step for the replicated data and bootstraps), so a rebuild on the same data and software reproduces every number.

Software. R package versions are recorded in `renv.lock` at the project root; Table 35 lists the main ones and CmdStan.

Data. The provider revises past data occasionally (late score corrections, reclassified games, new betting lines), so a rebuild months later need not reproduce this paper exactly. Every raw API response is cached as JSON under `data/raw/`, which is the snapshot the paper was built from; Table 36 records, for each season, what the models saw and a hash of it, so a later build can be checked against this one. This build read the data on 2026-10-25.

References

[1] D. Harville, "The use of linear-model methodology to rate high school or college football teams," *Journal of the American Statistical Association*, vol. 72, no. 358, pp. 278–289, 1977. DOI: 10.1080/01621459.1977.10480991

[2] K. Massey, "Statistical models applied to the rating of sports teams," Honors thesis, Bluefield College, 1997.

Table 36: The data behind this build, by season: games, completed games, games with an efficiency margin and with a closing line, and an xxHash64 hash of the season's data as the models read it.

Season	Games	Completed	With eff.	With line	Hash
2014	1,584	1,584	848	865	`fe508702dd22ccc4`
2015	1,536	1,536	863	832	`0084c4c6218fd29e`
2016	1,547	1,547	857	866	`9f77e9eb87993e05`
2017	1,549	1,549	869	873	`2b8e18d30686fb62`
2018	1,554	1,554	881	869	`e556d11d1e047a70`
2019	1,621	1,621	887	881	`6db960d055385d9a`
2020	850	850	568	567	`18167f03fb35b9c5`
2021	1,570	1,570	887	887	`eb6e433f88c4e3ef`
2022	1,598	1,598	1,457	1,459	`b3018234866220de`
2023	1,603	1,603	1,490	1,413	`5dd582b934de2ea5`
2024	1,670	1,669	1,606	1,557	`ff8cefb57c94cbde`
2025	1,687	1,687	1,658	1,597	`ddb34df864cbabe8`
2026	1,608	1,041	1,037	1,062	`2582e8bb8b5d5e54`

[3] R. A. Bradley and M. E. Terry, "Rank analysis of incomplete block designs: I. The method of paired comparisons," *Biometrika*, vol. 39, no. 3/4, pp. 324–345, 1952. DOI: 10.2307/2334029

[4] D. Mease, "A penalized maximum likelihood approach for the ranking of college football teams independent of victory margins," *The American Statistician*, vol. 57, no. 4, pp. 241–248, 2003. DOI: 10.1198/0003130032396

[5] J. P. Keener, "The Perron–Frobenius theorem and the ranking of football teams," *SIAM Review*, vol. 35, no. 1, pp. 80–93, 1993. DOI: 10.1137/1035004

[6] W. N. Colley, "Colley's bias free college football ranking method: The Colley matrix explained," Princeton University, Tech. Rep., 2002. [Online]. Available: https://www.colleyrankings.com/matrate.pdf

[7] H. S. Stern, "Statistics and the college football championship," *The American Statistician*, vol. 58, no. 3, pp. 179–185, 2004. DOI: 10.1198/000313004X1611

[8] A. N. Langville and C. D. Meyer, *Who's #1? The Science of Rating and Ranking*. Princeton, NJ: Princeton University Press, 2012.

[9] R. Yurko, S. Ventura, and M. Horowitz, "nflWAR: A reproducible method for offensive player evaluation in football," *Journal of Quantitative Analysis in Sports*, vol. 15, no. 3, pp. 163–183, 2019. DOI: 10.1515/jqas-2018-0010

[10] D. Harville, "Predictions for National Football League games via linear-model methodology," *Journal of the American Statistical Association*, vol. 75, no. 371, pp. 516–524, 1980. DOI: 10.1080/01621459.1980.10477504

[11] H. Stern, "On the probability of winning a football game," *The American Statistician*, vol. 45, no. 3, pp. 179–183, 1991. DOI: 10.1080/00031305.1991.10475798

[12] M. E. Glickman and H. S. Stern, "A state-space model for National Football League scores," *Journal of the American Statistical Association*, vol. 93, no. 441, pp. 25–35, 1998. DOI: 10.1080/01621459.1998.10474084

[13] K. L. Lange, R. J. A. Little, and J. M. G. Taylor, "Robust statistical modeling using the t distribution," *Journal of the American Statistical Association*, vol. 84, no. 408, pp. 881–896, 1989. DOI: 10.1080/01621459.1989.10478852

[14] M. J. Lopez, G. J. Matthews, and B. S. Baumer, "How often does the best team win? A unified approach to understanding randomness in North American sport," *The Annals of Applied Statistics*, vol. 12, no. 4, pp. 2483–2516, 2018. DOI: 10.1214/18-AOAS1165

[15] M. E. Glickman, "Parameter estimation in large dynamic paired comparison experiments," *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, vol. 48, no. 3, pp. 377–394, 1999. DOI: 10.1111/1467-9876.00159

[16] L. Knorr-Held, "Dynamic rating of sports teams," *Journal of the Royal Statistical Society: Series D (The Statistician)*, vol. 49, no. 2, pp. 261–276, 2000. DOI: 10.1111/1467-9884.00235

[17] D. H. Annis and B. A. Craig, "Hybrid paired comparison analysis, with applications to the ranking of college football teams," *Journal of Quantitative Analysis in Sports*, vol. 1, no. 1, 2005. DOI: 10.2202/1559-0410.1000

[18] A. E. Elo, *The Rating of Chessplayers, Past and Present*. New York: Arco, 1978.

[19] N. Silver, *How our NFL predictions work*, FiveThirtyEight, 2014. [Online]. Available: https://fivethirtyeight.com/methodology/how-our-nfl-predictions-work/

[20] B. Connelly, *A college football SP+ primer*, ESPN, 2023.

[21] ESPN Analytics, *Football power index (FPI): Methodology*, ESPN, 2023. [Online]. Available: https://www.espn.com/college-football/fpi

[22] K. Massey, *College football ranking composite*, masseyratings.com, 2026. [Online]. Available: https://masseyratings.com/cf/compare.htm

[23] NCAA, *NCAA evaluation tool (NET) and team sheets, division i men's basketball*, NCAA.org, 2024.

[24] A. Gelman et al., *Bayesian workflow*, 2020. arXiv: 2011.01808 [stat.ME].

[25] CollegeFootballData.com, *CollegeFootballData.com API*, Accessed September 2026, 2026. [Online]. Available: https://collegefootballdata.com

[26] D. B. Rubin, "Inference and missing data," *Biometrika*, vol. 63, no. 3, pp. 581–592, 1976. DOI: 10.1093/biomet/63.3.581

[27] O. Papaspiliopoulos, G. O. Roberts, and M. Sköld, "A general framework for the parametrization of hierarchical models," *Statistical Science*, vol. 22, no. 1, pp. 59–73, 2007. DOI: 10.1214/088342307000000014

[28] M. Betancourt, *A conceptual introduction to Hamiltonian Monte Carlo*, 2017. arXiv: 1701.02434 [stat.ME].

[29] R. J. Carroll, D. Ruppert, L. A. Stefanski, and C. M. Crainiceanu, *Measurement Error in Nonlinear Models: A Modern Perspective*, 2nd ed. Chapman and Hall/CRC, 2006. DOI: 10.1201/9781420010138

[30] Y. Hong, "On computing the distribution function for the Poisson binomial distribution," *Computational Statistics & Data Analysis*, vol. 59, pp. 41–51, 2013. DOI: 10.1016/j.csda.2012.10.006

[31] M. D. Hoffman and A. Gelman, "The No-U-Turn sampler: Adaptively setting path lengths in Hamiltonian Monte Carlo," *Journal of Machine Learning Research*, vol. 15, no. 47, pp. 1593–1623, 2014.

[32] B. Carpenter et al., "Stan: A probabilistic programming language," *Journal of Statistical Software*, vol. 76, no. 1, pp. 1–32, 2017. DOI: 10.18637/jss.v076.i01

[33] A. Vehtari, A. Gelman, D. Simpson, B. Carpenter, and P.-C. Bürkner, "Rank-normalization, folding, and localization: An improved $\widehat{R}$ for assessing convergence of MCMC," *Bayesian Analysis*, vol. 16, no. 2, pp. 667–718, 2021. DOI: 10.1214/20-BA1221

[34] W. M. Landau, "The targets R package: A dynamic Make-like function-oriented pipeline toolkit for reproducibility and high-performance computing," *Journal of Open Source Software*, vol. 6, no. 57, p. 2959, 2021. DOI: 10.21105/joss.02959

[35] R Core Team, *R: A language and environment for statistical computing*, R Foundation for Statistical Computing, Vienna, Austria, 2026. [Online]. Available: https://www.R-project.org/

[36] J. Gabry, D. Simpson, A. Vehtari, M. Betancourt, and A. Gelman, "Visualization in Bayesian workflow," *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, vol. 182, no. 2, pp. 389–402, 2019. DOI: 10.1111/rssa.12378

[37] N. Kallioinen, T. Paananen, P.-C. Bürkner, and A. Vehtari, "Detecting and diagnosing prior and likelihood sensitivity with power-scaling," *Statistics and Computing*, vol. 34, no. 1, p. 57, 2024. DOI: 10.1007/s11222-023-10366-5

[38] A. Vehtari, A. Gelman, and J. Gabry, "Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC," *Statistics and Computing*, vol. 27, no. 5, pp. 1413–1432, 2017. DOI: 10.1007/s11222-016-9696-4

[39] A. Vehtari, D. Simpson, A. Gelman, Y. Yao, and J. Gabry, "Pareto smoothed importance sampling," *Journal of Machine Learning Research*, vol. 25, no. 72, pp. 1–58, 2024.

[40] L. J. Tashman, "Out-of-sample tests of forecasting accuracy: An analysis and review," *International Journal of Forecasting*, vol. 16, no. 4, pp. 437–450, 2000. DOI: 10.1016/S0169-2070(00)00065-0

[41] G. W. Brier, "Verification of forecasts expressed in terms of probability," *Monthly Weather Review*, vol. 78, no. 1, pp. 1–3, 1950. DOI: 10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2

[42] T. Gneiting and A. E. Raftery, "Strictly proper scoring rules, prediction, and estimation," *Journal of the American Statistical Association*, vol. 102, no. 477, pp. 359–378, 2007. DOI: 10.1198/016214506000001437

[43] S. D. Levitt, "Why are gambling markets organised so differently from financial markets?" *The Economic Journal*, vol. 114, no. 495, pp. 223–246, 2004. DOI: 10.1111/j.1468-0297.2004.00207.x

[44] F. X. Diebold and R. S. Mariano, "Comparing predictive accuracy," *Journal of Business & Economic Statistics*, vol. 13, no. 3, pp. 253–263, 1995. DOI: 10.1080/07350015.1995.10524599

[45] J. E. Matheson and R. L. Winkler, "Scoring rules for continuous probability distributions," *Management Science*, vol. 22, no. 10, pp. 1087–1096, 1976. DOI: 10.1287/mnsc.22.10.1087

[46] D. J. Spiegelhalter, "Probabilistic prediction in patient management and clinical trials," *Statistics in Medicine*, vol. 5, no. 5, pp. 421–433, 1986. DOI: 10.1002/sim.4780050506

[47] D. Karlis and I. Ntzoufras, "Analysis of sports data by using bivariate Poisson models," *Journal of the Royal Statistical Society: Series D (The Statistician)*, vol. 52, no. 3, pp. 381–393, 2003. DOI: 10.1111/1467-9884.00366

[48] G. Baio and M. Blangiardo, "Bayesian hierarchical model for the prediction of football results," *Journal of Applied Statistics*, vol. 37, no. 2, pp. 253–264, 2010. DOI: 10.1080/02664760802684177

[49] D. Kahle, J. Seaman, and J. Stamey, "An overview of Bayesian computation," in *Case Studies in Bayesian Methods for Biopharmaceutical CMC*, ser. Chapman & Hall/CRC Biostatistics

Series, P. Faya and T. Pourmohamad, Eds., Boca Raton, FL: Chapman and Hall/CRC, 2022, pp. 5–24.

[50] E. Miyakawa and D. Kahle, "A practical comparison of Bayesian computing platforms in R," in *New Frontiers in Data Science (SDSC 2025)*, ser. Communications in Computer and Information Science, H. Han and J. Stamey, Eds., vol. 2556, Cham: Springer, 2025, pp. 58–74. DOI: 10.1007/978-3-031-99879-9_4

[51] M. Betancourt, *Diagnosing suboptimal cotangent disintegrations in Hamiltonian Monte Carlo*, 2016. arXiv: 1604.00695 [stat.ME].

[52] A. Gelman and D. B. Rubin, "Inference from iterative simulation using multiple sequences," *Statistical Science*, vol. 7, no. 4, pp. 457–472, 1992. DOI: 10.1214/ss/1177011136

[53] C. J. Geyer, "Practical Markov chain Monte Carlo," *Statistical Science*, vol. 7, no. 4, pp. 473–483, 1992. DOI: 10.1214/ss/1177011137

[54] D. J. Schad, M. Betancourt, and S. Vasishth, "Toward a principled Bayesian workflow in cognitive science," *Psychological Methods*, vol. 26, no. 1, pp. 103–126, 2021. DOI: 10.1037/met0000275

[55] T. Säilynoja, P.-C. Bürkner, and A. Vehtari, "Graphical test for discrete uniformity and its applications in goodness-of-fit evaluation and multiple sample comparison," *Statistics and Computing*, vol. 32, no. 2, p. 32, 2022. DOI: 10.1007/s11222-022-10090-6

[56] S. Talts, M. Betancourt, D. Simpson, A. Vehtari, and A. Gelman, *Validating Bayesian inference algorithms with simulation-based calibration*, 2018. arXiv: 1804.06788.

[57] M. Modrák et al., "Simulation-based calibration checking for Bayesian computation: The choice of test quantities shapes sensitivity," *Bayesian Analysis*, 2023, Advance publication. arXiv: 2211.02383.